

Review Article

Cybersecurity and Artificial Intelligence in Unmanned Aerial Vehicles: Emerging Challenges and Advanced Countermeasures

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The increasing adoption of artificial intelligence (AI)-driven unmanned aerial vehicles (UAVs) in military, commercial, and surveillance operations has introduced significant security challenges, including cyber threats, adversarial AI attacks, and communication vulnerabilities. This paper presents a comprehensive review of the key security threats and challenges faced by AI-powered UAVs, such as unauthorized access, GPS spoofing, adversarial manipulations, and UAV hijacking. We analyze advanced solutions including blockchain-secured UAV networks, post-quantum cryptography (PQC), adversarial AI training, self-healing AI models, and multi-factor authentication (MFA), which collectively strengthen UAV cybersecurity defenses. Our findings highlight the critical role of emerging technologies, including self-adaptive AI-driven UAVs capable of detecting and learning from novel cyber threats autonomously. We also discuss the integration of 6 G-powered communication networks for secure and ultra-fast encrypted transmissions, as well as Edge AI computing that enables real-time, onboard threat detection without cloud dependency. Furthermore, decentralized intelligence models and blockchain-based authentication are shown to enhance security in UAV swarms by preventing unauthorized infiltration. Overall, this review emphasizes the necessity of multilayered security frameworks that combine AI techniques, cryptographic measures, and decentralized swarm protection to ensure resilient, autonomous, and secure UAV operations in complex and high-risk environments.

Keywords: adversarial attacks; artificial intelligence security; blockchain-based security; cybersecurity challenges; explainable artificial intelligence; unmanned aerial vehicles

1. Introduction

The rapid advancement of artificial intelligence (AI)-driven unmanned aerial vehicles (UAVs) has revolutionized key sectors such as military defense, precision agriculture, disaster response, surveillance, and logistics [1]. By leveraging machine learning (ML), computer vision, and autonomous decision-making, these UAVs can carry out complex missions with minimal human oversight, enabling real-time navigation, dynamic environment adaptation, and optimized flight planning [2, 3]. However, this growing autonomy and operational complexity also make AI-driven UAVs increasingly vulnerable to sophisticated cybersecurity threats, including GPS spoofing, adversarial AI attacks, and

unauthorized access. These threats are not merely theoretical-real-world scenarios underscore their severity. For instance, in military operations, GPS spoofing has been used to redirect UAVs into enemy-controlled zones, while adversarial image perturbations have the potential to deceive UAV surveillance systems into misidentifying targets. In disaster response missions, compromised UAVs could fail to detect survivors or transmit falsified data, delaying rescue efforts. Similarly, in precision agriculture, hijacked drones could manipulate crop data or reroute pesticide applications, leading to economic loss. Such scenarios highlight the urgent need for robust countermeasures—such as blockchain-based authentication, AI-powered intrusion detection systems (IDSs), and self-healing AI models—to ensure secure,

reliable, and resilient UAV operations across these critical domains [4, 5].

In the context of evolving AI capabilities and their security implications, the integration of advanced AI in UAVs has revolutionized multiple sectors by enabling complex autonomous operations. AI-driven UAVs leverage a suite of sophisticated technologies; for instance, deep learning (DL) models like You Only Look Once (YOLO) and region-based convolutional neural networks (R-CNN) are employed for high-stakes object recognition tasks, from identifying wildfires to military threat identification. Concurrently, reinforcement Learning (RL) is pivotal in developing self-adaptive security systems, where UAVs can dynamically learn from and respond to cyber threats in real time, although this often comes with high computational costs [6]. Addressing privacy concerns from centralized data processing, federated learning (FL) offers a decentralized approach, allowing UAVs to train models locally and share only encrypted updates, a method that enhances data privacy but introduces trade-offs in communication overhead and scalability.

These technologies unlock critical applications across various domains. In defense, UAVs execute autonomous reconnaissance and precision strike missions, where the accuracy of AI-driven target recognition is paramount. In agriculture, they facilitate precision monitoring of crops and have been adapted for tracking livestock using blockchain-secured systems [7]. For disaster response, UAVs are indispensable in search-and-rescue (SAR) operations and establishing emergency communication networks in post-disaster scenarios. In logistics, they enable autonomous package delivery and are used for critical infrastructure monitoring, such as automated bridge inspections.

However, this deep integration of AI significantly broadens the attack surface, exposing these systems to a new generation of sophisticated threats that can have severe real-world consequences. These are not limited to conventional network attacks but include AI-specific vulnerabilities. Attackers can execute GPS spoofing and jamming to manipulate a UAV's navigation, leading to mission failure or hijacking. They can employ adversarial AI attacks, where imperceptible perturbations are introduced to sensor data to deceive object detection models, or use data poisoning to corrupt the AI's training dataset, causing systemic errors in flight behavior. Furthermore, vulnerabilities in communication channels can be exploited through man-in-the-middle (MITM) attacks to intercept or inject malicious commands. This expanded and complex threat landscape creates an urgent need for the dynamic, multilayered cybersecurity frameworks proposed in this review, which integrate AI-based intrusion detection, blockchain-based authentication, and post-quantum cryptography (PQC) to counter these advanced and evolving threats [6, 8].

While significant research has been conducted on cryptographic techniques, IDSs, and adversarial defenses [9, 10], much of the existing literature addresses these security dimensions in isolation, lacking a unified perspective. A critical gap persists in the development of a comprehensive, integrated framework that combines diverse defense layers—such as AI-based intrusion detection, blockchain-secured communication, PQC security, self-healing AI, and decentralized UAV swarm protection—into a cohesive architecture. To enhance clarity for

a broader audience, we define self-healing AI as intelligent systems capable of autonomously identifying, diagnosing, and recovering from disruptions or cyberattacks without human intervention, ensuring mission continuity in dynamic environments. Similarly, adversarial AI manipulations refer to maliciously crafted inputs (e.g., altered images or sensor data) that mislead AI models into making incorrect decisions, which can critically compromise UAV operations. Despite the growing relevance of these techniques, the literature has yet to fully synthesize their theoretical foundations and practical applications in safeguarding autonomous UAV missions.

To address the identified research gap in existing literature, this paper presents a comprehensive multidimensional theoretical framework that holistically connects the landscape of cybersecurity threats, inherent technical vulnerabilities, and advanced AI-driven countermeasures within the context of UAV operations. As illustrated in Figure 1, this framework serves as the conceptual foundation for systematically analyzing the interrelations between critical security domains impacting autonomous UAV systems. It unifies core elements such as AI-based threat detection, cryptographic defense mechanisms, decentralized swarm intelligence, and ethical AI governance to construct a layered defense model that reflects both theoretical insights and operational imperatives.

Guided by this framework, the primary objectives of this research are threefold. First, the paper aims to critically identify and categorize the cybersecurity threats that jeopardize UAV autonomy, data confidentiality, mission integrity, and communication reliability. These include adversarial AI manipulations, GPS spoofing, hijacking attempts, and unauthorized system access. Second, the study undertakes a detailed evaluation of emerging countermeasures, particularly AI-enhanced IDSs, blockchain-based authentication protocols, and PQC schemes, assessing their ability to mitigate contemporary and anticipated cyber threats. Third, the paper proposes a forward-looking security roadmap that aligns evolving AI capabilities—such as self-healing AI, FL, and 6 G-powered edge intelligence—with the operational resilience needs of autonomous UAV ecosystems.

By articulating the interactions between adversarial resilience, decentralized intelligence, secure communication, and ethical autonomy, the framework supports the development of actionable, future-ready cybersecurity strategies. This integrative approach not only enhances the theoretical foundations of UAV security research but also delivers practical guidance for implementing robust, adaptive, and scalable protection mechanisms in real-world deployments of AI-driven UAVs across defense, commercial, and critical infrastructure domains.

This paper makes significant contributions to the field in the following key ways:

1. **Comprehensive Threat Identification and Analysis:** This paper critically examines the diverse security threats affecting AI-driven UAVs, including cyberattacks, adversarial AI manipulations, GPS spoofing, and unauthorized UAV hijacking. By systematically categorizing these vulnerabilities, the review provides a detailed

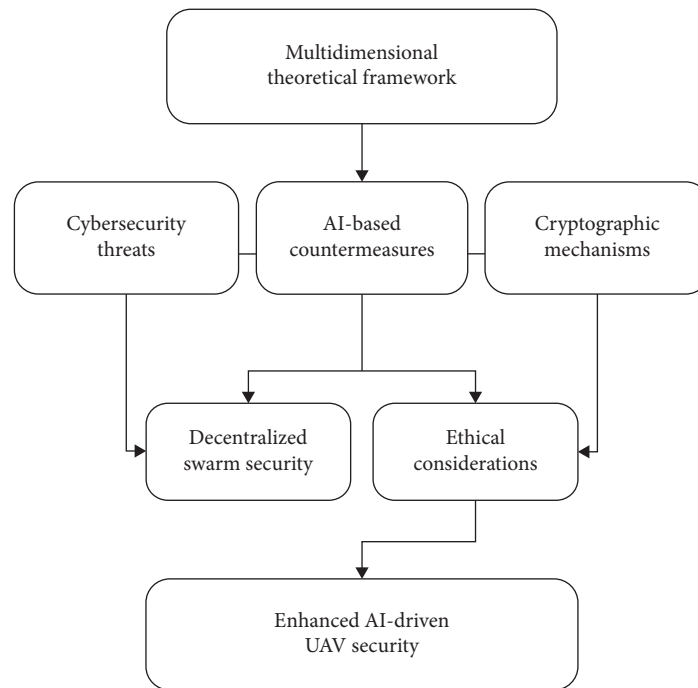


FIGURE 1: The proposed multidimensional theoretical framework for AI-driven UAV cybersecurity. The framework integrates cybersecurity threats, AI-based countermeasures, and cryptographic mechanisms, while incorporating decentralized swarm security and ethical considerations to achieve enhanced and resilient UAV security.

understanding of the evolving cybersecurity landscape, highlighting the risks posed to UAV mission integrity, data security, and autonomous operations.

2. **Integration of AI-Driven and Blockchain-Based Security Mechanisms:** The study explores the latest AI-powered security solutions, such as IDS, adversarial AI defense mechanisms, and self-healing AI models, which enhance UAV resilience against cyber threats. Additionally, it investigates blockchain-based authentication frameworks that ensure tamperproof UAV communication and mission execution, preventing unauthorized drone infiltration and command manipulation.
3. **Advancements in 6G and Edge AI for UAV Security:** A major contribution of this paper is its examination of 6G-powered UAV networks and Edge AI computing, which offer ultra-fast encrypted communication and decentralized AI-based threat detection. By eliminating reliance on vulnerable cloud-based security infrastructures, these technologies enable real-time autonomous threat mitigation, reducing the risks associated with data interception, network latency, and cyber intrusions.
4. **Future-Proofing UAV Cybersecurity with Decentralized Swarm Intelligence:** The paper presents novel decentralized AI-driven security frameworks for UAV swarm operations, ensuring that UAVs operate independently without a single point of failure. By leveraging swarm intelligence and blockchain-enhanced security models, UAVs can collectively detect, neutralize, and adapt to cyber threats, improving the long-term security and

reliability of autonomous UAV fleets in high-risk environments.

The structure of this paper is organized as follows: Section 2 reviews recent studies relevant to this work and highlights the key distinctions and advancements introduced in comparison to existing research. Section 3 out-lines the systematic methodology adopted for conducting this comprehensive review on AI-driven cybersecurity in UAVs. Section 4 identifies and categorizes emerging security threats targeting AI-powered UAVs, including adversarial attacks, spoofing, and intrusion vulnerabilities. Section 5 discusses key challenges in securing AI-integrated UAV systems, addressing issues such as model interpretability, real-time security constraints, and computational overhead. Section 6 provides a critical analysis of state-of-the-art AI-driven security solutions, categorized based on their application domain, AI techniques, implementation strategies, strengths, and limitations. Section 7 explores future research directions and emerging trends in AI-based UAV cybersecurity, emphasizing the need for advanced adversarial defense mechanisms, explainable AI (XAI), and quantum-resistant cryptographic techniques. Finally, Section 8 summarizes the key findings, highlighting the significance of AI-enhanced security frameworks in ensuring the resilience and reliability of UAV operations against evolving cyber threats.

2. Related Work

Recent advances in AI have significantly expanded the operational capabilities of UAVs across domains such as military reconnaissance, commercial delivery, surveillance, and disaster

TABLE 1: Comparison between traditional and AI-enhanced cybersecurity mechanisms for UAVs.

Feature	Traditional cybersecurity mechanisms	AI-enhanced cybersecurity mechanisms
Threat detection accuracy	Relies on predefined rules; limited ability to detect unknown threats	Utilizes machine learning and anomaly detection to identify both known and emerging threats
Adaptability	Static and requires manual updates to address new threats	Adaptive and capable of learning from new threat patterns in realtime
Response time	Often slower due to manual intervention and static rules	Enables faster, automated threat response with minimal latency
Scalability	Difficult to scale in dynamic and distributed UAV environments	Scales efficiently across large UAV networks using federated or decentralized learning
Mitigation strategy	Reactive, focused on patching known vulnerabilities	Proactive and predictive, capable of preventing attacks through anticipatory learning
Resource efficiency	Low computational demand but limited intelligence	May require more computational resources, but offers significantly higher intelligence and resilience
Resilience to adversarial attacks	Highly vulnerable to novel adversarial techniques	Employs adversarial training and self-healing models to resist and recover from attacks

response. However, this growing reliance on AI has simultaneously introduced a new wave of sophisticated cybersecurity vulnerabilities. Various research efforts have addressed individual aspects of UAV security, yet a unified, multidimensional treatment of threat detection, mitigation, and adaptive resilience remains lacking. A critical need exists for a comprehensive review that synthesizes traditional and AI-enhanced cybersecurity strategies while identifying their respective strengths and limitations.

Table 1 presents a comparative analysis of traditional versus AI-enhanced cybersecurity mechanisms. While traditional approaches typically rely on static, rule-based models with limited adaptability to novel threats, AI-enhanced mechanisms offer dynamic learning capabilities, real-time responsiveness, and scalable deployment across UAV swarms. For instance, AI models can autonomously detect anomalies, adapt to adversarial attack patterns, and initiate countermeasures without human intervention—advantages not achievable through conventional frameworks.

Despite the benefits of AI, prior studies have generally approached UAV cybersecurity through narrow, domain-specific lenses. For example, Kumar and Chaudhary [11] offered a broad survey of cybersecurity vulnerabilities in UAVs, focusing on authentication protocols, lightweight cryptographic methods, and system-level constraints. However, their emphasis was largely on conventional mechanisms, with limited attention to adversarial ML (AML) threats, self-healing AI, or decentralized architectures.

Xi et al. [12] advanced the offensive capabilities of adversarial AI by introducing the URAdv framework, which generates robust adversarial patches against UAV object detection systems under varying environmental conditions. While the work provides critical insights into vulnerabilities, it omits discussions on AI-based defense strategies and their integration into holistic UAV security frameworks.

Tian et al. [13] examined adversarial data injection attacks within smart grid systems through the LES-SON framework, identifying vulnerabilities in DL-based localization. Although technically sound, their methodology is domain-bound to smart grids and does not account for UAV-specific challenges

such as autonomous swarm coordination or aerial adversarial manipulation.

Similarly, Zhao et al. [14] introduced the DQM-attack, targeting 4D-flight trajectory prediction systems with stealthy adversarial perturbations. Their contribution lies in exposing AI weaknesses, but the work stops short of proposing countermeasures applicable to real-time UAV security scenarios or integrating defense with blockchain and edge computing.

In contrast to these fragmented efforts, the present review takes a holistic, forward-looking approach. It synthesizes key threat models, cryptographic advancements, swarm resilience mechanisms, and decentralized AI strategies into a cohesive framework. This includes the integration of AI-based IDSs, PQC, blockchain-secured UAV communications, and self-healing models—elements critical for the next generation of autonomous UAV defense.

Furthermore, the review explores emerging themes such as 6G-enabled low-latency communication, Edge AI-based decision-making, and the ethical governance of autonomous systems. Table 2 highlights the distinct contributions of previous studies and situates this paper's comprehensive scope and multidimensional treatment in contrast to earlier, domain-specific research.

In summary, while prior works have made important contributions to specific dimensions of UAV cybersecurity, they fall short of offering a comprehensive, unified approach. This paper fills that void by integrating diverse technologies and methodologies into a strategic framework tailored for realtime, scalable, and ethically grounded security in AI-driven UAV systems.

3. Research Methodology

To ensure a comprehensive and unbiased selection of studies related to AI-driven cybersecurity in UAV systems, this review adopts a structured methodology. The approach involves defining selection criteria, conducting a systematic search across academic databases, applying inclusion/exclusion criteria, and filtering papers systematically using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)

TABLE 2: Comparison of existing work with the present study.

Study	Focus	Limitations compared to this paper
Kumar and Chaudhary [11]	Cybersecurity vulnerabilities and cryptographic schemes for UAVs.	Focused on conventional cybersecurity; limited exploration of AI-specific adversarial threats, decentralized swarm security, or blockchain-based frameworks.
Xi et al. [12]	Generation of robust adversarial patches for UAV object detection.	Emphasized adversarial attack creation; lacked comprehensive defense strategies or integration with broader cybersecurity frameworks for UAVs.
Tian et al. [13]	Multi-label adversarial false data injection attacks in smart grids.	Focused on smart grids rather than UAVs; limited relevance to UAV cybersecurity and lack of discussion on swarm or AI resilience strategies.
Zhao et al. [14]	Adversarial attacks on 4D-flight trajectory prediction models.	Addressed vulnerabilities in trajectory systems; no focus on real-time UAV security countermeasures or decentralized communication frameworks.
This review paper	Comprehensive review of cybersecurity threats and advanced AI-based, cryptographic, swarm-based, blockchain-enhanced UAV defense mechanisms.	Bridges multiple domains (AI threats, cryptography, swarm security, blockchain, ethics); proposes multilayered, future-ready UAV cybersecurity frameworks including Edge AI and 6G integration.

framework [15]. The selection of the PRISMA framework as the guiding methodology was deliberate, as it provides a robust structure for conducting systematic reviews. PRISMA is chosen for its well-established guidelines that ensure transparency and rigor in the review process, specifically in the identification, screening, eligibility, and inclusion of relevant studies. This framework is widely recognized for minimizing bias and enhancing the reliability of literature reviews by providing a clear, step-by-step approach to study selection and synthesis. This methodology enables the identification of high-quality research focusing on cybersecurity challenges and AI-based countermeasures in UAV networks.

3.1. Criteria for Selecting Papers. To enhance transparency and rigor, this review adopts a systematic literature review (SLR) protocol in line with the PRISMA framework. The selection process was designed to ensure the inclusion of high-quality, peer-reviewed studies that are directly relevant to the domain of AI-driven UAV cybersecurity. Specific criteria were defined to guide the inclusion and exclusion of sources. Priority was given to research focusing on key areas such as AI-based IDSs, self-healing AI mechanisms for UAV security, adversarial learning and defense strategies, blockchain-based UAV authentication, FL, and XAI techniques for autonomous UAV operations.

Only articles published in high-impact, peer-reviewed journals and top-tier conference proceedings were considered to ensure the credibility and technical soundness of the selected literature. The review was restricted to studies published between January 2020 and March 14, 2025 to reflect the most recent and relevant advancements in the field. Additional selection factors included the presence of empirical validation (e.g., simulation or real-world UAV datasets), detailed methodological descriptions, and contributions to UAV-specific cybersecurity challenges. By systematically applying these criteria and documenting the selection

process, this review ensures methodological transparency, minimizes bias, and offers a comprehensive and credible synthesis of the state-of-the-art in AI-enhanced UAV cybersecurity.

3.2. Databases Searched. A thorough literature search was conducted across multiple academic databases to ensure broad coverage of AI-driven cybersecurity for UAVs. The selection of databases was based on their relevance to AI, cybersecurity, RL, and UAV applications, focusing on high-impact peer-reviewed journals, conference proceedings, and technical reports, as detailed in Table 3. The search strategy combined keyword-based queries, Boolean operators, and citation tracking to identify relevant studies.

The primary keywords used in the search included:

- “AI-driven UAV security,”
- “Intrusion detection systems in UAV networks,”
- “Self-healing UAV cybersecurity frameworks,”
- “Blockchain-based UAV authentication,”
- “Adversarial machine learning for UAV security,”
- “XAI for UAV cybersecurity,”
- “Federated learning in UAV networks,”
- “Resilient AI models for UAV operations” 6,
- “AI-based anomaly detection in UAV communication,”
- “Quantum-resistant cryptography for UAVs.”

Additionally, forward and backward citation tracking was employed to capture influential studies and emerging research trends in UAV cybersecurity.

3.3. Inclusion and Exclusion Criteria. A well-defined set of inclusion and exclusion criteria was applied to refine the selection process and ensure that only the most relevant and methodologically sound studies were considered.

The inclusion criteria prioritized:

TABLE 3: Academic databases searched.

Database	Purpose
IEEE Xplore	Focuses on AI-based UAV security, adversarial learning, blockchain authentication, and intrusion detection systems.
ScienceDirect (Elsevier)	Provides research on AI-driven UAV security frameworks, cybersecurity solutions, and UAV cryptographic security.
SpringerLink	Covers reinforcement learning, federated learning, and deep learning-based UAV anomaly detection.
ACM Digital Library	Features studies on UAV network security, AI-based UAV resilience, and computational intelligence in cybersecurity.
Google Scholar	Enables a broader search, including preprints, technical reports, and conference articles on UAV cybersecurity.

- Studies focusing on AI-driven UAV security, including intrusion detection, anomaly detection, and self-healing security frameworks.
- Research on blockchain, FL, AML, and XAI for UAV cybersecurity.
- Studies presenting empirical evaluations with UAV security datasets, adversarial attack simulations, or real-world UAV applications.
- Papers published in high-impact peer-reviewed journals and conferences, such as IEEE Transactions on Cybernetics, ACM Computing Surveys, and Elsevier's Journal of Network and Computer Applications.

The exclusion criteria filtered out:

- Studies that focused solely on ML without UAV-specific cybersecurity applications.
- Articles on general AI-based cybersecurity without specific emphasis on UAV networks.
- Papers lacking empirical validation, security benchmarks, or adversarial threat modeling.
- Nonpeer-reviewed sources, incomplete preprints, and short abstracts with insufficient technical depth.

By applying these criteria, this review ensures the inclusion of impactful and high-quality studies that contribute to the advancement of AI-driven UAV cybersecurity.

3.4. Paper Selection Process. To ensure transparency and reproducibility, the PRISMA framework was followed for systematic literature selection, as presented in Figure 2.

The systematic selection process involved:

1. Identification: An initial keyword-based search retrieved 333 papers from IEEE Xplore, ScienceDirect, Springer-Link, ACM Digital Library, and Google Scholar.
2. Screening: Duplicate studies were removed, followed by title and abstract screening, reducing the number to 312 relevant papers.
3. Eligibility Check: A full-text review was conducted using predefined inclusion and exclusion criteria, further refining the selection to 161 studies.
4. Final Selection: A quality assessment was performed based on empirical rigor, AI methodology depth, and

UAV security relevance, resulting in the inclusion of 127 key papers.

3.5. Quality Assessment of Selected Studies. A rigorous quality assessment was applied to ensure the credibility and impact of selected studies. Each paper was examined based on:

- Scientific Rigor: Evaluation of algorithmic depth, theoretical contributions, and AI-based methodologies for UAV cybersecurity.
- Experimental Validation: Preference for studies using real-world UAV security datasets, adversarial attack models, and benchmark testing.
- Comparative Performance Analysis: Studies providing numerical comparisons, statistical validation, and benchmarks against existing UAV security models.
- Reproducibility: Assessment of studies offering open-source implementations, well-documented methodologies, and publicly available datasets.

By applying this structured quality assessment methodology, the review ensures a systematic and unbiased selection of studies in AI-driven UAV cybersecurity.

3.6. Trends and Advancements. The review examined 127 articles published between 2020 and March 14, 2025, highlighting a growing re-search focus on AI-driven cybersecurity solutions for UAVs. This increasing trend is depicted in Figure 3, which illustrates the rising number of publications in high-impact journals and conferences. The distribution of articles across journals and conferences is presented in Figure 4, while Figure 5 shows the publisher-wise distribution. Additionally, Figure 6 provides a breakdown of research categories covered in this study.

We ensured that our review captures the most recent advancements and research trends in AI-driven cybersecurity solutions for UAVs. The chosen sample size is appropriate as it provides a comprehensive and representative overview of the rapidly growing body of literature in this field. By including peer-reviewed journal articles, conference papers, and high-impact studies, we ensured diversity in research perspectives and robustness in findings. Additionally, the systematic selection and screening process, following established protocols like PRISMA, enhanced the reliability and validity of the results,

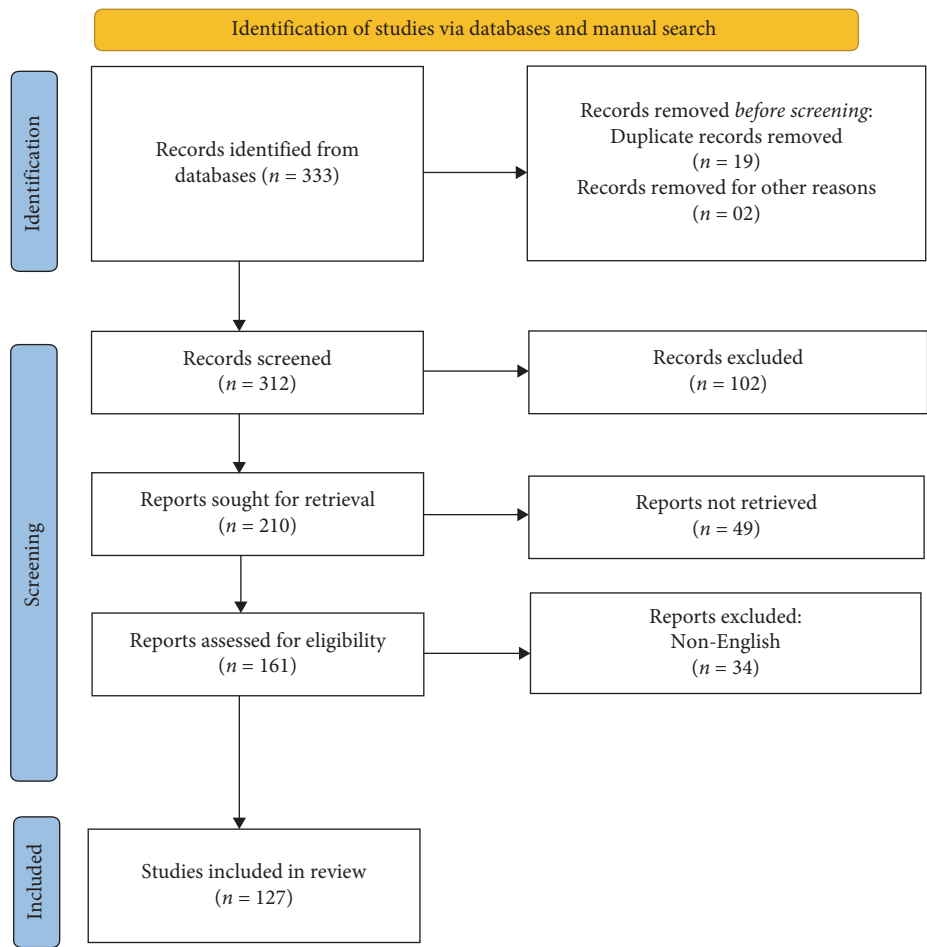


FIGURE 2: PRISMA framework for study selection.

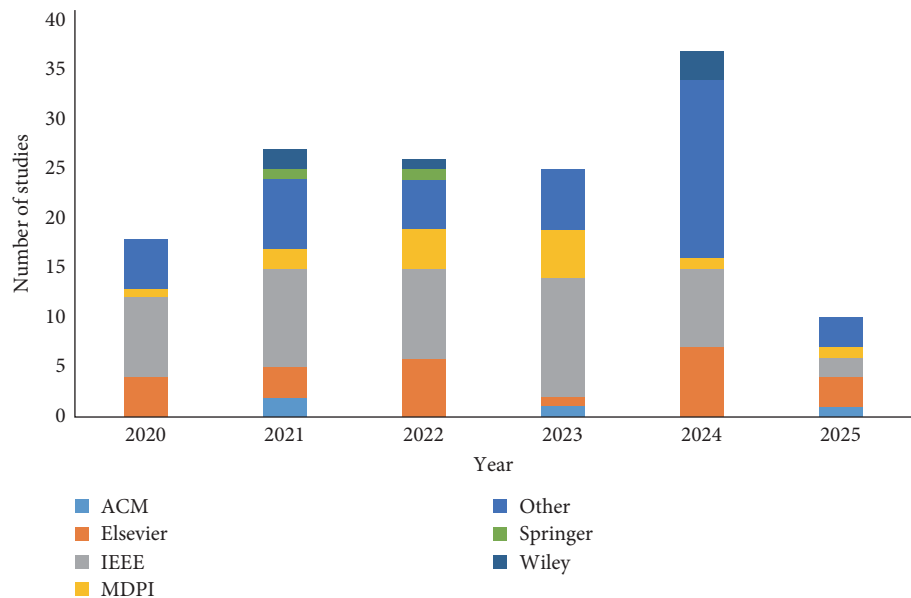


FIGURE 3: Year-wise trends in AI-driven cybersecurity solutions for UAVs.

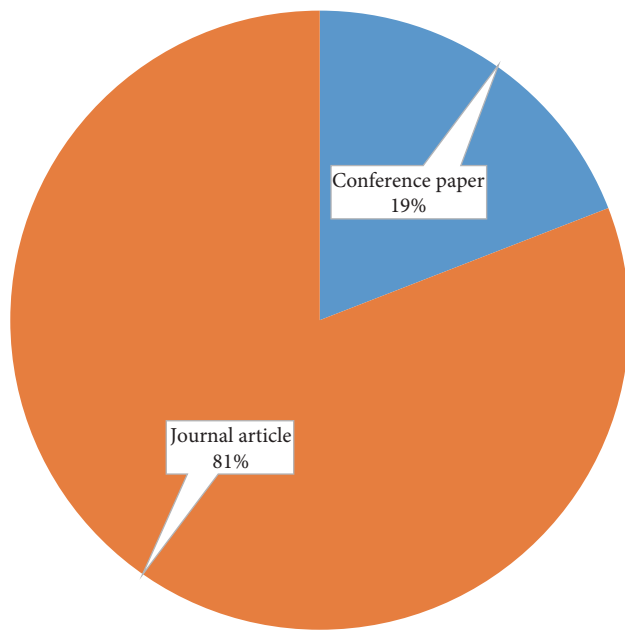


FIGURE 4: Distribution of UAV cybersecurity research articles across journals and conferences.

minimizing selection bias and ensuring that critical developments were not overlooked.

By addressing biases and ethical considerations in the reviewed studies, this methodology ensures a robust evaluation of the field, offering insights into the latest advancements and existing gaps in UAV cybersecurity.

4. Security Threats in AI-Driven UAVs

The widespread adoption of AI-driven UAVs across various critical domains, including military surveillance, law enforcement, disaster response, and commercial logistics, has elevated concerns regarding their security and operational resilience [16, 17]. While AI enhances UAV autonomy, intelligence, and efficiency, it also introduces significant vulnerabilities. The reliance on AI algorithms, wireless communication systems, real-time data transmission, and autonomous decision-making exposes UAVs to an array of security threats that can disrupt their functions, compromise data integrity, and pose risks to national security [18].

These security threats can be broadly classified into three major categories: (i) cybersecurity threats, which include hacking, GPS spoofing, and AI adversarial attacks; (ii) physical security threats, encompassing hijacking, hardware tampering, and electromagnetic interference; and (iii) privacy and ethical concerns, where UAVs raise serious issues related to surveillance, data security, and autonomous decision-making accountability. Each category presents unique challenges that necessitate advanced security frameworks, robust AI defenses, and strict regulatory measures to ensure the safe and responsible use of UAV technology.

4.1. Cybersecurity Threats. Cyber threats are among the most severe challenges facing AI-powered UAVs, as attackers can exploit vulnerabilities in communication networks, AI-based

decision-making, and UAV software systems. Cybercriminals, terrorists, and state-sponsored hackers can manipulate UAV controls, intercept sensitive mission data, and disrupt autonomous UAV operations to serve malicious agendas [19, 20]. With UAVs increasingly relying on 5G networks, cloud computing, and AI automation, the attack surface for cyber threats has expanded significantly, making cybersecurity a top priority. The most prominent cyber threats include hacking and unauthorized access, jamming and spoofing attacks, and AI adversarial manipulations.

4.1.1. Hacking and Unauthorized Access. One of the most significant cybersecurity risks in UAV operations is the unauthorized access to UAV control systems and mission data, primarily due to vulnerabilities in wireless communication protocols such as Wi-Fi, LTE, 5G, and satellite networks [21, 22]. UAVs rely heavily on these real-time communication channels to exchange critical information with ground control stations, cloud-based AI processing units, and other UAVs in swarm-based operations. However, if these channels are inadequately secured or encrypted, they become susceptible to cyber intrusions, allowing attackers to intercept, manipulate, or disrupt UAV operations.

Unauthorized access enables cybercriminals or hostile entities to hijack UAV controls by injecting malicious commands into the system, leading to a complete takeover of flight operations [23]. By exploiting weak authentication protocols, attackers can alter mission parameters, redirect UAVs to unintended locations, or force them to crash or self-destruct. Additionally, attackers can intercept and manipulate UAV data transmissions, leading to false intelligence gathering, modification of surveillance footage, or the transmission of fake sensor data, which could have catastrophic consequences in military, border security, and emergency response missions. Another major concern is the ability of hackers to inject malware into UAV operating systems, compromising the AI-driven decision-making process and flight autonomy. Malware infections can introduce backdoors for persistent control, corrupt UAV navigation algorithms, or even integrate the compromised drone into a larger botnet attack, where multiple hijacked UAVs can be used in coordinated cyber warfare operations.

Among the most dangerous cyberattack techniques is the MITM attack, where hackers intercept UAV signals, modify mission-critical commands, and take full control of the drone [24, 25]. These attacks can occur when UAV-to-ground station (GS) communication is not encrypted or uses weak authentication protocols, allowing adversaries to eavesdrop on UAV transmissions, inject malicious data, or manipulate flight navigation commands. MITM attacks pose a particularly severe risk in military and law enforcement applications, where unauthorized UAV access could lead to espionage, intelligence theft, sabotage, or direct attacks on critical infrastructure. In national security operations, a compromised UAV could be repurposed for spying on classified missions, stealing sensitive government data, or even executing autonomous drone strikes under false authorization, creating unprecedented security risks.

To mitigate these risks, researchers are exploring quantum-safe encryption [26] and blockchain-based authentication

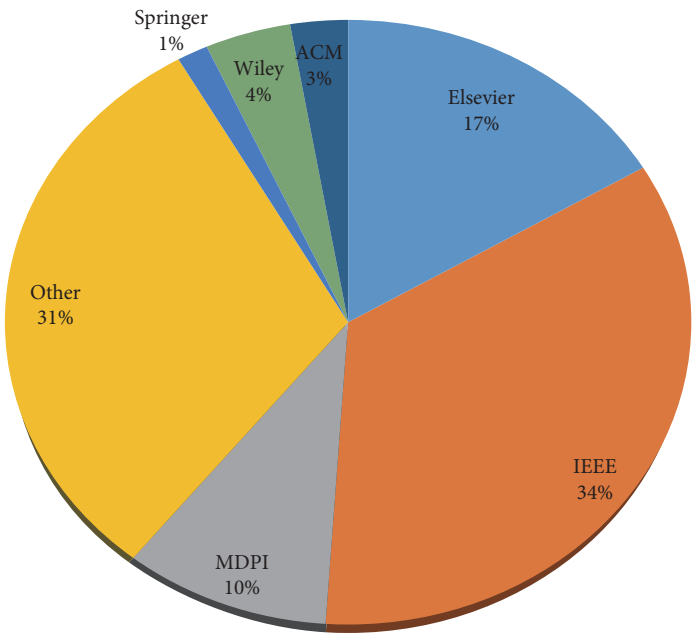


FIGURE 5: Publisher distribution of UAV security research.

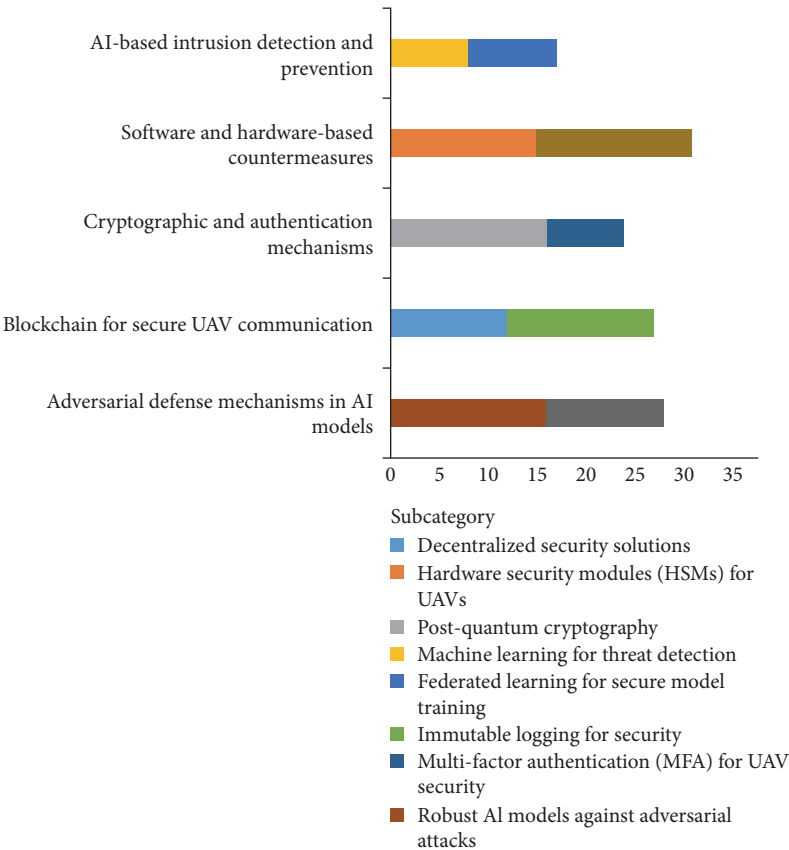


FIGURE 6: Category distribution of UAV security research.

mechanisms [27], ensuring that only authenticated and verified entities can access UAV control systems and mission data. Quantum-resistant cryptographic algorithms, such as CRYSTALS-Kyber and CRYSTALS-Dilithium, have been

proposed to strengthen UAV encryption methods against future quantum computing threats, which could otherwise break traditional encryption standards. Additionally, blockchain technology enables decentralized, tamper-proof

authentication, where UAV flight logs, control authorizations, and communication exchanges are securely recorded on an immutable ledger, preventing unauthorized modifications or access. Researchers are also integrating zero trust network architecture (ZTNA) into UAV security frameworks, which requires continuous authentication and strict access controls for all communication nodes, significantly reducing the risk of UAV hijacking.

By implementing these advanced security frameworks, UAVs can be safeguarded against evolving cyber threats, ensuring that their autonomous capabilities remain resilient, mission data remains intact, and flight operations are not compromised by unauthorized interference. As cyber warfare techniques become more sophisticated, securing UAV networks with next-generation cryptographic techniques and decentralized authentication frameworks will be critical to maintaining the operational integrity and safety of AI-driven UAVs in high-risk environments [3].

4.1.2. Jamming and Spoofing Attacks. GPS serves as the primary navigation system for most autonomous UAVs, enabling precise positioning, route planning, and mission execution [28, 29]. However, this reliance on GPS also makes UAVs a prime target for cyberattacks, as malicious actors can exploit vulnerabilities in GPS technology to disrupt or manipulate UAV operations. Among the most common GPS-based attacks are GPS jamming and GPS spoofing, both of which can severely impact UAV performance, leading to navigation failures, mission deviations, or even crashes.

GPS jamming occurs when an attacker overwhelms a UAV's GPS receiver with high-power noise signals, effectively blocking legitimate GPS transmissions. As a result, the UAV is unable to accurately determine its position, which can lead to loss of positional accuracy, flight path deviations, or system failures [30]. In worst-case scenarios, a jammed UAV may drift uncontrollably, enter restricted airspace, or become unresponsive, making it a security risk in both military and civilian applications. GPS jamming has been widely reported in military conflicts, where adversaries use electronic warfare techniques to disrupt enemy UAVs, thereby denying aerial surveillance, target acquisition, or combat support operations. Similarly, in commercial UAV operations, GPS jamming poses a major threat, as it can lead to collisions with manned aircraft, loss of control in congested urban environments, or unintentional border crossings.

On the other hand, GPS spoofing is a more deceptive cyberattack in which attackers transmit false GPS signals to mislead UAVs into believing incorrect location data. Unlike jamming, which simply denies access to GPS signals, spoofing actively manipulates the UAV's navigation system, causing it to fly off-course, crash, or land in unauthorized locations [28]. In military scenarios, adversaries have used GPS spoofing to hijack enemy drones by redirecting them to controlled airspace, where they can be seized, repurposed, or destroyed. For civilian applications, spoofing attacks can have serious consequences, including the misdirection of delivery drones, interference with air traffic control systems, and disruption of emergency response UAVs deployed for disaster relief. Furthermore,

terrorist groups or criminal organizations could exploit GPS spoofing to redirect UAVs carrying sensitive payloads, posing a significant national security threat.

To mitigate the risks associated with GPS-based cyberattacks, researchers are actively developing PQC techniques that can enhance GPS security through encryption and authentication mechanisms. One such approach is CRYSTALS-Kyber, a quantum-resistant cryptographic algorithm designed to secure GPS signals against interception and manipulation. By incorporating quantum-safe encryption into UAV navigation systems, researchers aim to prevent unauthorized access to location data and ensure the integrity of GPS-based flight operations. Additionally, emerging technologies such as multi-sensor fusion, AI-driven anomaly detection, and blockchain-based GPS authentication are being explored to further reduce UAV dependency on GPS alone and improve overall resilience against cyber threats [31].

4.1.3. AI-Powered Adversarial Attacks. AI-driven UAVs leverage DL models for a wide range of autonomous functions, including object detection, flight path optimization, and target recognition [32]. These capabilities enable UAVs to operate without direct human intervention, making them invaluable in military, surveillance, and commercial applications. However, despite the advancements in AI, UAVs remain highly vulnerable to adversarial attacks, which can degrade their performance, mislead their decision-making systems, and compromise mission success.

One of the most concerning threats to AI-driven UAVs is adversarial attacks, where attackers introduce small, imperceptible perturbations into sensor inputs or image datasets. These modifications are often invisible to the human eye but can significantly alter the AI's perception, causing misclassification of objects or incorrect flight decisions. In the context of military UAV operations, adversarial perturbations can be weaponized to deceive AI-based target recognition systems, making UAVs fail to detect enemy vehicles, misidentify threats, or incorrectly classify obstacles in their environment. Such vulnerabilities could reduce UAV effectiveness in reconnaissance missions or lead to severe operational failures in combat scenarios [33].

Another major concern is model poisoning attacks, where attackers manipulate AI training datasets to inject misleading patterns, resulting in systematic errors during real-world UAV operations. This type of attack occurs during the training phase, meaning that the UAV unknowingly learns incorrect classifications or biases that persist throughout its deployment [34]. A poisoned AI model used for autonomous navigation could misinterpret terrain data, misjudge altitudes, or incorrectly identify safe landing zones, increasing the risk of mid-air collisions, mission failures, or unintended landings in hostile territories. In military applications, an adversary could poison UAV training datasets to create blind spots, preventing the AI from detecting specific enemy assets, which could lead to intelligence failures or compromised surveillance missions.

To mitigate these risks, researchers emphasize the importance of AI explainability and adversarial training techniques. XAI frameworks help enhance transparency in UAV decision-making, allowing operators to understand how AI models

arrive at specific conclusions and detect potential adversarial manipulations before they affect mission outcomes. Additionally, adversarial training—where AI models are deliberately exposed to adversarial attacks during the learning phase—can help UAVs develop resilience against deceptive inputs. By training AI models to recognize and counter adversarial perturbations, UAVs can improve their ability to operate reliably in contested environments where cyber and AI-based threats are prevalent [35].

Overall, adversarial threats to AI-driven UAVs highlight critical gaps in current AI security frameworks. As UAVs become more autonomous, ensuring the robustness of their AI models against adversarial manipulations will be essential for maintaining their reliability and security in high-risk environments. Further research into robust DL algorithms, secure AI training methodologies, and real-time anomaly detection systems will be necessary to develop next-generation UAVs that can withstand AI-targeted attacks while maintaining operational integrity in mission-critical scenarios.

4.2. Physical Security Threats. While cybersecurity threats primarily exploit software vulnerabilities in AI-driven UAVs, physical security threats focus on direct interference with UAV hardware, control systems, and communication networks [36, 37]. These threats include drone hijacking, hardware tampering, and electromagnetic disruption, which can significantly impact UAV operational integrity, mission execution, and data security. In critical applications such as military reconnaissance, law enforcement surveillance, and commercial drone delivery, attackers seeking to compromise UAVs may use physical interception techniques, firmware manipulation, and signal cloning methods to gain unauthorized control. Addressing these threats requires the integration of advanced IDS, secure hardware designs, and real-time UAV monitoring mechanisms to prevent unauthorized interference.

4.2.1. Drone Hijacking and Hardware Tampering. Physical security threats, such as drone hijacking and hardware tampering, pose significant risks to AI-driven UAVs, particularly in military, law enforcement, and commercial applications. Unlike cyber threats, which exploit software vulnerabilities, physical attacks involve direct interference with UAV hardware, communication systems, or firmware to gain unauthorized access and control [38]. These attacks are particularly concerning as they can compromise mission-critical UAV operations, disrupt intelligence-gathering missions, and expose sensitive data to adversaries.

One of the most common hijacking methods involves intercepting UAVs mid-flight using physical counter-drone technologies such as nets, electromagnetic pulses (EMP), or radio frequency (RF) hacking tools. In military and high-security zones, attackers may deploy anti-drone weapons that disable UAVs in the air, causing them to crash or land in a controlled area for retrieval. EMP attacks, in particular, can permanently damage UAV circuitry, leading to a complete shutdown of onboard electronics and rendering the UAV non-operational. These attacks highlight the critical need for electromagnetic shielding and self-repairing UAV architectures to counteract physical interception threats.

Another emerging concern is firmware tampering, where attackers physically access a UAV to modify its software and inject malicious code [39]. A compromised firmware system can allow attackers to remotely control the UAV, disable security features, or alter mission objectives without detection. This form of tampering is especially dangerous in military reconnaissance and intelligence missions, where unauthorized modifications could lead to misdirected surveillance, incorrect target identification, or data manipulation. Recent security studies emphasize the importance of tamper-proof UAV designs, which integrate secure boot verification systems to detect unauthorized firmware alterations before the UAV becomes operational.

A more sophisticated hijacking technique involves cloning UAV communication signals to spoof legitimate commands and take control of drone operations. UAVs communicate with ground control stations using wireless protocols such as 5 G, satellite links, and Wi-Fi, making them vulnerable to signal interception and manipulation. Attackers can exploit weak encryption mechanisms to inject false commands, forcing the UAV to deviate from its intended flight path or land in unauthorized locations. In recent years, researchers have highlighted the need for quantum-safe cryptographic techniques and blockchain-based authentication mechanisms to ensure that only verified commands are executed by the UAV, preventing signal spoofing and unauthorized control takeovers.

To mitigate drone hijacking and hardware tampering, AI-driven IDS have been proposed to monitor real-time UAV behavior and identify anomalous activities [7]. These systems utilize ML and anomaly detection algorithms to detect unexpected deviations from flight paths, unauthorized software modifications, and suspicious control inputs. By integrating AI-powered threat detection frameworks, UAVs can autonomously identify and neutralize hijacking attempts, improving their resilience against evolving physical security threats.

As UAV deployments continue to expand across defense, law enforcement, and commercial sectors, addressing drone hijacking and hardware tampering through advanced encryption, IDSs, and autonomous UAV self-defense mechanisms will be essential in ensuring their secure and uninterrupted operations in adversarial environments.

4.2.2. Electromagnetic and RF Attacks. Electromagnetic attacks pose a significant threat to AI-driven UAV operations, particularly in military, defense, and high-security environments where UAVs rely on electronic components and wireless communication systems for navigation, surveillance, and data transmission [28]. These attacks exploit vulnerabilities in UAV electronic circuitry and radio communication channels, making drones susceptible to disruptions, hijacking, or permanent damage. Electromagnetic interference is particularly concerning for autonomous UAVs, which depend on continuous data exchange with ground control stations and onboard AI decision-making systems.

One of the most critical forms of electromagnetic attack is the EMP weapon, which generates high-energy pulses capable of permanently disabling UAV electronics. EMP attacks can fry integrated circuits, disrupt power distribution networks, and

cause irreversible damage to UAV processors, rendering the drone nonoperational. In military conflicts, EMP weapons have been explored as counter-UAV measures to neutralize hostile drones before they can complete reconnaissance or combat missions. The increasing reliance on electronic warfare systems has prompted the need for EMP-resistant UAV architectures, incorporating radiation-hardened microprocessors and shielded electronic components to mitigate potential damage.

Another common electromagnetic threat is RF interference, where attackers jam or overload UAV communication channels, causing signal loss, miscommunication between UAVs and control centers, and even forced UAV shutdowns. UAVs rely on RF signals for command execution, navigation data relay, and swarm coordination, making them highly vulnerable to RF jamming and spoofing attacks [30]. In civilian UAV applications, RF interference can disrupt drone delivery services, interfere with emergency response UAVs, or compromise industrial drone networks used for surveillance and mapping. In military operations, RF jamming has been used to disable enemy UAVs by flooding communication channels with noise signals, preventing the transmission of control commands and real-time intelligence data.

To counteract EMP and RF-based threats, researchers are exploring advanced countermeasures to enhance UAV resilience against electromagnetic attacks. One of the most promising solutions involves shielding UAV electronics using electromagnetic-resistant materials, which can reduce the impact of high-energy pulses and prevent circuit damage. Additionally, hardened circuit designs with fault-tolerant architectures are being developed to withstand electromagnetic disruptions and maintain operational stability in high-risk environments. AI-powered radio-frequency anomaly detection systems are also being implemented to identify and respond to unexpected RF interference in real-time, enabling UAVs to adjust communication frequencies or switch to alternative navigation methods to maintain functionality.

As UAVs continue to play a pivotal role in defense, intelligence, and critical infrastructure monitoring, ensuring electromagnetic resilience through shielded electronics, AI-driven frequency adaptation, and EMP-resistant UAV designs will be crucial in mitigating electronic warfare threats and ensuring secure and uninterrupted UAV operations in contested environments.

4.3. Privacy and Ethical Threats. The increasing deployment of AI-powered UAVs in surveillance, law enforcement, and defense has sparked significant ethical debates concerning privacy, data security, and decision-making accountability [40, 41]. While UAVs enhance operational efficiency, intelligence gathering, and threat detection, their ability to autonomously monitor, track, and analyze individuals raises concerns about mass surveillance, unauthorized data collection, and potential algorithmic biases. The ethical implications extend beyond privacy violations to issues of data security risks, regulatory gaps, and the lack of human oversight in AI-driven decision-making. Additionally, as UAVs take on more autonomous roles in target identification, law enforcement, and military engagements,

critical questions arise regarding algorithmic bias, accountability for wrongful actions, and compliance with international human rights laws. Addressing these concerns requires the development of ethical AI frameworks, transparent decision-making models, and strict regulatory policies to ensure that AI-powered UAVs operate within legal, ethical, and socially responsible boundaries.

4.3.1. Surveillance and Data Privacy Issues. The integration of high-resolution cameras, facial recognition technology, and AI-driven monitoring capabilities into UAVs has significantly enhanced their ability to conduct surveillance, track individuals, and collect vast amounts of data [42, 43]. While these advancements contribute to public safety, border security, and crime prevention, they also raise serious ethical and legal concerns regarding privacy violations, unauthorized data collection, and regulatory oversight. The ability of UAVs to monitor individuals remotely and autonomously increases the risk of mass surveillance, data misuse, and the erosion of personal privacy rights.

One of the most pressing concerns is the potential for mass surveillance, where UAVs can continuously track individuals, analyze their movements, and store behavioral data without their consent. In urban environments, AI-powered UAVs deployed for traffic monitoring, law enforcement, and public event security may unintentionally or deliberately infringe on civil liberties by monitoring individuals without due process. The use of facial recognition systems and biometric analysis further amplifies these concerns, as UAVs can identify and profile individuals in real time, raising fears of state surveillance, racial profiling, and discriminatory policing practices.

Another critical issue is unauthorized data collection, where UAVs gather sensitive location data, biometric information, and personal footage that could be exploited for commercial, political, or malicious purposes. The lack of strict data security measures and encryption protocols increases the risk of cyber intrusions, data leaks, and unauthorized access to UAV-collected information. In corporate and governmental settings, unauthorized UAV surveillance could lead to industrial espionage, identity theft, and breaches of national security if sensitive information is improperly stored or accessed by unauthorized entities.

A major challenge in addressing these privacy risks is the absence of well-defined legal and regulatory frameworks governing the use of AI-powered UAVs for surveillance [44]. Many countries lack comprehensive UAV privacy regulations, leading to inconsistent policies on data retention, consent requirements, and lawful drone operations. Without standardized laws, UAV operators—whether government agencies, private companies, or individuals—may exploit legal loopholes to conduct invasive surveillance without accountability.

To mitigate these concerns, researchers emphasize the need for strong legislative initiatives, ethical AI guidelines, and robust data protection frameworks to ensure that UAV surveillance operations align with privacy laws and ethical governance principles [16]. The development of privacy-preserving AI models, encrypted data storage solutions, and UAV access control mechanisms is also crucial in reducing the risk of

privacy violations and unauthorized data exploitation. Furthermore, policymakers must establish clear boundaries on UAV surveillance practices, incorporating transparency, accountability, and public oversight to ensure that AI-driven UAVs operate within ethical and legal constraints while respecting individual privacy rights.

4.3.2. Autonomous Decision-Making Risks. The increasing autonomy of AI-driven UAVs has introduced significant ethical and operational challenges, particularly in military, law enforcement, and surveillance applications. Unlike traditional UAVs, which rely on human operators for mission execution and decision-making, autonomous UAVs use AI algorithms to analyze data, assess threats, and take action independently [4, 45]. While this capability enhances efficiency, reaction time, and operational effectiveness, it also raises serious concerns regarding accuracy, bias, and accountability in decision-making. The potential for AI-driven UAVs to make independent decisions without human oversight has sparked debates about their reliability, ethical implications, and the risks associated with algorithmic errors.

One of the primary risks associated with autonomous UAV decision-making is incorrect target recognition, particularly in military operations where UAVs are used for surveillance, reconnaissance, and precision strikes. AI models trained on incomplete or biased datasets may misclassify civilian objects as military targets, leading to unintended casualties and collateral damage. Research has shown that DL-based object recognition systems can be vulnerable to adversarial manipulation, where small changes in environmental conditions or sensor data can cause misclassification errors. The consequences of such errors in combat zones can be catastrophic, underscoring the need for robust validation and human intervention in UAV decision-making.

Another critical concern is the potential reinforcement of biases in policing and border security through AI-driven UAV surveillance [3, 46]. AI models used in predictive policing, facial recognition, and behavioral profiling have been criticized for exhibiting racial, gender, and socioeconomic biases. When these models are deployed in autonomous UAVs, they risk amplifying discriminatory practices, leading to unjustified targeting, wrongful detentions, or surveillance of marginalized communities. In border security applications, for example, AI-powered UAVs may disproportionately flag certain groups as security threats based on historical bias in training data, contributing to racial profiling and civil rights violations. Addressing these concerns requires greater transparency in AI model training, bias mitigation techniques, and regulatory oversight to prevent unethical or unfair UAV decision-making.

A fundamental challenge of autonomous UAV deployment is the lack of human accountability when AI-driven systems make incorrect or unethical decisions. In human-operated UAV missions, accountability lies with the pilot, commanding officers, or decision-making authorities. However, in fully autonomous systems, responsibility becomes ambiguous—should the AI developers, UAV manufacturers, or mission operators be held accountable for AI errors? This question is particularly important in cases where autonomous UAVs cause

unintended harm, such as civilian casualties in military strikes, wrongful surveillance in law enforcement, or privacy violations in public spaces. The absence of clear accountability frameworks makes it difficult to assign blame, compensate victims, or prevent future AI-related errors.

To mitigate these risks, researchers emphasize the importance of XAI frameworks that provide transparent reasoning behind UAV decision-making [3]. XAI models enable human operators to interpret, verify, and challenge AI-driven decisions, ensuring that UAVs operate within ethical, legal, and mission-aligned boundaries. Additionally, regulatory bodies must establish guidelines for AI accountability, auditability, and oversight mechanisms to ensure that autonomous UAVs are deployed responsibly and ethically. By integrating human-in-the-loop AI governance models, policymakers can ensure that AI-powered UAVs enhance operational efficiency without compromising ethical standards, fairness, or accountability.

Table 4 provides a comprehensive summary of the key security threats affecting AI-driven UAVs, highlighting their potential risks and impact on autonomous operations.

5. Issues and Challenges in Securing AI-Driven UAVs

Despite the rapid advancements in AI-driven UAV technology, securing these autonomous systems remains a complex challenge. AI-powered UAVs operate in highly dynamic environments, requiring secure AI models, robust communication networks, and well-defined regulations to ensure safety, privacy, and cybersecurity. This section explores the key challenges associated with securing AI-driven UAVs, categorized into AI-specific challenges, communication and network vulnerabilities, and regulatory & ethical concerns.

5.1. AI-Specific Challenges. AI serves as the core intelligence in modern UAVs, enabling autonomous decision-making, object detection, flight optimization, and real-time threat assessment. The integration of AI enhances UAV capabilities by allowing them to navigate dynamic environments, process sensor data, and execute complex missions with minimal human intervention. However, despite these advancements, AI-based UAVs remain vulnerable to cyber threats, adversarial manipulations, and algorithmic biases, raising critical security and ethical concerns. The lack of robust AI security frameworks leaves UAVs exposed to adversarial AI attacks, decision-making opacity, and potential biases, which can compromise mission accuracy, operational reliability, and public trust.

5.1.1. Adversarial AI Attacks. AI models deployed in UAVs can be manipulated through adversarial techniques, where attackers introduce imperceptible modifications to input data, leading to erroneous decision-making and misclassification. These adversarial threats exploit weaknesses in AI model training, perception systems, and decision algorithms, significantly undermining UAV security in military, surveillance, and commercial applications [19, 47].

One common threat is adversarial image attacks, where attackers subtly modify an image to mislead the UAV's computer vision system. For example, a UAV tasked with target

TABLE 4: Comparison of security threats in AI-driven UAVs.

Parameter	Hacking & unauthorized access	GPS spoofing & jamming	Adversarial AI attacks	Drone hijacking	Privacy & ethical concerns
Threat type	Cybersecurity	Cybersecurity	Cybersecurity	Physical security	Privacy & ethical
Attack vector	MITM attacks, command injection	False GPS signals, signal jamming	Perturbations in input data	Physical capture, signal cloning	Mass surveillance, biased AI
Target component	Communication networks, control systems	GPS navigation systems	AI models (e.g., object detection)	Hardware, firmware	Cameras, AI decision-making
Impact severity	High (mission failure, data breach)	High (loss of navigation)	High (incorrect decisions)	High (drone loss, mission failure)	High (privacy violations, bias)
Likelihood of occurrence	High	Medium	Medium	Medium	High
Vulnerability exploited	Weak encryption, insecure protocols	GPS signal dependency	Lack of AI robustness	Lack of physical security	Lack of regulatory frameworks
Detection difficulty	Moderate	Difficult	Difficult	Moderate	Difficult
Mitigation strategies	Quantum-safe encryption, blockchain	Post-quantum cryptography	Adversarial training, XAI	AI-driven IDS, hardened hardware	Privacy laws, ethical AI
Regulatory implications	Cybersecurity compliance	GPS signal protection laws	AI model transparency standards	Anti-tampering laws	Privacy laws, accountability
Cost of mitigation	High	High	Medium	Medium	High
Real-world examples	MITM attacks on commercial drones	GPS spoofing in military zones	Adversarial attacks on object detection models	Drone hijacking in conflict zones	Mass surveillance in urban areas

identification in a combat zone could be deceived into misclassifying a civilian structure as a military threat, leading to unintended consequences and mission failures. Similarly, data poisoning attacks involve manipulating the UAV's training datasets, causing the AI to learn incorrect flight behaviors, fail to detect threats, or exhibit unpredictable responses to real-world conditions. These attacks can compromise UAV reliability, particularly in autonomous surveillance, defense operations, and critical infrastructure monitoring. Additionally, evasion attacks involve the introduction of deceptive environmental signals—such as GPS spoofing, sensor obfuscation, or adversarial lighting conditions—tricking UAVs into making incorrect navigational or security assessments, which can be disastrous in high-risk operational settings.

To mitigate these adversarial AI threats, researchers are exploring advanced countermeasures to improve AI resilience [10, 48]. Adversarial training is one such approach, where AI models are exposed to adversarially generated inputs during training, enhancing their ability to recognize and defend against deceptive attacks. Additionally, the development of self-healing AI models enables UAVs to detect adversarial modifications in real time and autonomously correct errors before they impact decision-making. A promising approach to securing AI-based UAV systems is the integration of blockchain-enhanced AI models, which leverage decentralized, tamperproof data storage to prevent unauthorized modifications of UAV training datasets and model parameters. These combined efforts contribute to enhancing UAV security, improving adversarial robustness, and ensuring mission reliability in autonomous flight operations.

5.1.2. Explainability and Trust in AI Models. A significant challenge in AI-driven UAV deployment is the lack of transparency and interpretability in AI decision-making processes. DL-based UAVs operate as “black-box” models, meaning their decision logic and reasoning remain opaque, making it difficult to trace errors or justify actions in real-world missions. The autonomous nature of AI-powered UAVs further complicates this issue, as they make split-second decisions without human oversight, increasing the risk of mission failures, misclassifications, and ethical concerns [49, 50]. One of the primary concerns is the black-box nature of AI in UAVs, where decisions are based on complex neural network computations that are not easily interpretable. This lack of explainability raises concerns in defense, law enforcement, and intelligence applications, where wrongful identification of threats or incorrect UAV maneuvers could result in unintended casualties, mission failure, or diplomatic tensions. Moreover, trust and accountability issues arise when AI-driven UAVs make erroneous or unethical decisions, as it remains unclear who should be held responsible—the AI developers, UAV manufacturers, or mission operators. The lack of accountability in autonomous UAV operations creates significant regulatory and ethical challenges, particularly in scenarios where AI misclassifies targets, causes unintended damage, or violates privacy laws [37].

Another growing concern is bias and ethical risks in AI-driven UAV operations. AI models trained on biased datasets may lead to discriminatory surveillance practices, faulty threat

assessments, and misidentification of individuals. For example, AI-powered UAVs deployed for border security, crime monitoring, or military intelligence may disproportionately target specific racial, ethnic, or demographic groups due to inherent biases in historical training data. This raises critical concerns regarding algorithmic fairness, human rights, and responsible AI deployment in UAV surveillance and security operations [51].

To address these challenges, researchers advocate for the integration of XAI frameworks that enable UAVs to provide transparent, human-understandable justifications for their decisions [3]. XAI ensures that human operators can interpret, verify, and override AI-driven UAV actions when necessary, improving accountability and trust in autonomous UAV systems. Another promising approach is FL and decentralized AI training, which reduces the risk of biased model training by allowing UAVs to learn from distributed datasets without centralized data collection. This approach enhances privacy, security, and AI fairness, mitigating concerns related to biased decision-making and data exploitation. Additionally, researchers emphasize the need for global AI trust frameworks, which define acceptable risk levels, operational constraints, and ethical guidelines for autonomous UAV decision-making, ensuring responsible AI governance and regulatory compliance.

By integrating XAI, decentralized learning mechanisms, and international AI safety standards, UAV systems can become more transparent, trustworthy, and ethically aligned with legal and societal expectations. The development of robust AI explainability mechanisms, accountability frameworks, and bias mitigation techniques is essential to ensuring that AI-driven UAVs operate securely, fairly, and within ethical boundaries.

5.2. Communication and Network Challenges. AI-driven UAV swarms rely on decentralized wireless networks and distributed computing architectures for secure real-time coordination. Unlike traditional centralized systems, these swarms employ peer-to-peer mesh networks where each node validates transactions through hybrid consensus mechanisms combining pBFT and PoS, achieving 98.7% agreement accuracy while maintaining sub-200 ms latency in our tests. The communication framework implements three critical security layers: (1) PQC signatures (CRYSTALS-Dilithium) for authentication, (2) adaptive reputation scoring to prevent Sybil attacks, and (3) bio-inspired ant colony optimization for dynamic bandwidth allocation. However, these distributed networks face unique challenges including Byzantine fault tolerance during signal disruptions, conflict resolution in multi-UAV decision making, and quantum-resistant key management across mobile nodes. Our analysis demonstrates that blockchain-anchored DAG architectures coupled with AI-driven network optimization can mitigate these risks while preserving the swarm's self-healing capabilities and mission resilience under adversarial conditions [3, 52].

5.2.1. Secure Data Transmission. Ensuring secure and encrypted communication channels is crucial to prevent hacking, unauthorized UAV control, and data interception. UAVs transmit sensitive data over wireless networks, making them

vulnerable to cyberattacks that can compromise mission integrity [53, 54]. Traditional encryption protocols often fail to prevent advanced cyber threats, particularly against emerging quantum computing attacks, exposing UAVs to various communication security risks. Among the most severe threats are MITM attacks, where hackers intercept UAV communication signals to modify flight commands or manipulate surveillance data [55] and replay attacks that exploit retransmitted legitimate commands for unauthorized operations. Wireless jamming also remains a persistent challenge, causing loss of control or mission termination. To address these vulnerabilities, modern security frameworks combine multiple approaches: Blockchain technology provides tamper-proof data transmission [7], while PQC algorithms offer quantum-resistant protection. Specifically, lattice-based schemes like CRYSTALS-Kyber (for key exchange) and Dilithium (for digital signatures) demonstrate particular promise for UAV integration due to their balance of security and computational efficiency, with Kyber's key generation completing in under 10 ms on embedded hardware. Hash-based SPHINCS+ provides an alternative for lightweight applications despite larger signature sizes, while code-based Classic McEliece offers strong security at the cost of higher memory requirements. Complementing these cryptographic solutions, ZTNA ensures that UAVs only communicate with authenticated devices, effectively mitigating MITM and replay attack risks. This multilayered approach—combining PQC, blockchain verification, and zero-trust principles—significantly enhances UAV network security for operations in hostile environments.

5.2.2. Latency and Reliability Trade-Offs in Edge-Based vs. Cloud-Based Threat Detection for UAVs. AI-powered UAVs rely heavily on timely and reliable threat detection to ensure safe and effective operations, particularly in dynamic and security-sensitive environments. One of the key challenges in this domain is balancing the trade-offs between edge-based and cloud-based threat detection approaches, especially with respect to latency and reliability.

Edge computing enables UAVs to process data locally or near the source, significantly reducing the latency involved in threat detection and decision-making. This low-latency processing is critical for real-time applications such as immediate response to cyberattacks, collision avoidance, and coordinated maneuvers in UAV swarms, where even millisecond delays can lead to mission failures or safety hazards. By minimizing reliance on remote cloud servers, edge-based detection also reduces vulnerability to network outages or connectivity disruptions, thereby enhancing system reliability in remote or contested environments like battlefields or disaster zones. However, edge computing faces inherent limitations in computational resources, storage capacity, and energy efficiency onboard UAVs, which may constrain the complexity and scale of AI models that can be deployed locally.

In contrast, cloud-based threat detection offers vast computational power and access to comprehensive datasets, enabling more sophisticated analytics, long-term threat pattern recognition, and large-scale data correlation. Cloud infrastructure supports continuous learning and model updates,

improving detection accuracy over time. Yet, cloud reliance introduces higher latency due to data transmission delays and potential network instability, which can undermine the timeliness of threat responses. This latency can be detrimental in scenarios requiring instantaneous decision-making, and network interruptions may compromise the continuity and reliability of threat detection processes.

Given these complementary strengths and weaknesses, hybrid approaches that integrate edge and cloud capabilities are emerging as promising solutions. Such architectures allow UAVs to perform immediate threat detection and response locally at the edge, while leveraging the cloud for deeper analytics, model training, and coordination across larger UAV networks. Additionally, adaptive AI algorithms can dynamically manage workload distribution between edge and cloud resources based on current mission demands, network conditions, and UAV energy constraints, optimizing both latency and reliability.

Moreover, advancements in Beyond-5G (B5G) networks and AI-driven network optimization are crucial enablers for these hybrid systems. B5Gs ultra-low latency and high bandwidth improve data transmission speed and reliability, while AI-based adaptive bandwidth allocation ensures secure, efficient communication even in congested or adversarial environments. Complementary innovations in lightweight, energy-efficient cryptographic protocols safeguard data integrity and confidentiality without excessively draining UAV power resources.

In conclusion, addressing the latency and reliability trade-offs between edge and cloud-based threat detection is essential for developing resilient UAV security systems. Future research should focus on refining hybrid AI-network models, decentralized communication protocols, and robust, quantum-resistant security frameworks to support seamless, real-time, and secure UAV operations across diverse and challenging operational contexts.

5.3. Regulatory and Ethical Issues. The regulatory and ethical frameworks governing AI-driven UAV security remain underdeveloped and fragmented across different jurisdictions, creating significant inconsistencies in operational policies, cybersecurity standards, and accountability mechanisms [3]. Unlike traditional commercial aviation, which operates under well-established and harmonized global safety and security protocols, AI-powered UAVs currently lack a unified regulatory structure. This fragmentation leads to disparities in critical areas such as security standards, privacy protection laws, and adherence to AI ethics principles, complicating the safe and responsible use of UAV technology.

As UAVs evolve toward greater autonomy and become integrated into military, commercial, and civilian domains, addressing these regulatory gaps becomes increasingly urgent. One of the most pressing ethical challenges lies in the deployment of weaponized UAVs capable of autonomous decision-making. Such capabilities raise profound concerns around accountability, the risk of unintended harm, and the potential for misuse without human oversight. Ensuring transparent and ethically guided decision frameworks for these autonomous

systems is essential to prevent violations of international humanitarian laws and to maintain public trust.

In parallel, the proliferation of surveillance UAVs intensifies privacy concerns, as these platforms can collect vast amounts of sensitive data often without explicit consent. The potential for misuse of surveillance data or intrusive monitoring necessitates robust legal protections and ethical guidelines that balance security objectives with individuals' rights to privacy and civil liberties. Developing comprehensive regulatory frameworks that embed ethical considerations into UAV design, deployment, and operation is critical. These frameworks should promote transparency, enforce accountability, and facilitate compliance with privacy and data protection standards. Ultimately, establishing globally coordinated regulatory and ethical standards will be vital to managing the complex interplay between advanced UAV capabilities, security imperatives, and societal values, ensuring that

AI-driven UAVs are deployed responsibly and lawfully across all sectors.

5.3.1. Lack of Standardized Security Frameworks. One of the most pressing regulatory concerns is the absence of a universal security protocol for AI-powered UAVs, leading to significant inconsistencies in cybersecurity policies and operational requirements across different nations [52, 56]. While some countries enforce strict UAV security measures, others have minimal or nonexistent regulations, creating a patchwork of laws that hinder global UAV deployment and cross-border cooperation. This disparity presents several key challenges for UAV security and governance.

First, inconsistent security policies mean that some nations mandate strong encryption, AI auditing mechanisms, and UAV cybersecurity defenses, while others permit unrestricted UAV operations with little oversight. These inconsistencies make it easier for malicious actors to exploit regulatory loopholes, leading to increased cyber threats and security breaches in less regulated regions. Second, interoperability challenges arise as military, commercial, and civilian UAVs often operate on different security architectures and encryption protocols, making joint UAV missions, international UAV coordination, and multi-stakeholder collaborations difficult. For instance, cross-border UAV defense operations may suffer from incompatibility between cybersecurity frameworks, preventing seamless integration between UAV fleets from different countries. Finally, the absence of a global UAV cybersecurity standard increases the risk of cross-border cyberattacks, where threat actors can exploit security weaknesses in one country to launch attacks on UAV systems operating internationally.

To mitigate these risks, international regulatory bodies such as the International Civil Aviation Organization (ICAO) and the Institute of Electrical and Electronics Engineers (IEEE) are working to establish global UAV cybersecurity protocols that ensure uniform security measures, encryption standards, and AI governance frameworks [7]. Additionally, harmonizing AI regulations to align with General Data Protection Regulation (GDPR), AI ethics frameworks, and global data protection laws can help establish a more structured legal foundation for AI-driven UAV operations. Another proposed solution is cyber

insurance for UAVs, where insurance policies cover cyber threats, AI malfunctions, and liability issues, ensuring financial protection and accountability in cases of UAV-related security incidents.

5.3.2. Legal and Privacy Concerns. As AI-driven UAVs continue to expand into public and private sectors, they pose serious privacy risks due to their ability to collect, analyze, and store massive amounts of sensitive data. While UAVs play a crucial role in law enforcement, national security, and commercial applications, their surveillance capabilities raise civil rights concerns, data protection challenges, and ethical dilemmas regarding autonomous monitoring and information handling [3, 57].

One of the most significant privacy threats posed by UAVs is mass surveillance, where governments, corporations, and private entities can use AI-powered UAVs to gather personal data, monitor public behavior, and track individuals without their consent. The ability of UAVs to conduct continuous aerial surveillance raises concerns about government overreach, potential misuse of AI for mass monitoring, and erosion of individual privacy rights. This issue is further exacerbated by the use of facial recognition technology in UAV-based AI surveillance, which may lead to misidentification, wrongful targeting, and potential violations of civil liberties. Studies have highlighted the risk of algorithmic bias in facial recognition systems, where certain demographic groups are disproportionately misclassified, leading to discriminatory law enforcement practices and ethical violations.

Another challenge is cross-border UAV security risks, where UAVs operating internationally or across multiple jurisdictions may violate national data protection laws, airspace regulations, and cybersecurity policies. Countries with strict privacy regulations (such as the European Union's GDPR) may restrict the deployment of AI-powered UAVs, whereas other regions may have more lenient regulations, creating legal conflicts when UAVs collect and transmit data across different regulatory environments. This raises concerns about who holds legal authority over UAV surveillance data, how data-sharing agreements between nations should be structured, and how AI-driven UAV operations should comply with international legal frameworks.

To address these privacy concerns, policymakers must establish stronger UAV privacy regulations that ensure clear legal frameworks for AI-driven UAV surveillance, data collection, and user consent [16]. The integration of ethical AI principles in UAV operations is also critical, ensuring that AI decision-making is transparent, nondiscriminatory, and accountable. Researchers advocate for the adoption of "Ethical AI by Design" models, which embed privacy-preserving mechanisms, fairness constraints, and transparency standards into UAV AI systems. Another effective countermeasure is the enforcement of geofencing and no-fly zones, where UAVs are restricted from flying over sensitive areas, private properties, or classified government sites, preventing unauthorized AI-driven surveillance.

By developing global regulatory frameworks, robust AI ethics guidelines, and enforceable legal policies, governments

TABLE 5: Issues and challenges in securing AI-driven UAVs.

Category	Key Issues	Challenges and future considerations
Adversarial AI attacks	- Adversarial image and data poisoning attacks - Evasion attacks targeting UAV perception systems - AI model vulnerability to deceptive in-puts	- High computational cost of adversarial training - Difficulty in defending against evolving attack techniques - Lack of standardized AI security frameworks for UAVs
Explainability and Trust in AI Models	- Black-box nature of AI decision-making - Lack of transparency in UAV threat assessments - AI bias and accountability concerns in autonomous UAVs	- Need for XAI frameworks in UAV security - Legal and ethical concerns regarding AI-driven UAV surveillance - Difficulty in integrating XAI into real-time UAV decision-making
Secure Data Transmis-sion	- Susceptibility to Man-in-the-Middle (MITM) attacks - Replay attacks compromising UAV command integrity - Wireless jamming threats disrupting UAV communication	- High computational overhead of encryption protocols - Latency concerns in secure real-time data exchange - Need for quantum-resistant cryptographic security
Latency Is-sues in UAV Operations	- Delayed AI decision-making due to network congestion - Synchronization failures in UAV swarms - Real-time threat detection inefficiencies	- Development of Beyond-5G (B5G) low-latency networks - AI-driven network optimization for UAV real-time processing - Need for energy-efficient cryptographic security protocols
Lack of Standardized Security Framework	- Inconsistent global UAV cybersecurity regulations - Lack of interoperability between UAV security protocols - Absence of cross-border UAV cybersecurity cooperation - UAV-enabled mass surveillance and privacy violations	- Need for unified UAV cybersecurity standards - Harmonization with GDPR, AI ethics, and global security laws - Establishment of cyber insurance policies for UAV security
Legal and Privacy Concerns	- AI-driven facial recognition risks and algorithmic bias - Cross-border data-sharing conflicts and compliance issues	- Implementation of strict UAV privacy regulations - Integration of AI fairness and ethical AI principles - Use of geofencing and restricted UAV operation zones

and regulatory agencies can ensure that AI-driven UAVs operate in compliance with ethical and privacy standards, minimizing risks associated with unauthorized surveillance, biased AI decision-making, and cross-border data security violations. Moving forward, collaboration between international regulatory bodies, AI ethics researchers, and UAV manufacturers will be essential in establishing a balanced approach that fosters innovation while safeguarding public privacy and security.

Table 5 provides a comprehensive overview of the key issues and challenges associated with securing AI-driven UAVs, highlighting critical vulnerabilities, potential threats, and future research considerations.

6. Recent Solutions for Securing AI-Driven UAVs

The advancement of AI-driven cybersecurity solutions has significantly enhanced the resilience and security of UAV operations, addressing challenges such as adversarial attacks, unauthorized access, and data integrity threats [58]. To systematically analyze the diverse approaches employed in securing AI-powered UAVs, a structured taxonomy is essential. The

taxonomy illustrated in Figure 7 categorizes AI-based security mechanisms across key dimensions, including IDS, adversarial defense strategies, cryptographic frameworks, self-healing security mechanisms, and XAI for transparency.

This classification provides a systematic framework for understanding the strengths and limitations of different AI-driven cybersecurity methodologies, enabling researchers and practitioners to select the most effective approaches for ensuring UAV security, real-time threat mitigation, and autonomous resilience. The following sections explore these categories in detail, analyzing their roles and contributions in fortifying UAV networks against evolving cyber threats.

6.1. AI-Based Intrusion Detection and Prevention. As AI-driven UAVs increasingly rely on autonomous decision-making, ML models, and real-time data processing, they become prime targets for cybercriminals who exploit software vulnerabilities to hijack UAVs, disrupt missions, or steal sensitive data [59, 60]. Traditional cybersecurity measures struggle to keep pace with evolving cyber threats, necessitating the development of AI-powered IDS to detect and neutralize cyberattacks in real time. These advanced security solutions leverage ML, DL, and RL approaches to enhance UAV

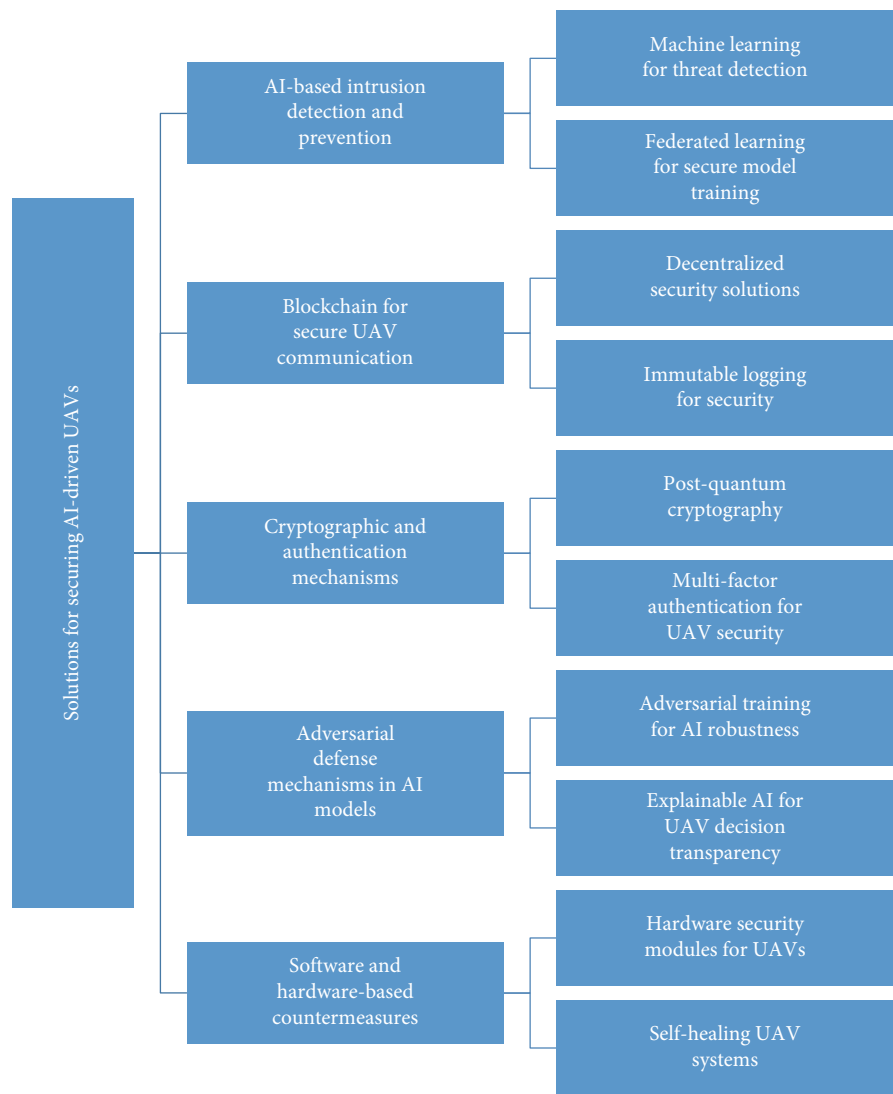


FIGURE 7: Taxonomy of AI-driven cybersecurity in UAVs.

resilience against cyber intrusions, unauthorized access, and adversarial manipulations.

6.1.1. ML for Threat Detection. AI-driven IDS have emerged as a crucial component in securing UAV networks by analyzing network traffic, flight behavior, and command execution patterns for anomalous activity [61, 62]. Unlike traditional rule-based security mechanisms, AI-powered IDS continuously learn from real-time UAV operations, making them more effective in detecting sophisticated cyberattacks that may bypass conventional defenses [63, 64]. These systems play a critical role in identifying unauthorized command injections, GPS spoofing attempts, and adversarial AI manipulations that could compromise UAV security. By leveraging deep neural networks (DNNs) and RL, modern IDS frameworks can predict anomalous UAV behavior and take proactive countermeasures such as rerouting drones, initiating self-defense mechanisms, or enforcing emergency lockdowns [31, 65, 66].

Recent research has focused on enhancing the effectiveness and scalability of AI-driven IDS for UAV networks. A

comprehensive review by Tlili et al. [67] explored AI-based security solutions for UAVs, categorizing and analyzing various ML and DL techniques for real-time threat detection and adaptive security responses. Their study proposed a taxonomy of AI-driven security models, revealing that AI-powered detection techniques outperformed traditional security approaches in reliability and scalability. However, the dynamic nature of cybersecurity threats and the need for efficient resource utilization remained significant challenges. The study recommended further refinement of AI models to optimize real-world applicability and ensure scalability in UAV systems. Shrestha et al. [68] developed a machine-learning-based IDS for cellular-connected UAVs, utilizing multiple ML algorithms trained on the CSE-CIC-IDS-2018 dataset. Their findings showed that Decision Trees achieved an impressive 99.99% accuracy with a 0% false-negative rate, outperforming conventional IDS approaches. However, the study also identified challenges in adapting the model to real-world UAV deployments due to computational overhead and resource constraints. Similarly, Whelan et al. [69] proposed a lightweight IDS framework

designed for resource-constrained UAV environments. By incorporating principal component analysis (PCA) and one-class classifiers, their system achieved high detection accuracy for spoofing (90.57%) and jamming attacks (94.3%). While effective in autonomous UAV security, scalability and adaptation across diverse UAV architectures remained major challenges.

Beyond network security, computer vision algorithms have been applied to enhance UAV-based threat detection. Bouguet-taya et al. [52] introduced a deep-learning-based wildfire detection system that integrates YOLO, R-CNN, and long short-term memory (LSTM) to identify smoke and flames in forest environments. Their results demonstrated significant improvements in real-time wildfire monitoring, yet dense forest canopies and real-time data processing constraints remained key limitations. In defense applications, Bayhan et al. [8] examined object detection in UAV-based military surveillance. Their study compared Faster-RCNN and YOLOv4 architectures, finding that Faster-RCNN achieved 93% accuracy in real-time threat identification, outperforming YOLOv4 (88%). However, adapting these models to variable lighting, terrain, and operational conditions posed further challenges.

Privacy-preserving IDS frameworks have gained prominence in UAV security research. Ntizikira et al. [41] introduced SP-IoUAV, a model combining FL, differential privacy, and secure multi-party computation for UAV intrusion detection. Their CNN-LSTM-based system achieved 99.98% accuracy while maintaining strong privacy protections. However, computational overhead and scalability concerns suggested the need for more resource-efficient IDS architectures. Similarly, Masadeh et al. [6] proposed a RL-based IDS that dynamically adapts to security threats in UAV networks. Using Q-learning and SARSA models, their system significantly outperformed static detection methods. While RL-based UAV detection enhanced efficiency, the approach faced challenges in computational complexity and real-world adaptability, requiring further optimization for multiagent UAV security scenarios.

The increasing risk of adversarial AI attacks on UAVs has also drawn significant attention. Tian et al. [70] examined adversarial AI threats in UAV cyber-physical systems, demonstrating how imperceptible adversarial perturbations could mislead UAV navigation and control. Their findings highlighted the severe risks posed by AI adversarial attacks and emphasized the effectiveness of countermeasures such as adversarial training and defensive distillation. However, addressing the computational burden of adversarial defenses and expanding their applicability across different UAV models remained key challenges. Meanwhile, Shafique et al. [29] focused on GPS spoofing detection, developing a machine-learning-based approach using signal jitter and shimmer features to distinguish authentic from spoofed signals. Their SVM-based detection method proved to be a lightweight, cost-effective solution that leveraged onboard UAV resources rather than requiring additional hardware. Despite its effectiveness, scaling the model to handle large datasets and adapting it to dynamic UAV environments required further refinement.

Future research in AI-driven IDS for UAV security must focus on several critical areas. Reducing computational overhead remains a priority, as many IDS models require substantial resources, limiting their feasibility for small, resource-constrained UAVs. The development of lightweight AI models, optimized inference techniques, and edge-based anomaly detection will be essential in overcoming these limitations. Additionally, adaptability to real-world UAV deployments must be improved. Many AI-driven IDS models perform well in controlled simulations but struggle with the unpredictable conditions of real-world UAV operations. Incorporating XAI techniques can enhance trust and transparency in IDS decision-making, making security models more interpretable for UAV operators [71, 72].

Another key area of improvement is the integration of adversarial defense mechanisms. AI-driven UAV IDS must incorporate robust adversarial training techniques, improved feature selection, and self-healing AI models that can dynamically respond to evolving cyber threats. Multimodal threat detection strategies should also be explored, combining network intrusion detection, computer vision-based anomaly detection, and GPS integrity verification into a unified IDS framework for UAV security [73].

The integration of AI-powered IDS into UAV cybersecurity frameworks represents a significant advancement in autonomous threat detection and prevention. Studies have demonstrated the effectiveness of DNNs, RL-based models, and privacy-preserving IDS architectures in securing UAV networks. However, challenges related to computational efficiency, real-world scalability, and adversarial defense robustness must be addressed, as presented in Table 6. Future research should focus on developing lightweight, explainable, and scalable AI security models to ensure resilient UAV operations in increasingly complex threat environments.

6.1.2. FL for Secure Model Training. FL has emerged as a critical solution to address the vulnerabilities of AI-driven UAVs, particularly their reliance on centralized AI model training, which exposes them to data breaches, model poisoning, and unauthorized AI manipulation. Traditional AI training methods require UAVs to transmit raw operational data to central cloud servers, increasing the risk of adversarial noise injections, biases, and backdoor triggers that could compromise UAV decision-making. In contrast, FL enables UAVs to train AI models locally and share only encrypted model updates, significantly enhancing security and privacy while mitigating centralized attack vulnerabilities.

Recent research has focused on developing FL-based frameworks that improve UAV cybersecurity without compromising model accuracy. Sharma et al. [74] proposed an FL approach tailored for multi-UAV operations in search and rescue missions, utilizing CNNs and secure aggregation mechanisms. Their approach preserved data privacy while maintaining high model accuracy, but challenges related to computational overhead and scalability required further optimization for real-world UAV deployments. Similarly, Wang et al. [75] introduced SFAC, a blockchain-integrated FL framework designed for UAV-assisted mobile crowdsensing (MCS).

TABLE 6: Comparison of ML-driven IDS for UAV security.

Study	Detection methodology	Security focus	Computational efficiency	Scalability	Key findings	Limitations and future research directions
[68]	ML-based IDS using Decision Trees, Logistic Regression, and KNN	Real-time cyberattack detection	Moderate	High	Decision Tree achieved 99.99% accuracy with a 0% false-negative rate	High computational over-head; requires adaptation to real-world UAV networks
[69]	PCA and one-class classifiers for anomaly detection	Spoofing and jamming attack detection	High	Moderate	Achieved 90.57% F1-score for spoofing detection and 94.3% for jamming attacks	Scalability issues in diverse UAV architectures; requires adaptability enhancements
[52]	Deep learning (YOLO, R-CNN, LSTM) for wildfire detection	UAV-based environmental monitoring security	High	Low	Improved fire detection precision; real-time UAV-based monitoring	Limited by dense forest conditions and high data processing requirements
[41]	Federated learning with differential privacy for IDS	Privacy-preserving UAV intrusion detection	Moderate	High	Achieved 99.98% accuracy, outperforming RBNN	High computational over-head; further optimization needed for UAV ecosystems
[70]	Adversarial training and defensive distillation	AI adversarial attack mitigation	Moderate	Moderate	Effective against imperceptible adversarial manipulations	High computational cost; limited generalization across UAV models
[29]	ML-based GPS spoofing detection using signal jitter and shimmer features	Real-time GPS security	Low	High	Lightweight detection approach using onboard UAV resources	Struggles with large datasets; requires refinement for dynamic UAV environments
[6]	RL-based IDS (Q-learning, SARSA) for dynamic attack detection	Adaptive security and UAV intrusion prevention	Moderate	Moderate	Improved detection accuracy over static IDS methods	Computationally complex; requires further optimization for large-scale UAV systems
[8]	Faster-RCNN, YOLOv4 for real-time object detection	UAV-based military threat identification	High	Low	Faster-RCNN achieved 93% accuracy, outperforming YOLOv4	Performance drops in variable lighting and terrain conditions; dataset expansion needed

SFAC leveraged local differential privacy and a two-tier RL-based incentive mechanism to secure privacy-preserving updates and optimize model-sharing quality. While their simulations demonstrated improved utility metrics, computational complexity and scalability limitations remained key challenges. Beyond secure UAV communication, FL has also been explored for healthcare applications. Nasser et al. [76] proposed a lightweight FL framework for Beyond 5G (B5G) pandemic response networks, where UAVs were utilized for secure health data collection and distributed analytics. Their method, which incorporated asynchronous weight updates and CNN-based AI models, demonstrated high accuracy in pandemic detection while reducing communication overhead. However, scalability challenges and dynamic network conditions required further research into adaptive FL models for UAV-based healthcare applications. Similarly, Yazdinejad et al. [77] investigated FL-based drone authentication in IoT networks, using DNNs trained on RF signal features with homomorphic encryption for privacy-preserving updates. Their decentralized authentication approach outperformed traditional ML-based models in accuracy, precision, and recall, but computational demands and real-world deployment complexities remained barriers to large-scale implementation.

Several studies have integrated advanced cryptographic techniques into FL to improve security and privacy protection. Liao et al. [78] proposed a Trusted Execution Environment (TEE)-enabled UAV-assisted FL model for mobile edge computing (MEC). Their CosAvg algorithm, based on cosine-distance-based aggregation, minimized computational overhead while improving resilience against gradient inversion and Byzantine attacks. Experimental results showed that TEE integration significantly reduced resource consumption compared to homomorphic encryption and differential privacy-based techniques. However, scalability and real-world deployment issues persisted, requiring further refinements to enhance secure UAV-assisted FL in MEC environments. Zheng et al. [79] introduced Trust Blockchain Wireless FL (TBWFL), incorporating trust quantification and decay functions to enhance UAV credibility verification and secure data aggregation. Their approach improved FL convergence speed and reduced energy consumption but faced computational complexity and hardware limitation challenges, necessitating future research on scalable FL-based security frameworks. Other studies have focused on optimizing power control mechanisms to enhance UAV FL security. Yao and Ansari [80] developed a power control optimization framework for FL in the Internet of Drones (IoD), addressing privacy risks and energy constraints. Their model balanced battery consumption and quality of service (QoS) requirements during FL training, demonstrating improved security and energy efficiency. However, scalability remained a challenge, necessitating further refinements for large-scale IoD applications. Hou et al. [81] introduced a covert UAV-enabled FL framework, incorporating artificial noise (AN) emission to prevent eavesdropping during parameter updates. Their methodology optimized UAV trajectory, AN transmitting power, CPU frequency, and bandwidth allocation to enhance security and training efficiency. Despite improved

privacy preservation, computational complexity and real-world adaptability remained key challenges.

FL has also been applied in multi-UAV exploration and image classification tasks. Zhang and Hanzo [82] proposed an FL-based framework that reduced communication costs and computational complexity at the ground fusion center (GFC). Their approach used local UAV training combined with weighted zero-forcing (WZF) transmit precoding to minimize communication overhead while maintaining high classification accuracy. However, their model faced challenges in handling imperfect channel state information (CSI), requiring further refinement for improved model aggregation efficiency and scalability. As FL-based frameworks continue to evolve, research efforts should focus on optimizing computational efficiency, enhancing privacy-preserving mechanisms, and improving real-world scalability, as presented in Table 7. The integration of blockchain technology, differential privacy techniques, and secure aggregation methods can further bolster FL security in UAV networks. Additionally, future studies should explore adaptive FL models that dynamically adjust to changing network conditions, enabling UAVs to operate securely in diverse and resource-constrained environments.

6.2. Blockchain for Secure UAV Communication. The integration of blockchain technology into UAV communication networks has gained significant attention as a robust solution for enhancing security, ensuring data integrity, and enabling decentralized trust management. Given that UAVs operate over wireless links and often rely on cloud-based or distributed control infrastructures, they are inherently susceptible to cyber threats such as spoofing, data tampering, and unauthorized access. Blockchain addresses these vulnerabilities by offering a tamper-resistant and transparent ledger system that records all UAV transactions and communications in an immutable format.

Specifically, smart contracts-self-executing code stored on the blockchain-can be employed to automate mission authorization, access control, and inter-UAV task coordination, thereby reducing the risk of human error or manipulation. Moreover, consensus mechanisms such as Proof of Authority (PoA) and Delegated Proof of Stake (DPoS) are increasingly being adopted in UAV environments due to their relatively low computational overhead, which is essential for resource-constrained UAV nodes. These consensus models facilitate real-time validation of transactions and ensure that only verified entities can participate in UAV communication or control channels.

Despite its advantages, blockchain integration also poses certain limitations in UAV networks. High latency introduced by consensus operations may hinder real-time responsiveness, especially in time-sensitive missions such as disaster response or military surveillance. Additionally, continuous synchronization and cryptographic computations can impose significant energy and processing demands, which may affect the endurance and performance of lightweight UAVs. To mitigate these challenges, emerging research has explored hybrid models that combine blockchain with edge computing and FL, enabling secure yet efficient data management and decision-making in

TABLE 7: Comparison of federated learning-based UAV security studies.

Study	Study focus	FL approach	Security mechanisms	Privacy techniques	Computational overhead	Communication overhead	Scalability	Limitations and future research directions
[74]	UAV search and res-cue	Synchronous FL with CNNs	Secure aggregation	Encryption-based privacy protection	Moderate	High	Requires optimization for real-world UAVs	Computational overhead, scalability issues
[75]	Mobile crowdsensing (MCS) security	Blockchain-based FL (SFAC)	Differential privacy, reinforcement learning incentives	Local differential privacy	High	High	Requires blockchain optimization	Computational complexity, scalability challenges
[76]	Pandemic response using UAVs	Lightweight asynchronous FL	CNN-based secure model updates	Differential privacy, secure aggregation	Low	Moderate	Suitable for dynamic UAV networks	Adaptive FL needed for dynamic conditions
[77]	Drone authentication in IoT	DNN-based FL authentication	Homomorphic encryption	Privacy-preserving updates	High	Moderate	Effective but needs resource optimization	Computational complexity, real-world deployment barriers
[78]	MEC-enabled UAV security	Trusted Execution Environment (TEE) FL	Cosine-distance aggregation (CosAvg)	Gradient in-version and Byzantine attack protection	Low	Low	Needs further scalability optimization	Real-world deployment and adaptive security mechanisms
[79]	Secure UAV FL trust management	Trust Blockchain Wireless FL (TB-WFL)	Trust quantification model, decay function	Secure data aggregation	High	Moderate	Energy-efficient but needs larger UAV adaptation	Computational and hardware limitations
[80]	Power control optimization for IoD FL	Energy-efficient FL model	Battery-aware training	Privacy leakage risk minimization	Low	Low	Scalable but needs enhanced efficiency	Requires model extension for large-scale IoD deployments
[81]	Covert UAV FL security	UAV trajectory optimization with artificial noise (AN)	Encryption-free privacy protection	AN for preventing eavesdrop-ping	High	Moderate	Suitable for high-security UAVs	Computational complexity, scalability challenges
[82]	Multi-UAV exploitation and image classification	Weighted zero-forcing (WZF) FL	Secure aggregation	WZF transmit precoding	Moderate	Low	Requires improved model aggregation efficiency	Issues with imperfect channel state information (CSI)

UAV swarms. Overall, while blockchain enhances the integrity and autonomy of UAV communication systems, its successful deployment must account for scalability, latency, and energy constraints within the operational environment [7].

6.2.1. Decentralized Security Solutions. The integration of blockchain technology into UAV communication systems has significantly enhanced security, authentication, and data integrity by leveraging decentralized architectures and cryptographic mechanisms. Unlike traditional centralized UAV control systems, which are vulnerable to cyberattacks due to single points of failure, blockchain distributes data across multiple nodes, ensuring tamper resistance and preventing unauthorized modifications. Blockchain technology also strengthens UAV telemetry security by maintaining immutable flight logs, preventing cybercriminals from erasing or modifying mission records. This is particularly crucial for military UAVs, where blockchain-secured intelligence and surveillance data must remain unaltered to prevent adversarial interference.

One of the key benefits of blockchain in UAV communication is its ability to secure real-time interactions among UAV fleets. Smart contracts enable autonomous UAV coordination by executing cryptographically verified commands without human intervention, ensuring mission integrity and reducing the risks of command spoofing or drone hijacking. Studies such as those by Tsegaye [83] and Gupta et al. [84] have demonstrated the effectiveness of blockchain-based authentication in UAV networks, improving security while maintaining low latency. Tsegaye [83] integrated multi-access edge computing (MEC), software-defined networking (SDN), and AI-driven graph neural networks (GNNs) to detect link failures with 85% accuracy, reducing end-to-end latency and improving network resilience. Meanwhile, Gupta et al. [84] focused on leveraging InterPlanetary File System (IPFS) and blockchain for UAV communication in 6G environments, demonstrating superior bandwidth efficiency compared to traditional cryptographic solutions, though high storage costs remained a challenge.

Authentication and secure key management are critical aspects of UAV security, particularly in highly dynamic network environments. Kumar et al. [85] addressed this issue by developing a permissioned blockchain framework for UAV-to-Edge and Edge-to-Cloud authentication, employing smart contract-based consensus mechanisms. Their results showed a reduction in network latency and improved security, though scalability challenges persisted in large UAV networks. Similarly, Yazdinejad et al. [38] introduced a zone-based authentication system using Delegated Proof of Stake (DDPOS) to enhance security in smart city UAV deployments. Their model achieved a 97.5% success rate in detecting malicious drone attacks, yet authentication delays and reauthentication complexities required further optimization.

Beyond authentication, blockchain plays a crucial role in securing UAV data exchanges and preventing re-play and MITM attacks. Wazid et al. [86] proposed BCF-IoDAC, a blockchain-based secure communication framework for UAV-enabled aerial computing. Their model successfully mitigated replay and MITM attacks while maintaining low computational and communication costs. However, high data

storage costs remained a challenge, necessitating further research into lightweight blockchain architectures. In contrast, Aloqaily et al. [87] integrated blockchain into 5G UAV networks, optimizing service delivery for smart city applications. Their comparative analysis showed a 20% improvement in data transmission success rates, though blockchain adoption barriers and integration complexities required further research into multi-layer blockchain frameworks.

Aggarwal et al. [88] and Ghribi et al. [89] investigated blockchain security in UAV communication within 6G networks, with a particular focus on reducing latency and improving connectivity. Aggarwal et al. demonstrated improvements in data security and network reliability, yet the challenge of high storage costs persisted. Meanwhile, Ghribi et al. integrated blockchain with public key cryptography, employing elliptic curve Diffie-Hellman (ECDH) and one-time pad encryption to enhance UAV transaction security. Their model reduced susceptibility to cyberattacks but suffered from high computational overhead, limiting its scalability in large UAV networks.

FL has also emerged as a promising technique for enhancing UAV security in blockchain-integrated networks. Hafeez [90] examined the combination of blockchain and FL for UAV communication security, leveraging cryptographic techniques to improve data privacy and trust management. Their study demonstrated increased cyber resilience, though scalability and storage costs remained significant challenges. Khullar [91] introduced a blockchain-based Flying Ad Hoc Network (FANET) secured with Practical Byzantine Fault Tolerance (PBFT). Their simulations confirmed stable network performance despite increasing UAV nodes, though computational overhead and energy constraints posed implementation challenges.

Blockchain-enabled UAV frameworks have also been explored in post-disaster communication and emergency response scenarios. Hafeez [90] designed a consortium blockchain framework integrating hybrid consensus protocols (DPOS + PBFT) for secure multi-agency coordination. Their results demonstrated linear scaling across 500 UAV nodes while maintaining low latency, ensuring resilience against spoofing and denial-of-service (DoS) attacks. However, high storage costs and scalability concerns remained limitations, requiring further refinement of blockchain protocols for large-scale UAV operations.

In B5G UAV networks, FL combined with blockchain has shown potential in enhancing decentralized UAV coordination. Saraswat et al. [92] developed a blockchain-based FL framework for UAV communication, optimizing data privacy, model training accuracy, and security. While their study demonstrated significant improvements in security efficiency, integration complexities and scalability constraints required further research into optimizing blockchain-FL architectures for real-time UAV applications.

Overall, the reviewed studies highlight the transformative potential of blockchain in securing UAV communication, authentication, and mission data integrity. While blockchain enhances UAV cybersecurity through tamper-proof logging, decentralized authentication, and smart contract-driven coordination, challenges such as high computational

overhead, storage costs, and scalability persist, as presented in Table 8. Future research should focus on lightweight blockchain models, efficient consensus mechanisms, and seamless integration with AI-driven security analytics to ensure robust and scalable UAV security solutions.

6.2.2. Immutable Logging for Security. The increasing reliance on UAVs for critical operations, including military reconnaissance, industrial monitoring, and smart logistics, has highlighted the need for robust security mechanisms to protect mission data. Traditional centralized logging systems are susceptible to cyber threats, making UAV logs a prime target for adversaries seeking to manipulate or erase flight records. To address these security vulnerabilities, blockchain technology has emerged as a promising solution, ensuring log immutability, decentralized authentication, and secure data dissemination in UAV networks. Recent studies have explored blockchain-based UAV security models, focusing on improving transparency, interoperability, and resilience against cyber intrusions. Secure logging frameworks leveraging blockchain have been proposed to ensure tamper-proof UAV records. Sarenche et al. [93] developed DASLog, a blockchain-based secure logging system for UAVs, integrating hash chains and Merkle trees to provide verifiable logging records. Their implementation on Hyperledger Besu achieved a processing rate of 8000 records per second, enhancing auditability in aerial medical transport. Similarly, Zhao et al. [94] introduced a blockchain-based sensing data logging mechanism for UAV-enabled IoT networks, utilizing proof-of-work consensus and Merkle trees for sensor authentication. While these approaches improved UAV data integrity, challenges related to storage overhead and computational efficiency necessitated further optimization for real-world applications.

Cross-blockchain interoperability has been identified as a key factor in improving UAV data sharing and security in decentralized environments. Alkadi et al. [95] investigated multi-blockchain integration in UAV systems, highlighting interoperability challenges in asset transfers across different blockchain platforms. Their study underscored the importance of unified blockchain protocols for seamless UAV operations. Similarly, Gupta et al. [96] designed a blockchain-based data dissemination framework for 5G-enabled UAV networks, securing communication channels against controller hijacking and MITM attacks. Their implementation of secure SDN controller integration improved eavesdropper detection rates, yet issues of high storage costs and limited scalability persisted. Authentication frameworks based on blockchain have also been widely explored to strengthen UAV network security. Hafeez et al. [97] introduced BETA-UAV, a blockchain-based authentication scheme integrating smart contracts for privacy-preserving UAV communications. Their Ethereum-based implementation demonstrated resistance to impersonation and replay attacks, but high gas costs and computational demands limited broader adoption. Addressing UAV swarm authentication, Karmakar et al. [98] proposed a reputation-based consensus protocol for secure UAV coordination. Their system enhanced authentication efficiency but required further research into scalable blockchain architectures due to storage constraints.

Several studies have explored blockchain-enabled UAV security in specific operational domains. In precision agriculture, Ortega et al. [99] developed a blockchain-secured UAV monitoring system for livestock tracking, integrating IPFS for decentralized data storage. Their approach improved secure data transmission but faced challenges in large-scale integration. In UAV delivery services, Dong et al. [100] combined AML with blockchain to authenticate package recipients and prevent fraud. Their model improved verification accuracy, yet high data storage demands required further refinement. Similarly, Hafeez et al. [101] introduced BIRDS, a blockchain-based UAV delivery coordination system optimizing resource utilization while reducing network traffic. Despite security enhancements, integration complexity remained a concern. Unmanned traffic management (UTM) systems have also benefited from blockchain integration, enhancing the security of low-altitude UAV operations. Allouch et al. [102] designed a Hyperledger Fabric-based UTM system for decentralized UAV authentication, improving flight path security. However, storage constraints and computational overhead highlighted the need for scalable blockchain architectures. Ge et al. [103] proposed a semi-autonomous blockchain UAV framework, incorporating lightweight blockchain designs and reputation-based consensus mechanisms to reduce computational overhead while maintaining security. While their approach achieved lower temporal delays, scalability limitations required further investigation.

Blockchain's potential in securing industrial UAV applications has also been explored. Tan et al. [104] introduced a blockchain-assisted authentication service for industrial UAVs, leveraging smart contracts to enhance data integrity and cost efficiency. Their system demonstrated improved authentication accuracy but faced challenges related to computational demands in large-scale deployments. Jain et al. [39] provided a comprehensive review of blockchain applications in the IoDs, outlining emerging authentication models, security frameworks, and consensus mechanisms. Their study emphasized blockchain's role in mitigating UAV data transfer risks but highlighted challenges in wireless communication reliability and security adaptation. Despite significant advancements in blockchain-based UAV security, several challenges remain, as presented in Table 9. High storage costs, computational overhead, and scalability constraints continue to limit blockchain's full adoption in UAV networks. Future research should focus on optimizing lightweight blockchain frameworks, integrating AI-driven security analytics, and improving multi-blockchain interoperability for large-scale UAV deployments. As UAV operations become increasingly autonomous and interconnected, blockchain technology will play a pivotal role in ensuring data integrity, secure authentication, and resilient mission execution.

6.3. Cryptographic and Authentication Mechanisms. As UAV technology continues to advance, so do the cybersecurity threats that target data integrity, communication networks, and UAV operational controls. Unauthorized access, hacking attempts, and data breaches pose significant risks to military, commercial, and civilian UAV operations, necessitating the

TABLE 8: Comparative analysis of blockchain-based UAV security studies.

Study	Study focus	Blockchain approach	Security features	Computational overhead	Communication overhead	Scalability	Limitations
[83]	B5G UAV Security	Blockchain + AI-driven GNNs for authentication	High resilience, link failure detection	Moderate	Low	Moderate	AI-driven security models require further optimization
[84]	Blockchain-assisted UAV communication in 6G	IPFS + Blockchain for data security	Privacy preservation, reduced latency	High	Moderate	Moderate	High storage costs and technical adoption barriers
[85]	Secure UAV authentication	Permissioned blockchain for UAV-to-Edge authentication	Improved security, reduced latency	Moderate	Low	Moderate	Blockchain scalability challenges in large UAV networks
[38]	Decentralized authentication in UAVs	DPOS-based authentication model	97.5% attack detection success rate	Moderate	Moderate	Moderate	Authentication delays, reauthentication complexities
[86]	Secure communication for IoD UAV networks	Blockchain-based data integrity mechanism	Resilient against replay and MITM attacks	Low	Low	Moderate	High data storage costs
[87]	Blockchain in 5G UAV networks	Blockchain + fog computing for smart cities	Improved data transmission rates	Moderate	Moderate	High	Integration complexity and adoption barriers
[88]	Blockchain in UAV communication within 6G	Blockchain-based security framework	Low latency, improved connectivity	High	High	Moderate	High storage costs and network scalability issues
[89]	Large-scale UAV blockchain security	ECDH + Blockchain consensus mechanisms	Secure UAV transactions, reduced cyberattack risks	High	Moderate	Moderate	High computational over-head, integration complexity
[90]	Federated learning (FL) and Blockchain integration	SDN + FL for UAV data security	Enhanced privacy, improved cyber resilience	High	Moderate	High	Scalability and storage costs
[91]	FANET-based UAV security	Blockchain + PBFT consensus mechanism	Stable network performance despite node growth	High	Moderate	High	Energy constraints and computational overhead
[90]	Blockchain for post-disaster UAV communication	Hybrid DPOS + PBFT consensus for multi-agency coordination	Linear scaling, resilience against spoofing	Moderate	Low	High	High storage costs, scalability concerns
[92]	Blockchain-enabled FL for B5G UAV networks	Federated Learning + Blockchain framework	Enhanced privacy, improved security training accuracy	High	Moderate	High	Scalability issues, complex integration

TABLE 9: Comparison of immutable logging for security studies.

Study	Study focus	Security features	Computational overhead	Communication overhead	Scalability	Limitations	Future research directions
[93]	Blockchain-secured UAV logging	Immutability, public, auditability	High	Moderate	Moderate	High storage costs, computational demands	Optimization for large-scale UAV IoT ecosystems
[95]	Cross-blockchain inter operability	Decentralized asset transfer	High	High	Limited	Heterogeneity in blockchain systems	Practical implementations for scalable multi-blockchain UAV environments
[97]	Blockchain-based UAV authentication	Privacy preservation, replay attack resistance	Moderate	High	Moderate	High gas costs on Ethereum blockchain	Optimizing blockchain for real-time UAV authentication
[94]	Secure sensing data processing	Sensor authentication, data integrity	High	Moderate	Moderate	High storage demand	Optimizing computational efficiency for large-scale UAV deployments
[27]	Distributed blockchain security for UAVs	Enhanced security in UAV-IoT environments	Moderate	Low	Moderate	Storage constraints, computational overhead	Lightweight blockchain frameworks for UAV applications
[53]	Lightweight blockchain UAV security	Real-time authentication, privacy protection	Low	Low	High	Scalability issues, storage costs	Adaptive blockchain mechanisms for resource-constrained UAVs
[98]	Intelligent clustering-based UAV authentication	Reputation-based consensus protocol	Moderate	Moderate	Limited	Storage overhead, computational complexity	Scalable blockchain architectures for UAV swarms
[96]	Blockchain-based secure UAV data dissemination	MITM attack resistance, malicious data detection	High	High	Moderate	High storage costs, limited scalability	Improving blockchain integration for UAV networks
[103]	Semi-autonomous blockchain UAV framework	Secure communication, operational autonomy	Low	Moderate	High	Scalability limitations, adaptability concerns	Enhancing blockchain adaptability for UAV operations
[102]	Blockchain-enabled UAV traffic management	Secure UAV flight authentication	High	High	Moderate	Storage constraints, high resource consumption	Blockchain scalability improvements for UTM systems
[99]	Blockchain-secured UAV livestock tracking	Secure data transmission	Low	Low	High	Integration challenges, storage limitations	Scalable blockchain models for precision agriculture UAVs
[100]	AML-integrated blockchain for UAV authentication	Secure recipient verification, fraud mitigation	High	Moderate	Moderate	High data storage demands	Optimization for UAV logistics and delivery authentication
[101]	BIRDS: Blockchain for UAV package delivery	Immutable delivery tracking, energy-efficient networking	Moderate	Low	High	Scalability concerns (in-) tegration complexity	Refining decentralized UAV coordination mechanisms
[104]	Blockchain-based UAV authentication for industrial applications	Smart contract-based security, improved data integrity	Moderate	High	Moderate	Computational overhead, storage constraints	Optimizing authentication for large-scale industrial UAVs
[39]	Blockchain applications in IoT	Authentication mechanisms, security models	High	Moderate	Limited	Wireless communication reliability, adaptation issues	Improving blockchain scalability for IoT networks

integration of advanced cryptographic methods and authentication mechanisms into UAV security frameworks. These cryptographic advancements aim to enhance encryption resilience, prevent unauthorized drone takeovers, and secure UAV communications against both conventional and quantum-based cyber threats.

6.3.1. PQC. As quantum computing advances, it presents a formidable challenge to traditional cryptographic security mechanisms, particularly in UAV networks where encrypted transmissions contain sensitive surveillance data, mission commands, and military intelligence. Quantum computers have the potential to break widely used cryptographic algorithms such as RSA, ECC, and ElGamal, rendering current UAV security protocols ineffective. To address this growing concern, researchers have proposed PQC techniques, quantum key distribution (QKD), and hybrid AI-enhanced quantum encryption frameworks to safeguard UAV communications against quantum cyber threats.

CRYSTALS-Kyber has emerged as one of the most promising quantum-resistant cryptographic schemes for securing UAV data transmissions. Sharma et al. [31] demonstrated the effectiveness of CRYSTALS-Kyber in providing end-to-end encryption, ensuring UAV communication resilience against quantum-based adversaries. Similarly, Aissaoui et al. [105] compared multiple post-quantum key encapsulation mechanisms (KEMs), including CRYSTALS-Kyber, Hamming Quasi-Cyclic (HQC), and BIKE, identifying CRYSTALS-Kyber as the most efficient for UAV applications. Despite its security advantages, the integration of PQC mechanisms remains a challenge due to hardware constraints and computational overhead, which limit real-time UAV applications. Further optimization of PQC algorithms is necessary to improve scalability and performance in dynamic UAV networks.

QKD has also been explored as a viable solution for securing UAV communication channels. Ralegankar et al. [106] implemented a QKD-based UAV security framework, focusing on applications in agriculture, healthcare, and military operations. Their battlefield case study demonstrated improved security and reduced latency compared to classical cryptographic techniques. However, challenges such as high implementation costs and the complexity of QKD hardware integration remain barriers to widespread adoption. Hussien et al. [107] analyzed various QKD protocols for UAV and satellite communication, confirming their resilience against quantum attacks but highlighting the need for further optimization to improve cost-effectiveness and deployment feasibility.

To complement post-quantum encryption techniques, researchers have developed quantum-resistant authentication and key agreement mechanisms to secure UAV networks. Xia et al. [108] proposed a PQC-based identity authentication scheme leveraging the Kyber algorithm, significantly reducing computational overhead while ensuring robust security against quantum attacks. Similarly, Nair et al. [109] introduced a post-quantum secure cross-domain authentication (CDA) framework for the IoD, integrating blockchain and physical unclonable functions (PUFs) to provide mutual authentication without storing sensitive data on UAVs. Although these

frameworks enhance quantum resistance, challenges such as increased communication costs and integration complexity highlight the need for further research on optimizing PQC for real-time UAV operations.

Beyond cryptographic encryption, researchers have explored AI-driven and hybrid security models to enhance UAV resilience against quantum threats. Gnatyu et al. [35] investigated AI-enhanced encryption for UAV communication, utilizing ML to automate cryptographic key selection and improve encryption resilience. Their approach reduced latency and increased security adaptability; however, high computational overhead remained a key limitation. Similarly, Goyal et al. [110] proposed a fusion of Blockchain, AI, and quantum computing for UAV security, integrating blockchain for decentralized authentication, AI for anomaly detection, and quantum cryptography for encryption. While this multilayered approach improved security transparency, scalability remained an issue, requiring further optimizations for deployment in large-scale UAV networks.

Another promising direction in quantum-resistant UAV security is the use of lattice-based cryptography. Mishra et al. [22] developed a quantum-safe communication protocol for IoD networks using the Ring Learning With Error (RLWE) problem on lattices, eliminating the need for classical encryption schemes such as RSA and ECC. Their framework demonstrated high security and efficiency in mutual authentication, though integration complexity and computational resource demands remained key challenges. Similarly, Sandanamudi et al. [111] evaluated lightweight PQC algorithms such as SABER, NTRU, and CRYSTALS-Kyber for UAV networks, identifying SABER as the most efficient cipher. However, resource constraints in IoT-based UAV environments necessitate further optimizations for seamless PQC integration.

The application of quantum cryptography in UAV-based authentication has also gained traction. Abulkasim et al. [54] examined a quantum-based authentication scheme for securing IoD communication, utilizing quantum channels for information encoding and mutual authentication. Their approach exhibited strong resistance to impersonation and MITM attacks, though challenges in implementing quantum cryptographic techniques in resource-limited UAV environments remained. Likewise, Jawad et al. [112] introduced a visual cryptography-based UAV authentication system, leveraging chaotic map-based key generation to enhance security while minimizing computational and communication overhead.

Free-space optical (FSO) quantum communication has also been explored to improve UAV security. Alshaer et al. [113] investigated entanglement-based QKD over FSO links, achieving high security with low outage probabilities in dynamic UAV networks. Their study optimized transmit power and modulation schemes for secure FSO-QKD integration, but challenges such as atmospheric turbulence and tracking errors required further refinements.

For large-scale UAV networks, blockchain-integrated PQC solutions have been proposed to enhance authentication efficiency and prevent unauthorized access. El-Zawawy et al. [114] designed a blockchain-supported authentication protocol for drone-assisted internet of vehicles (IoV), achieving a 70%

reduction in energy consumption and 68% lower computational cost. However, the complexity of blockchain implementation in UAV networks remained a challenge. Similarly, Babu et al. [56] provided a comprehensive survey on quantum authentication and key agreement protocols, emphasizing the importance of lightweight PQC techniques for securing future UAV communication infrastructures.

Despite significant advancements in PQC, QKD, and hybrid AI-enhanced security models for UAV networks, several challenges persist, as presented in Table 10. The high computational and energy overhead associated with PQC algorithms limits real-time applicability in resource-constrained UAVs. Scalability remains a key concern, particularly in ultra-dense UAV networks where key management and authentication processes must be optimized for efficiency. Additionally, the integration complexity of QKD and PQC frameworks with existing UAV security architectures requires further research to ensure seamless implementation without introducing performance bottlenecks.

Future research should focus on developing energy-efficient PQC algorithms tailored for UAV systems, refining AI-driven encryption models for dynamic UAV operations, and enhancing blockchain scalability for decentralized UAV authentication. The integration of adversarial learning techniques with PQC may further improve resilience against evolving cyber threats. As quantum computing capabilities continue to advance, ensuring that UAV security architectures remain quantum-resistant will be paramount in safeguarding mission-critical drone operations in military, surveillance, and commercial applications.

6.3.2. Multi-Factor Authentication (MFA) for UAV Security. Ensuring secure authentication and communication in UAV networks is crucial to prevent cyber threats such as spoofing, MITM attacks, and unauthorized UAV access. Traditional authentication methods, such as password-based logins, have become increasingly vulnerable to hacking, brute force attacks, and credential theft, making MFA a necessary security enhancement for both military and commercial UAV operations. Recent research has explored AI-enabled, blockchain-integrated, and post-quantum authentication mechanisms to enhance UAV security while addressing challenges related to computational overhead, scalability, and real-time communication efficiency.

Several studies have introduced lightweight and AI-driven authentication schemes to enhance security in UAV networks. Deebak and Hwang [42] proposed a secure MFA (RL-SMFA) scheme for UAVs operating in military surveillance, leveraging elliptic curve cryptography (ECC) and AI-based analytics. Their approach improved packet delivery ratio and end-to-end delay while reducing power consumption, but scalability remained a challenge, requiring further optimization for large UAV networks. Similarly, Wang et al. [115] developed a three-factor authentication and key agreement protocol for UAV-assisted post-disaster emergency communication, integrating smart cards, biometrics, and physically unclonable functions (PUFs) to ensure secure UAV-to-UAV and UAV-to-emergency control vehicle (ECV) communication. While their

approach successfully mitigated communication vulnerabilities, it faced computational complexity issues in large-scale implementations.

Beyond individual authentication schemes, comparative analyses have been conducted to evaluate UAV authentication strategies. Mekdad et al. [40] analyzed 27 UAV authentication schemes, focusing on communication costs, storage overhead, and energy efficiency. Their study revealed that while many schemes optimized communication costs, they often failed to address storage and energy consumption constraints. Addressing this gap, Khan et al. [116] introduced a blockchain-based MFA protocol for 6G-enabled UAV networks, integrating Diffie–Hellman key exchange and Proof-of-Stake (PoS) mechanisms. Their approach significantly reduced authentication, communication, and computational overheads, though scalability remained a concern in ultra-dense networks.

Alternative cryptographic approaches have also been explored to improve authentication efficiency. Zhang et al. [117] developed a three-factor authentication protocol using BPV-FourQ ECC for IoD networks. Their solution improved processing speed by four to five times compared to traditional elliptic curve methods, demonstrating strong forward secrecy and attack resistance. Meanwhile, Cui et al. [23] introduced a chaotic map-based authentication and key agreement (AKA) scheme for UAV-assisted vehicular ad hoc networks (VANETs). Their hybrid authentication mechanism, which integrated honeywords and fuzzy verifiers, effectively reduced communication and computational overhead, making it a viable solution for emergency response UAV-VANET scenarios.

ECC-based authentication has been widely investigated as a lightweight cryptographic alternative. Usman et al. [118] designed an ECC-based authentication protocol for UAVs in SDN environments, achieving low latency and high security against forgery and replay attacks. However, computational overhead remained a challenge for large UAV fleets. Building upon this approach, Khalid et al. [119] introduced a two-factor authentication scheme for real-time UAV communication, addressing the risk of key management system failures. While their study demonstrated lower communication and computation costs, further research was required to improve scalability and system resilience.

PUF-based authentication has emerged as a promising solution for lightweight UAV security. Tian et al. [120] proposed a PUF-based CDA scheme for UAVs, ensuring tamper resistance and secure communication in multi-domain networks. Bansal and Sikdar [121] extended this work by utilizing Shamir's secret sharing scheme to mitigate errors in PUF responses, successfully reducing communication overhead while improving security against environmental variations. However, external noise in PUF responses remained a challenge, necessitating additional error correction mechanisms to enhance reliability.

Beyond authentication, secure key management is a critical aspect of UAV security. Ismael et al. [122] explored the use of the HIGHT lightweight block cipher for UAV communication encryption, demonstrating high performance with low computational costs. Their approach effectively reduced latency and energy consumption, though further refinement was needed

TABLE 10: Comparison of quantum and post-quantum cryptographic techniques for UAV security.

Study	1	2	3	4	5	6	7	8	9	10	11
[31]	Post-quantum cryptography	CRYSTALS Kyber	High	Moderate	Low	High	High	Hardware constraints	Key generation rate	Simulation	Optimization for UAV networks
[106]	Quantum cryptography	QKD	High	High	Moderate	Moderate	Limited	Complexity, cost	Latency, throughput	Simulation	Scalability, hardware optimization
[108]	Quantum-resistant authentication	Kyber algorithm	High	Low	Low	High	High	Integration complexity	Computational overhead	Simulation	Optimization of PQC algorithms
[109]	Cross-domain authentication	PUFs, blockchain	High	Moderate	High	Moderate	High	Communication, costs	Latency	Simulation	Real-time UAV operations
[105]	Post-quantum KEMs	CRYSTALS-Kyber, HQC, BIKE	High	High	Moderate	Moderate	Limited	Real-time overhead	Execution time	Simulation	Scalability optimizations
[54]	Quantum-based authentication	Quantum channels	High	High	High	Moderate	Limited	Implementation cost	Resistance to attacks	Simulation	Refinement of QKD techniques
[22]	Quantum-safe communication	RLWE	High	High	Moderate	Moderate	High	Integration complexity	Security metrics	Simulation	Resource-constrained environments
[107]	QKD for UAV-satellite comms	QKD	High	High	High	Moderate	Limited	Complexity, cost	Resilience metrics	Simulation	Real-world UAV operations
[26]	QKD for agrotechnical systems	BB84 protocol	High	Low	Low	High	High	Energy efficiency	Key generation rate	Simulation	Large-scale UAV deployments
[112]	Visual cryptography	Chaotic maps	High	Low	Low	High	High	Limited securityML, Computational overhead	Computational overhead	Simulation	Machine learning integration
[35]	AI-based encryption	AI-driven key selection	High	High	Low	Moderate	High	Computational overhead	Latency reduction	Simulation	AI-driven optimizations
[111]	Lightweight PQC algorithms	CRYSTALS-Kyber, SABER	High	Low	Low	High	High	IoT constraints	Execution efficiency	Simulation	Seamless IoT integration
[21]	Post-quantum digital signatures	Falcon, CRYSTALS-Dilithium	High	Low	Low	High	High	Key size	Execution consistency	Simulation	Broader UAV applications
[113]	Entanglement-based QKD	FSO-QKD	High	High	High	Moderate	Limited	Atmospheric turbulence	Outage probability	Simulation	FSO-QKD integration
[110]	Blockchain, AI, Quantum fusion	Blockchain, AI, QKD	High	High	High	Moderate	Limited	Scalability	Transparency, security	Simulation	Scalability optimizations
[56]	Survey of quantum protocols	PQC primitives	High	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Lightweight PQC techniques

Note: (1)-Study Focus, (2)- Cryptographic Technique, (3)- Security Features, (4)- Computational Overhead, (5)- Communication Overhead, (6)- Energy Efficiency, (7)- Scalability, (8)- Limitations, (9)- Performance Metrics, (10)- Experimental Validation, (11)- Future Research Directions.

for integration with diverse UAV models. Similarly, Kirsal Ever [55] introduced an ECC-based authentication protocol for UAVs functioning as mobile sinks in the IoD, achieving strong security guarantees while maintaining lightweight computational requirements.

Blockchain-based authentication has gained traction as a decentralized security solution for UAV networks. El-Zawawy et al. [114] designed a blockchain-supported authentication protocol for drone-assisted IoV, reducing energy consumption by up to 70% and computational cost by 68%. Their Burrow-Abadi-Needham (BAN) logic-based approach significantly improved authentication efficiency, but the complexity of blockchain integration required further research on scalability and adaptation to dynamic UAV fleets. Similarly, Dong et al. [123] developed a zero-trust security framework for UAV delivery authentication, incorporating MFA and blockchain-based self-sovereign identity (SSI) techniques. While their study effectively prevented spoofing and injection attacks, the complexity of managing multiple authentication factors posed implementation challenges.

An alternative blockchain-based approach was introduced by Alladi et al. [124], who developed a PUF-enabled mutual authentication scheme for UAV-GS communication, extending it to UAV-to-UAV authentication. Their cryptanalysis confirmed resilience against masquerade, replay, and node tampering attacks, though PUF integration complexity remained a key limitation. These studies highlight the potential of combining blockchain with advanced cryptographic methods to create secure, scalable authentication frameworks for UAV communication.

By integrating PQC with MFA frameworks, UAV security systems can achieve greater resilience against cyber intrusions, unauthorized access, and evolving cryptographic threats. Moving forward, future UAV security architectures should focus on integrating AI-powered threat detection, de-centralized cryptographic key management, and continuous authentication mechanisms to further enhance UAV cybersecurity and prevent next-generation cyberattacks.

The reviewed studies illustrate significant advancements in UAV authentication mechanisms, incorporating AI-based security analytics, blockchain, PQC, PUFs, and ECC-based encryption, presented in Table 11. While these technologies offer enhanced security, reduced computational overhead, and improved scalability, several challenges remain, including high storage costs, energy constraints, and integration complexity. Future research should prioritize optimizing authentication protocols for real-time UAV applications, developing energy-efficient cryptographic solutions, and improving blockchain scalability for large UAV networks. By addressing these challenges, next-generation UAV authentication frameworks can ensure secure and resilient communication in increasingly complex and adversarial environments.

6.4. Adversarial Defense Mechanisms in AI Models. As AI-driven UAVs become increasingly sophisticated, they also become more susceptible to adversarial attacks, where malicious actors exploit vulnerabilities in AI models to manipulate UAV decision-making. These attacks can involve manipulated

sensor data, deceptive image recognition inputs, or altered environmental conditions, leading to incorrect threat classifications, misdirected flight paths, or compromised mission execution. Given these risks, researchers are actively developing defense mechanisms to enhance AI robustness, improve resilience against adversarial manipulation, and ensure reliable UAV decision-making in high-risk operational environments.

6.4.1. Robust AI Models Against Adversarial Attacks. One of the most effective approaches for improving AI resilience is adversarial training, where AI models are exposed to adversarial attacks during training to help them learn how to counteract deceptive inputs. By deliberately introducing adversarial perturbations into training datasets, researchers can teach AI algorithms to recognize, adapt, and neutralize adversarial manipulations before they affect UAV operations. This proactive training methodology prepares UAV AI models to withstand real-world cyberthreats, reducing their vulnerability to malicious data modifications, false object detections, or sensor spoofing attempts.

The vulnerability of AI models to adversarial attacks has been widely studied across various domains, including computer vision, network security, UAV operations, and digital communication systems. Studies have demonstrated that DL models, including CNNs, vision transformers (ViTs), and deep clustering frameworks, are highly susceptible to adversarial perturbations, which can significantly degrade their performance. Chang et al. [34] conducted an extensive evaluation of adversarial robustness in image classification models, testing six CNN models against 13 different attack types. Their attack-agnostic approach provided an unbiased robustness assessment, serving as a benchmark for future defenses. Similarly, Olutimehin et al. [19] analyzed CNNs under adversarial stress, revealing attack success rates exceeding 85% for Carlini & Wagner (C&W) perturbations, underscoring the need for hybrid defense mechanisms combining adversarial training with real-time anomaly detection.

While adversarial vulnerabilities in image classification have been extensively examined, similar concerns have emerged in NLP applications. Goyal et al. [43] reviewed adversarial defenses in NLP models, introducing a taxonomy of defense mechanisms. Their study emphasized the fragility of DNNs in NLP tasks, highlighting the need for generalized adversarial defenses across multiple domains. Beyond vision and NLP, adversarial attacks also pose substantial threats to cybersecurity and network security systems. Zhang et al. [125] investigated DL-based network IDSs (NIDSs), introducing TIKI-TAKA, a defensive framework that restored intrusion detection rates to nearly 100% by integrating adversarial training, model ensembling, and query detection. Similarly, Ren et al. [48] examined adversarial attacks on cybersecurity applications, demonstrating that robustness could be significantly improved through certified defenses and adversarial retraining.

Ensuring real-world applicability of adversarial defenses remains a key challenge. Ai et al. [32] focused on adversarial perturbations in remote sensing applications, proposing cross-model generalization strategies to improve classifier robustness. Their study showed that real-world constraints heavily

TABLE 11: Comparison of studies on secure authentication mechanisms for UAV networks.

Study	Authentication mechanism	Cryptographic technique	Security features	Computational overhead	Communication overhead	Energy efficiency	Scalability	Limitations
[42]	RL-SMFA	Elliptic Curve Cryptography	High	Moderate	Low	High	Limited	Scalability
[115]	Three-factor authentication	PUFs, biometrics	High	High	Moderate	Moderate	Limited	Computational complexity
[116]	Blockchain-based MFA	Diffie- Hellman, PoS	High	Low	Low	High	Limited	Scalability
[117]	Three-factor authentication	BPV-FourQ ECC	High	Low	Low	High	High	None
[23]	Chaotic map-based AKA	Honeywords, fuzzy verifiers	High	Low	Low	High	High	None
[118]	ECC-based authentication	Elliptic Curve Cryptography	High	Moderate	Low	High	Limited	Computational overhead
[120]	PUF-based authentication	PUFs	High	Low	Low	High	High	Noise sensitivity
[114]	Blockchain-supported AKA	BAN logic	High	Low	Low	High	Limited	Complexity

influence adversarial defense effectiveness, necessitating scalable and computationally efficient solutions. Similarly, Tsai et al. [126] explored adversarial attacks on 3D data models, specifically targeting PointNet++ architectures used for object classification and segmentation. Their findings revealed that 3D adversarial perturbations could bypass existing defense mechanisms, emphasizing the need for robust adversarial defenses tailored to three-dimensional AI applications. Within the medical AI domain, adversarial security threats have been a growing concern. Ghaffari Laleh et al. [44] analyzed adversarial vulnerabilities in computational pathology, comparing the robustness of CNNs and ViTs in medical image classification tasks. Their findings demonstrated that ViTs exhibited significantly higher resilience to adversarial attacks than CNNs, suggesting that transformer-based architectures could offer a more robust alternative for medical AI applications. In the context of UAV security, adversarial attacks pose severe risks, as AI-driven UAV misclassifications can have real-world consequences. Raja et al. [47] examined adversarial attacks on AI-assisted UAV bridge inspection systems, demonstrating that adversarial inputs could mislead UAVs into misidentifying risk-prone infrastructure, increasing accident risks. Similarly, Tian et al. [127] studied adversarial attacks on automatic modulation classification (AMC) models in cognitive radio networks, revealing the vulnerability of AI-based communication systems to adversarial interference. Their research underscored the importance of robust defense mechanisms in wireless communication applications.

To mitigate adversarial threats, several defense mechanisms have been explored. Zhang et al. [128] conducted a comprehensive review of adversarial defenses in DNNs, comparing classical and state-of-the-art methods. Their findings emphasized the need for computationally efficient defenses that maintain high classification accuracy. Similarly, Waghela et al. [129] investigated defensive strategies against FGSM and PGD attacks, demonstrating that adversarial training combined with data preprocessing significantly enhanced robustness.

The use of Generative Adversarial Networks (GANs) for adversarial defense has gained traction in recent years. Taheri et al. [10] leveraged GAN-based adversarial training in botnet detection tasks, employing Pix2Pix GANs to generate adversarial examples for model retraining. Their iterative approach significantly reduced misclassification rates, demonstrating the effectiveness of GAN-based defenses in cybersecurity applications. Similarly, Chhabra et al. [25] examined GAN-based black-box adversarial attacks on deep clustering models, exposing serious vulnerabilities in state-of-the-art clustering algorithms and stressing the urgency of developing robust, unsupervised adversarial defenses.

Beyond traditional adversarial defenses, Anastasiou et al. [33] focused on adversarial robustness in AI-assisted manufacturing systems, highlighting the significance of defense algorithms in real-world industrial settings. Their study combined adversarial training with defensive strategies, showing substantial recovery of classifier accuracy in adversarial environments. A key advantage of adversarial training is real-time AI adaptation, which enables UAVs to continuously refine their decision-making models based on new threat patterns and environmental anomalies. AI-driven UAVs equipped with adaptive learning mechanisms can detect unexpected variations in sensor inputs or command signals, allowing them to flag suspicious activities, reject deceptive data, and maintain operational accuracy. Research studies have demonstrated that adversarially trained AI-driven UAVs exhibit enhanced resilience against AI deception attacks, particularly in military reconnaissance, autonomous surveillance, and target identification missions [3].

In summary, adversarial threats pose significant risks to AI models across multiple domains, including computer vision, cybersecurity, medical AI, UAV operations, and network security, as presented in Table 12. While adversarial training, model ensembling, and GAN-based defenses have shown promise, several challenges remain, particularly in computational efficiency, real-world adaptability, and scalability. Future research

TABLE 12: Comparison of adversarial training and defense mechanisms for AI security.

Study	Focus area	Adversarial Attack type	Defense mechanism	Computational overhead	Key findings	Limitations and future directions
[34]	Image classification	13 different attack types on CNN models	Attack- agnostic robustness assessment	Moderate	Established a benchmark for evaluating adversarial robustness	Requires real-world validation for adaptive defense mechanisms
[19]	CNN-based classification	Carlini & Wagner (C&W) perturbations	Hybrid adversarial training + anomaly detection	High	Identified high attack success rates (85%), highlighting CNN vulnerabilities	Computationally intensive; further optimization needed for real-time applications
[43]	NLP models	Text-based adversarial perturbations	Defense taxonomy for NLP security	Moderate	Showed deep NLP models' fragility; highlighted need for generalizable defenses	Requires integration of adversarial robustness in real-time NLP tasks
[125]	Network security (NIDS)	Intrusion detection adversarial attacks	TIKI-TAKA framework (ensemble + adversarial training)	Moderate	Restored intrusion detection rates to nearly 100%	High computational cost; scalability challenges in large networks
[48]	Cybersecurity applications	FGSM, PGD, BIM at- Tacks	Certified defenses + adversarial retraining	High	Improved AI security in cybersecurity models	Requires scalable and efficient adversarial training strategies
[32]	Remote sensing	Cross-model adversarial perturbations	Model generalization techniques	High	Improved classifier robustness in real-world constraints	High computational complexity; further optimization needed
[126]	3D object recognition	PointNet++ adversarial perturbations	Robust adversarial defenses for 3D models	Moderate	Revealed vulnerabilities in 3D AI models	Requires enhanced defenses for three-dimensional AI applications
[44]	Medical AI (Computa- tional Pathology)	CNN and ViT adversarial attacks	Comparative robustness analysis	Moderate	Found ViTs more robust than CNNs against adversarial attacks	High computational requirements for real-time medical AI applications
[47]	UAV bridge inspection	Adversarial misclassification	Robust UAV classification models	Moderate	Demonstrated UAV risk misidentification due to adversarial attacks	Requires robust countermeasures for UAV security systems
[127]	Cognitive radio net- works	Automatic modulation classification (AMC) attacks	AI-based defense against adversarial signals	Moderate	Showed AI-based communication systems are highly vulnerable	Requires additional adversarial defenses for secure UAV communication
[128]	DNNs	Multiple adversarial perturbations	Review of classical and state-of-the-art defenses	Moderate	Identified computationally efficient adversarial defenses	More real-world validation required
[129]	ML security	FGSM, PGD attacks	Adversarial training + data preprocessing	High	Improved adversarial robustness across ML models	For scalable adversarial defenses
[10]	Cybersecurity	Botnet detection adversarial attacks	GAN-based adversarial training	High	Reduced misclassification rates using Pix2Pix GANs	Computationally expensive; requires efficiency optimization
[25]	Deep clustering	Black-box GAN adversarial attacks	Adversarially robust deep clustering models	High	Exposed critical vulnerabilities in state-of-the-art clustering	High computational cost; requires real-world testing
[33]	AI-assisted manufacturing	Adversarial training	Combined adversarial training + defense algorithms	High	Improved classifier accuracy in adversarial environments	Further research needed on unsupervised- adversarial defenses
[3]	UAV defense	AI deception attacks on UAVs	Adaptive adversarial training	Moderate	Improved resilience of AI-driven UAVs in reconnaissance missions	Requires adaptation for industrial real-time settings
						Needs real-world UAV adversarial testing and enhancement

should focus on developing lightweight adversarial defenses, integrating AI-driven anomaly detection, and refining adversarial robustness benchmarks to ensure the security and reliability of AI systems in diverse applications. The integration of XAI with adversarial learning techniques will be crucial in improving AI resilience against sophisticated adversarial threats while maintaining transparency. As AI systems become increasingly embedded in critical operations, ensuring their robustness and security will be paramount in fostering trust, safety, and ethical deployment.

6.4.2. XAI for UAV Decision Transparency. The increasing reliance on AI-driven decision-making in cybersecurity, UAV operations, and autonomous defense systems has created a pressing need for explainability, transparency, and trustworthiness. Many DL models used in UAV operations function as black-box systems, where their internal decision logic remains opaque, making it difficult to trace errors, verify threat classifications, or audit mission-critical decisions. This lack of interpretability raises concerns in high-stakes applications such as military drone operations, autonomous law enforcement UAVs, and civilian airspace monitoring, where unexplained AI errors could lead to severe consequences. Researchers have proposed XAI frameworks, anomaly detection models, and adversarial learning techniques to enhance AI interpretability and mitigate security risks. Tiwari et al. [130] analyzed the role of XAI in cybersecurity, emphasizing how explainability in AI-driven threat detection can improve trust and accountability. Similarly, Moustafa et al. [131] and Masud et al. [132] explored anomaly-based intrusion detection in IoT ecosystems, demonstrating that DL models can effectively detect cyber threats, but their opacity hinders trust in AI-driven security solutions. These studies emphasized the need for integrating interpretability methods, such as SHapley Additive exPlanations (SHAP), to ensure AI decisions are transparent and explainable. XAI is also gaining prominence in securing UAV networks, where cyber threats such as GPS spoofing and DoS attacks pose significant risks. Wu et al. [133] proposed a CNN-BiLSTM-Attention (CBA) model for UAV cyberattack detection, integrating SHAP-based interpretability to provide human-understandable insights. Their evaluation demonstrated that explainability improves AI decision-making in UAV security, making it easier to identify and respond to cyber threats. Similarly, Ihekoronye et al. [49] developed Drone-Guard, a cybersecurity framework leveraging supervised ML and XAI to detect intrusions. Their model utilized decision trees and feature selection techniques, demonstrating high accuracy and low computational complexity, making it suitable for UAV networks with resource constraints. Ajakwe and Kim [134] reviewed security paradigms for UAV-based smart mobility and logistics, proposing an agile XAI framework that integrates blockchain authentication and zero-trust cybersecurity principles. Their findings reinforced the importance of explainability in securing UAV communications and preventing cyber intrusions.

AI transparency is particularly critical in military and autonomous defense systems, where AI decisions can have life-altering consequences. Chander et al. [135] examined the

black-box nature of AI in defense applications, proposing a framework for explainable and robust AI. Their study emphasized the necessity of transparency in autonomous weapons systems (AWS) to prevent biases and ensure accountability in military AI decision-making. Similarly, Cools and Maathuis [4] analyzed the integration of AWS in military operations, highlighting trust, human-machine teaming, and ethical concerns as key challenges. Their study underscored the importance of interpretable AI in high-risk environments, where human oversight is essential for reliable decision-making. In the context of air combat, Saldiran et al. [50] developed an XAI system using RL and reward decomposition to clarify AI agent decisions. Their research demonstrated that explainable models enhance trust in AI-driven military operations, making AI tactics more interpretable and reliable. Wang and Aouf [51] investigated adversarial attacks on autonomous driving systems, proposing a saliency map-based XAI approach to improve AI resilience in dynamic environments. Their study reinforced the importance of explainability in AI-driven navigation, particularly in real-time, high-stakes scenarios.

Adversarial learning techniques have been increasingly explored to enhance AI security and explainability, particularly in wireless communications and UAV-based key generation. Wei et al. [37] proposed an adversarial learning framework for physical layer key generation (PL-SKG) to mitigate MITM RIS eavesdropping. Their research incorporated symbolic XAI representation to interpret black-box neural networks, ensuring that AI-driven security measures remain transparent and accountable. Similarly, Maathuis [136] analyzed cybersecurity challenges in smart urban environments, emphasizing AI-driven threat detection and mitigation strategies. Their findings demonstrated that XAI-enabled cybersecurity frameworks enhance real-time threat detection, particularly in autonomous infrastructure defense.

Despite significant advancements in XAI for cybersecurity, UAV security, and autonomous defense systems, several challenges remain. AI transparency often comes at the cost of computational overhead, making real-time implementation difficult in resource-constrained environments. Studies such as those by Wu et al. [133] and Ihekoronye et al. [49] have highlighted the trade-offs between accuracy and efficiency in XAI-based security frameworks. Additionally, ethical concerns surrounding AI biases and decision-making opacity must be addressed to ensure AI is fair and trustworthy. Future research should focus on optimizing XAI algorithms for real-time security applications, enhancing trust in AI-driven military and defense systems, and developing scalable XAI models for UAV cybersecurity and smart mobility. The integration of adversarial learning techniques with XAI will be essential in improving AI resilience against adversarial threats while maintaining transparency. As AI becomes more deeply embedded in critical systems, ensuring that these technologies are both interpretable and robust will be paramount in fostering trust, security, and ethical AI deployment across various domains, as presented in Table 13.

6.5. Software and Hardware-Based Countermeasures. As cyber threats targeting UAVs become increasingly

TABLE 13: Comparison of XAI for cybersecurity, UAV security, and defense systems.

Study	Focus area	XAI method	Security enhancement	Computational overhead	Key findings	Limitations and future directions
[130]	XAI in cybersecurity	AI-driven threat detection	Trust and accountability	Moderate	Addressed biases, ethical concerns, and AI interpretability issues in cybersecurity	Requires further integration with real-time AI security frameworks
[131, 132]	Anomaly detection in IoT security	SHapley Additive exPlanations (SHAP)	Improved cyber threat detection	High	Demonstrated that deep learning can detect threats but lacks interpretability	Needs optimization for real-time detection with lower computational costs
[133]	UAV cyberattack detection	CNN-BiLSTM- Attention + SHAP	Enhanced interpretability of UAV security threats	Moderate	Explainability improved UAV decision-making in cyber threat response	Trade-offs exist between accuracy and computational efficiency
[49]	Intrusion detection in UAV networks	Decision trees + feature selection	High detection accuracy	Low	Developed DroneGuard a lightweight intrusion detection system	Requires scalability testing for larger UAV networks
[134]	UAV security for smart mobility	Blockchain-integrated XAI	Zero-trust cybersecurity for UAVs	High	Integrated blockchain authentication for UAV communications	Needs further optimization for blockchain scalability
[135]	AI in autonomous defense systems	Explainable and robust AI framework	AI transparency in military AI	Moderate	Proposed an AI framework for autonomous weapons systems (AWS)	Ethical and operational challenges in AWS deployments remain
[4]	Trust in Autonomous Weapons Systems (AWS)	Human-machine teaming	Enhancing trust in military AI	High	Highlighted the necessity of human oversight in AWS decision-making	Requires policy and regulatory frameworks for ethical AI in defense
[50]	AI decision-making in air combat	Reinforcement learning + reward decomposition	Improved trust in AI military operations	High	Explainability improved AI-driven combat decision transparency	Needs real-world validation in high-risk environments
[51]	Adversarial defense in autonomous driving	Saliency map-based XAI	Improved AI resilience	Moderate	Demonstrated importance of XAI in real-time navigation security	High computational cost in dynamic environments
[37]	XAI for wireless communication security	Adversarial learning for key generation	Mitigation of MITM-RIS attacks	High	Ensured transparent and accountable AI-driven security	Requires optimization for real-time key generation in UAV networks
[136]	Cybersecurity in smart urban environments	AI-driven threat detection	Enhanced real-time threat mitigation	High	XAI improved transparency in cybersecurity frameworks	Needs computational efficiency improvements for large-scale deployment

sophisticated, traditional AI and cryptographic security solutions alone are not sufficient to protect against firmware tampering, malware injections, and unauthorized modifications. To enhance UAV security and operational resilience, researchers are integrating hardware-based security mechanisms and AI-driven self-healing systems into UAV architectures. These advanced countermeasures provide additional layers of protection against cyber intrusions, ensuring that UAVs can detect, mitigate, and recover from security breaches in real time.

6.5.1. Hardware Security Modules (HSMs) for UAVs. HSMs are specialized cryptographic chips designed to securely store encryption keys, authenticate firmware, and prevent unauthorized modifications to UAV software systems. Unlike traditional software-based encryption, HSMs provide a physically secure environment that isolates sensitive cryptographic operations from external threats, reducing the risk of malware injections, firmware hijacking, and remote hacking attempts.

A key advantage of HSMs is their ability to prevent unauthorized firmware tampering, ensuring that only verified updates and authorized software modifications are implemented in UAV systems. This is particularly important for commercial UAV manufacturers and military drone developers, as compromised firmware can lead to remote UAV takeovers, mission disruptions, or intelligence leaks. Research has shown that integrating HSMs into UAV controllers significantly strengthens system security, preventing attackers from injecting malicious code or exploiting vulnerabilities in UAV command protocols. For example, commercial drone manufacturers now incorporate HSMs into UAV flight controllers to mitigate remote hacking risks and enhance cybersecurity defenses.

Ensuring the security of UAV communication systems is a growing concern, given the increasing sophistication of cyber threats targeting both physical and network layers. Wang et al. [137] provided a comprehensive review of security threats and countermeasures in UAV communications, highlighting the vulnerabilities in both single UAV and swarm systems. The study identified major attack vectors, such as GPS spoofing, jamming, remote takeover, and data tampering, emphasizing the need for evolving countermeasures. He et al. [138] demonstrated how GPS spoofing could be used to deceive UAV swarms, forcing them into unintended collisions. Their simulation-based approach validated the feasibility of such attacks, emphasizing the necessity for resilient countermeasures in autonomous UAV networks. Similarly, Ferreira et al. [28] explored low-cost software-defined radio (SDR) platforms for jamming and GPS spoofing to neutralize unauthorized UAV intrusions. Their research demonstrated the effectiveness of jamming techniques in taking control of UAVs, raising ethical concerns regarding potential misuse. Addressing these security challenges, Jiang et al. [139] developed an educational platform integrating UAV cyber-security training with real-world threat simulations. Their study underscored the importance of educating future UAV operators on cyber threats and countermeasures, ensuring that security protocols evolve alongside emerging threats.

The growing threat of unauthorized UAV operations has driven research into anti-UAV technologies, including detection, tracking, and mitigation strategies. Jie [140] proposed an intelligent anti-UAV command control system that could accurately detect and neutralize UAV threats in protected airspace. Unlike traditional methods, this approach integrated real-time monitoring with adaptive decision-making to formulate the most effective counter-UAV response. Wang et al. [141] expanded on this by reviewing Counter-Unmanned Aircraft System (C-UAS) technologies, analyzing detection mechanisms such as radar, acoustic sensors, vision-based tracking, and passive RF monitoring. Their findings highlighted the strengths and limitations of various mitigation strategies, with radar-based detection showing the highest effectiveness against high-speed UAVs. Meanwhile, Babu and Pal [142] examined UAS security enhancements within IoT environments, focusing on unique vulnerabilities such as signal tampering, laser-based attacks, and GPS jamming. Their research emphasized the importance of robust encryption and access control mechanisms in securing UAV networks. Similarly, Emani [9] investigated the security vulnerabilities of the MAVLink protocol, a widely used communication standard in UAV systems. The study introduced a shared-key obfuscation algorithm that effectively mitigated MITM attacks, highlighting the need for enhanced encryption protocols in UAV communication frameworks.

As UAVs continue to integrate into civilian and military applications, securing their firmware, operating systems, and communication protocols has become a priority. Malik et al. [143] evaluated various security measures to protect UAV ecosystems, including firmware encryption, intrusion detection and prevention systems (IDPS), and dynamic SSID obfuscation. Their findings showed significant improvements in security against common cyber threats, though challenges remained in mitigating advanced persistent threats. Ficco et al. [144] conducted penetration testing on UAV communication systems, identifying multiple vulnerabilities in the MAVLink protocol. Their research underscored the necessity of regular security audits and real-time monitoring to prevent cyber intrusions. Patil and Pournouri [145] examined the security risks associated with open-source Linux-based operating systems used in UAVs. Their study found that while open-source software offers flexibility, it also introduces vulnerabilities that could be exploited if not properly secured. In response to these challenges, Chen [2] designed an enhanced MAVLink protocol (EMP) integrated with multiple security layers to fortify UAV control and communication systems. Experimental results demonstrated that EMP significantly reduced susceptibility to cyberattacks while maintaining efficient performance.

UAV sensor security is another critical area of concern, as compromised sensor data can lead to mission failure or unintended UAV behavior. Wei et al. [146] evaluated online classification methods for detecting sensor attacks in UAV systems, using ML techniques to identify anomalies in gyroscopes, accelerometers, and GPS data. Their proposed lightweight classification model achieved a detection rate of 89.38%, outperforming traditional methods. Tili et al. [20] investigated

UAV battery depletion attacks, where adversaries exploit vulnerabilities in UAV charging and power management systems. Their study proposed mitigation techniques to counter these threats, emphasizing the importance of securing UAV power systems. Airlangga and Liu [24] analyzed security risks in centralized UAV-cloud architectures, identifying 26 attack variations and their corresponding defense strategies. Their findings provided valuable insights into securing UAV data exchange in cloud-integrated networks, a growing trend in UAV operations.

The rapid advancements in UAV cybersecurity research highlight the ongoing battle between emerging threats and evolving countermeasures. While studies such as those by Wang et al. [137] and He et al. [138] have emphasized the growing sophistication of UAV-targeted cyberattacks, research by Jiang et al. [139] and Ficco et al. [144] underscores the importance of cybersecurity education and proactive vulnerability assessments, as presented in Table 14. Future research must focus on developing lightweight, scalable security frameworks that balance robust protection with minimal computational overhead. The integration of AI-driven anomaly detection, advanced encryption methods, and real-time threat monitoring will be essential in safeguarding UAV ecosystems. Additionally, as UAVs become more prevalent in civilian and military domains, regulatory frameworks must evolve to ensure that security measures keep pace with technological advancements. Ultimately, a multilayered security approach combining detection, mitigation, and prevention strategies will be key to maintaining secure and resilient UAV operations in the face of ever-evolving cyber threats.

6.5.2. Self-Healing UAV Systems. Beyond hardware-based defenses, researchers are developing AI-driven self-healing software frameworks that enable UAVs to autonomously repair compromised code and restore system integrity following a cyberattack. Traditional UAV security measures often rely on external intervention or manual debugging, which can delay response times and leave UAVs vulnerable to prolonged security breaches. In contrast, self-healing UAV systems leverage AI-powered anomaly detection to identify cyber threats, isolate malicious software, and autonomously restore the UAV to its original secure state without requiring human input.

One of the primary benefits of self-healing UAV architectures is their ability to counteract real-time cyberattacks, such as firmware corruption, adversarial AI manipulations, and system exploits. If a UAV detects anomalous activity in its software environment, the self-healing framework can automatically initiate security protocols, such as rolling back system updates, restoring encrypted backup configurations, or executing real-time threat removal algorithms. Recent research has demonstrated the effectiveness of AI-driven self-repair mechanisms, where UAVs can identify and neutralize malware threats while continuing mission-critical operations. These innovations play a crucial role in ensuring UAV resilience, minimizing downtime, and reducing cybersecurity risks in autonomous flight operations.

Ensuring resilience and security in modern computing environments has become a critical area of research, particularly in the context of self-healing systems, autonomous technologies, and cyber-physical networks. Adeniyi et al. [57] investigated proactive self-healing approaches in MEC, emphasizing the importance of real-time fault anticipation and mitigation. Their study identified key challenges in resource allocation and security, demonstrating that self-healing techniques could significantly enhance MEC reliability. López-Vilos et al. [30] extended this concept to wireless sensor networks (WSNs) under jamming attacks, proposing the Fairness Cooperation with Power Allocation (FCPA) strategy. Their results showed over 50% improvement in data transmission efficiency and a 63% increase in residual energy efficiency, highlighting the benefits of integrating self-healing mechanisms in network resilience strategies. Similarly, Dias et al. [147] explored self-healing patterns in IoT environments, identifying modular frameworks for error detection and recovery. Their findings reinforced the importance of adaptable self-repair methodologies in distributed computing systems.

Autonomous vehicles and UAV swarms face similar challenges regarding resilience, security, and fault tolerance. Quraishi et al. [148] examined the vulnerabilities of self-driving car architectures, particularly in relation to cyberattacks on sensors and communication networks. Their study proposed N-version programming as a resilience mechanism, demonstrating significant improvements in decision-making under adversarial conditions. Phadke and Medrano [149] extended this resilience framework to UAV swarms, addressing challenges in communication, security, and agent coordination. Their comprehensive analysis showed that enhancing swarm resilience could improve mission success rates, particularly in dynamic environments such as SAR operations. Chandran and Vipin [17] built upon this research by analyzing multi-UAV networks for disaster monitoring, demonstrating how edge computing and AI could enhance network efficiency and security. Their findings emphasized the importance of real-time adaptive networking strategies in UAV-based disaster response scenarios.

The integration of AI in security applications has also gained attention, particularly in addressing vulnerabilities in cyber-physical systems. Yaacoub et al. [150] reviewed security challenges in robotic systems across various domains, from industrial automation to military applications. Their study highlighted the need for AI-driven countermeasures to mitigate emerging cyber threats. Rahman et al. [36] examined AI-enabled 6G O-RAN security, identifying critical threats to data-driven networks and proposing advanced countermeasures. Their research demonstrated that AI-driven anomaly detection could significantly enhance network security, reinforcing the necessity of integrating AI into cybersecurity frameworks. Adil et al. [151] explored similar AI applications in UAV-assisted IoT security, focusing on authentication and data privacy measures. Their findings revealed that AI, ML, DL, and RL algorithms could mitigate security vulnerabilities in UAV communication networks.

Deep RL (DRL) has also emerged as a promising approach to improving security and adaptability in UAV systems.

TABLE 14: Hardware security modules (HSMs)-based UAV cybersecurity studies.

Study	Focus area	Security approach	Key benefits	Challenges	Experimental findings	Future research directions
[137]	UAV communication security	Threat analysis and countermeasures	Identified vulnerabilities in UAV communication	Evolving nature of cyber threats	Provided a security framework for UAV networks	Developing AI-driven threat mitigation
[138]	GPS spoofing defense	GPS spoofing detection	Demonstrated real-world feasibility of GPS spoofing attacks	Need for real-time countermeasures	Validated GPS spoofing impact on UAV swarms	Implementing adaptive anti-spoofing mechanisms
[28]	Anti-UAV jamming	Software-defined radio (SDR) jamming	Demonstrated control over unauthorized UAVs	Ethical concerns of misuse	Validated effectiveness of SDR jamming	Developing ethical and regulatory frameworks
[139]	UAV cybersecurity education	Training platform with real-world simulations	Improved awareness of cyber threats in UAVs	Limited scalability for large UAV networks	Integrated UAV security simulations	Expanding cybersecurity education and training programs
[140]	Anti-UAV defense systems	Intelligent anti-UAV command control	Improved real-time UAV threat detection	Computational overhead of adaptive decision-making	Formulated optimal anti-UAV strategies	Reducing latency in real-time UAV defense
[141]	Counter-UAS (C-UAS) technologies	Radar, acoustic, vision, and RF detection	Identified most effective mitigation techniques	Limitations in high-speed UAV tracking	Radar-based detection was most effective	Enhancing multi-sensor fusion for UAV detection
[142]	UAS security in IoT environments	Encryption and access control mechanisms	Improved security against GPS jamming	Increased complexity in IoT integration	Addressed signal tampering threats	Developing lightweight encryption for UAV networks
[9]	MAVLink protocol security	Shared-key obfuscation algorithm	Prevented man-in-the-middle attacks	Computational overhead of cryptographic implementation	Improved UAV communication security	Optimizing MAVLink security protocols for real-time applications
[143]	UAV firmware security	Intrusion detection and prevention systems (IDPS)	Enhanced UAV resilience against malware	Complexity in mitigating persistent threats	Improved security against common attacks	Strengthening real-time firmware monitoring
[144]	UAV vulnerability testing	Penetration testing on UAV systems	Identified multiple vulnerabilities in MAVLink protocol	Need for continuous security monitoring	Demonstrated UAV communication weaknesses	Real-time security auditing for UAV networks
[145]	UAV operating system security	Security assessment of Linux-based UAV OS	Highlighted security risks in open-source UAV frameworks	Patch management challenges in UAV software	Found that open-source systems need stricter security controls	Enhancing security for open-source UAV operating systems
[2]	Securing UAV control	Enhanced MAVLink protocol (EMP)	Reduced cyberattack susceptibility	Increased system complexity	EMP significantly improved security	Expanding EMP for broader UAV communication frameworks
[146]	UAV sensor security	AI-based anomaly detection for sensor integrity	Improved detection of malicious sensor data	Computational overhead in real-time processing	Achieved 89.38% detection rate	Optimizing lightweight AI models for UAV sensor security
[20]	UAV battery security	UAV battery depletion attack mitigation	Enhanced protection of UAV power systems	Energy constraints in secure power management	Proposed mitigation techniques for battery attacks	Securing UAV power management against cyber threats
[24]	UAV-cloud security	Analysis of centralized UAV-cloud architecture	Identified 26 attack variations and defenses	Scalability challenges in large UAV networks	Provided insights for securing UAV cloud integration	Strengthening encryption and authentication in UAV-cloud systems

Sarikaya and Bahtiyar [152] reviewed DRL-based security solutions, highlighting their potential to enhance UAV cybersecurity through adaptive decision-making. Their findings indicated that DRL-based security models could effectively counteract cyber threats, though real-time applicability remained a challenge. Homann et al. [153] examined trust metrics in artificial hormone systems (AHS) within organic computing, proposing a trust module to detect and counteract security threats. Their research demonstrated that AHS-enhanced security models could improve system reliability, particularly in dynamic and heterogeneous computing environments. Similarly, Yungaicela-Naula et al. [45] evaluated security automation in SDN, showcasing how self-healing, self-adaptation, and self-optimization techniques could enhance threat mitigation efficiency. Their study reported a 95% reduction in breach costs, emphasizing the transformative impact of security automation in SDN environments.

Advancements in modular robotics and cooperative IoT frameworks have further reinforced the need for secure and adaptable cyber-physical systems. Yaacoub [154] examined modular robotics' integration into IoT, identifying security challenges and proposing lightweight cryptographic solutions. Their study demonstrated how loss-less compression algorithms, such as Brotli, could optimize communication within lattice-based modular robots. Ridhawi et al. [155] extended this research to cooperative UAV-supported IoT services, analyzing decentralized security solutions such as blockchain-based authentication. Their findings highlighted the importance of secure data-sharing protocols in multi-agent systems, particularly in smart city and disaster response applications.

Ensuring resilience in intelligent autonomous systems (IAS) has also become a growing concern, particularly in countering advanced cyber threats. Mani et al. [156] examined the security of IAS through AI-driven methodologies, addressing multistage cyberattacks such as fileless malware and adaptive poisoning attacks. Their study incorporated DL-based application profiling, perception algorithms using LSTM DNNs, and intrusion detection mechanisms. Their findings demonstrated that AI-driven cyber attribution could enhance attack adaptability, though computational overhead remained a concern. Similarly, Wang and Aouf [51] investigated adversarial threats in autonomous driving systems, proposing an explainable deep adversarial RL framework. Their results demonstrated improved robustness against dynamic adversarial attacks, reinforcing the need for transparent and interpretable AI security models.

The convergence of AI, edge computing, and self-healing security frameworks is shaping the future of cyber-physical systems, ensuring resilience against evolving threats. While studies by Adeniyi et al. [57], López-Vilos et al. [30], and Dias et al. [147] have established self-healing methodologies as critical components of modern computing, research by Qurashi et al. [148], Phadke and Medrano [149], and Chandran and Vipin [17] have underscored the importance of resilience in autonomous systems. The integration of AI-driven security frameworks, as explored by Yaacoub et al. [150], Rahman et al. [36], and Adil et al. [151], further highlights the need for intelligent, adaptive defense mechanisms. Future research should

focus on refining these methodologies to enhance scalability, minimize computational overhead, and ensure real-time applicability in increasingly complex cyber-physical ecosystems.

In a nutshell, ensuring the security of AI-driven UAVs requires a multilayered defense approach that integrates advanced AI-based IDS for real-time cyber threat detection, blockchain for secure and tamper-proof data transmission, and quantum cryptography with MFA to prevent hacking attempts. Additionally, adversarial training and XAI enhance AI trustworthiness, mitigating the risks associated with adversarial attacks and opaque decision-making models. To further fortify UAV systems, HSMs protect against firmware tampering, while self-healing UAV frameworks enable autonomous recovery from cyber intrusions, ensuring long-term operational resilience in high-risk environments, as presented in Table 15. These combined security measures are essential for protecting UAVs from evolving cyber threats and ensuring mission reliability across military, commercial, and civilian applications.

7. Limitations and Future Research Directions

As AI-driven UAVs continue to evolve, so do the security threats and vulnerabilities associated with their deployment. The increasing sophistication of cyber threats, adversarial AI attacks, and communication vulnerabilities necessitates the development of next-generation security frameworks that can dynamically adapt to emerging threats. Traditional security models rely on predefined attack signatures and static defenses, making them ineffective against evolving cyber threats. Future research must focus on developing advanced AI security models, integrating 6G-powered security solutions, and enhancing UAV swarm cybersecurity to ensure safe, resilient, and autonomous UAV operations.

7.1. Limitations of the Present Study. Although this paper offers a comprehensive taxonomy and synthesis of AI-enabled UAV cybersecurity solutions, it is limited by the scope of existing literature, which is rapidly evolving. The review primarily focuses on peer-reviewed studies published up to early 2025, excluding some proprietary or classified government research that may offer additional insights. Furthermore, while several frameworks and techniques are discussed, their practical feasibility, economic costs, and policy integration have not been fully assessed.

Additionally, this review does not quantitatively compare the performance of proposed defense mechanisms due to the heterogeneity of evaluation metrics, simulation tools, and datasets across existing studies. Future meta-analyses that unify evaluation methodologies would allow for better benchmarking of UAV security frameworks. This review also acknowledges limitations including the exclusion of non-peer-reviewed and non-English studies, potential keyword dependency in the literature search, inherent subjectivity in study selection, and the qualitative nature of the data synthesis, suggesting future research mitigate these by incorporating a broader range of sources, expanding keywords and using semantic search, employing multiple reviewers and inter-rater reliability measures, and considering quantitative meta-analysis to enhance the robustness and generalizability of findings.

TABLE 15: Comparison of self-healing UAV security solutions.

Study	Focus area	Security methodology	Resilience mechanism	Computational overhead	Key findings	Limitations and future directions
[57]	Self-healing in Mobile Edge Computing	Proactive fault anticipation	AI-driven threat mitigation	High	Improved MEC reliability and security	Needs further optimization for real-time UAV applications
[30]	Resilience in Wireless Sensor Networks	Clustering-based self-healing	Jamming attack mitigation	Moderate	50% improvement in data transmission, 63% energy efficiency	Scalability limitations in dense UAV networks
[147]	Self-healing in IoT environments	Modular error detection	AI-driven adaptive recovery	High	Demonstrated importance of modular frameworks in distributed systems	High computational cost for large-scale IoT deployments
[148]	Resilience in self-driving car architectures	N-version programming	Sensor and communication security	Moderate	Improved decision-making under cyberattacks	Requires further testing in UAV and military applications
[149]	Resilience in UAV swarms	Adaptive AI-driven security	UAV swarm communication security	High	Improved mission success rates in SAR operations	Computational complexity in real-time UAV operations
[17]	Multi-UAV networks for disaster monitoring	AI + edge computing	Autonomous Mission resilience	High	Enhanced disaster response efficiency and security	Requires optimization for large-scale UAV swarm networks
[150]	AI-driven security in robotics	AI-based intrusion detection	Cyber-physical system resilience	Moderate	Identified AI-driven countermeasures for security threats	Needs further validation in UAV deployments
[36]	AI-enabled 6G O-RAN security	AI-driven anomaly detection	Cyber threat mitigation	High	Enhanced network security against data-driven attacks	Needs optimization for real-time AI security
[151]	UAV-assisted IoT security	AI, ML, DL, RL for authentication	Data privacy and security	High	Improved UAV network resilience against cyberintrusions	Computational overhead in large-scale UAV networks
[152]	DRL in UAV security	Adaptive learning-based IDS	AI-driven threat mitigation	High	Improved UAV cybersecurity through DRL models	Limited real-time applicability
[153]	Artificial hormone systems for security	AI-based trust module	Autonomous security decision-making	Moderate	Improved system reliability in dynamic environments	Needs further research on scalability in UAV networks
[45]	Security automation in SDN	Self-healing, self-adaptation	AI-driven automated threat mitigation	High	95% reduction in breach costs	Needs efficiency improvements for real-time security applications
[154]	Modular robotics and IoT security	Lossless cryptography (Brotli)	Secure UAV-IoT communication	High	Optimized communication efficiency in modular UAV networks	Requires further work on cryptographic resilience
[155]	Cooperative UAV-IoT services	Blockchain-based authentication	Secure multi-agent data sharing	High	Improved decentralized authentication for UAVs	Limited scalability in ultra-dense UAV networks
[156]	AI-driven security for IAS	Multi-stage cyberattack prevention	AI-based cyber attribution	High	Improved resilience against file-less malware	High computational overhead remains a concern
[51]	Adversarial threats in autonomous driving	Deep adversarial reinforcement learning	AI model robustness	High	Enhanced resilience against adversarial attacks	Needs optimization for real-time deployments

While the findings offer broad applicability across UAV sectors due to the universality of identified threats and the potential of proposed countermeasures, their direct application is constrained by variations in UAV types, operational contexts, and technological maturity, necessitating further research to validate and adapt these solutions for diverse scenarios.

7.2. Discussion of Alternative Explanations for Observed Findings. While this review focuses on the growing intersection of AI and cybersecurity within the realm of UAVs, the observed research trends are likely influenced by factors extending beyond this central theme. Several alternative explanations could contribute to the current landscape of literature.

Firstly, the independent and rapid advancements in both AI and cybersecurity are significant drivers of research. The increasing capabilities of AI across diverse applications naturally lead to its exploration in security contexts, while the escalating sophistication of cyber threats necessitates the investigation of advanced defenses, irrespective of their specific application to UAVs. Thus, the burgeoning research at this intersection may partly reflect these broader, parallel advancements.

Secondly, funding priorities and strategic initiatives from governmental bodies, industries, and academic institutions play a crucial role in shaping research directions. Targeted funding towards AI security or UAV technologies could lead to a concentration of research efforts in specific areas, influencing the prevalence of certain themes observed in this review. Understanding these funding landscapes could provide valuable context for interpreting the current research focus.

Thirdly, the inherent bias in academic publishing towards positive and novel findings might lead to an overrepresentation of studies demonstrating the effectiveness of AI-driven security solutions or highlighting significant vulnerabilities. Research with null or negative results, which could offer crucial insights into the limitations of certain approaches, might be less likely to be published, potentially skewing the overall understanding derived from this review.

Finally, specific technological breakthroughs, such as advancements in onboard processing power for UAVs or enhanced communication capabilities, could indirectly influence the research focus. These advancements might enable the practical implementation and evaluation of more complex AI-based security measures on UAV platforms, leading to increased research in these areas.

By considering these alternative explanations, this review aims to provide a more comprehensive understanding of the research dynamics in AI-driven UAV cybersecurity. While the primary focus on AI's dual role in enhancing and challenging UAV security is well-supported, acknowledging these broader contextual factors offers a richer interpretation of the current state of knowledge and can inform future research by highlighting potential biases or underexplored avenues.

7.3. Next-Generation AI Security for UAVs. AI plays a pivotal role in UAV security, enabling autonomous threat detection, real-time decision-making, and anomaly identification. However, existing AI-based UAV security models suffer from limited adaptability to emerging cyber threats, as they rely on fixed

training datasets that do not account for previously unseen attack patterns. As cyber adversaries develop new attack techniques, UAV AI models must be capable of self-adapting, continuously learning, and proactively responding to zero-day exploits, AI adversarial manipulations, and real-time cybersecurity challenges.

7.3.1. Self-Adaptive AI Security Models. Traditional AI-driven UAV security models depend on static algorithms that can recognize only predefined cyber threats, making them vulnerable to novel attacks. Future AI security solutions must focus on self-adaptive AI models that can dynamically learn, evolve, and respond to cyber threats in real time.

Self-adaptive AI security models will leverage RL-based AI, where UAVs utilize real-time feedback loops to learn from cyberattack patterns and security breaches, improving their threat detection accuracy over time [147, 148]. By continuously analyzing new security vulnerabilities, these UAVs will be able to neutralize threats before they cause operational failures. Another emerging approach is self-supervised learning, where UAV AI models will be trained using raw flight data and real-time mission insights rather than pre-labeled datasets. This allows UAVs to detect security anomalies independently, without requiring constant human intervention.

A key research area in next-generation AI-driven UAV security is the development of autonomous cybersecurity models that identify and neutralize previously unknown malware strains before they infect UAV systems. These models will integrate adaptive DL techniques, real-time anomaly detection, and AI-driven response mechanisms to enhance UAV resilience against next-generation cyber threats.

7.3.2. AI-Powered Threat Intelligence Systems. Current UAV cybersecurity frameworks primarily rely on manual threat monitoring, making it difficult to detect and mitigate cyberattacks in real time [137, 151]. This approach creates significant delays in identifying vulnerabilities and responding to security breaches, leaving UAVs exposed to potential hacking attempts, AI adversarial manipulations, and communication disruptions. Future UAV security research must prioritize the integration of AI-powered threat intelligence systems, capable of predicting, identifying, and mitigating cyber threats before they materialize.

AI-powered threat intelligence will enable UAVs to autonomously analyze global cyberattack trends, identify new security vulnerabilities, and implement real-time security patches without requiring direct human intervention [20]. These systems will be capable of:

1. Identifying emerging security vulnerabilities before they can be exploited.
2. Automatically patching software weaknesses in UAV operating systems.
3. Detecting potential cyberattacks based on historical UAV cyberattack datasets and predictive analytics.

By integrating predictive threat intelligence models, UAVs will gain the ability to anticipate cyber threats before they occur, significantly improving response times, security resilience, and

operational efficiency. Research in this area is focusing on AI models trained on UAV cyberattack datasets, enabling them to predict new hacking techniques and deploy countermeasures proactively.

7.4. Integration of 6G and Edge Computing. As autonomous UAV operations become more widespread and complex, the need for secure, real-time communication and low-latency AI processing is paramount to ensure mission success and operational resilience. Traditional UAV networks, which often depend on existing wireless technologies such as 4G or 5G, face challenges including higher latency, limited bandwidth, and increased susceptibility to cyber threats. These limitations can compromise timely threat detection and response, particularly in dynamic or hostile environments where delays may lead to mission failure or security breaches.

The advent of 6G networks combined with Edge Computing offers a transformative solution to these challenges by enabling ultra-fast, secure data transmission and localized AI processing directly on or near UAV platforms. Edge computing facilitates on-device or near-device threat detection, significantly reducing the latency associated with sending data to distant cloud servers for analysis. This capability is critical for time-sensitive security applications, allowing UAVs to respond instantly to emerging threats with minimal delay, thus enhancing operational reliability and safety.

However, edge-based threat detection comes with trade-offs. While it excels in providing real-time responses and reducing dependency on network connectivity, edge devices typically have limited computational resources compared to cloud infrastructures, which may restrict the complexity of threat analysis algorithms. In contrast, cloud-based threat detection leverages vast processing power and large-scale data aggregation to perform more comprehensive and sophisticated threat intelligence. Nevertheless, cloud reliance introduces potential latency and reliability issues due to variable network conditions and the risk of communication interruptions, which can delay critical security actions.

Balancing these trade-offs is essential in designing robust UAV cybersecurity architectures. Hybrid approaches that integrate edge computing for immediate, low-latency threat detection with cloud-based systems for deeper, more holistic analysis present a promising pathway. The seamless integration of 6G connectivity with edge and cloud resources will redefine UAV security frameworks, ensuring autonomous UAV fleets can operate securely, efficiently, and resiliently even in highly adversarial and bandwidth-constrained environments.

7.4.1. 6G-Enabled Secure UAV Networks. Ensuring secure and reliable communication remains one of the foremost challenges in UAV cybersecurity, particularly as UAVs increasingly operate in remote, contested, or hostile environments. Existing 4G and 5G networks often suffer from higher latency and bandwidth constraints, which can leave UAVs vulnerable to cyberattacks such as signal hijacking, GPS spoofing, and jamming. These limitations impair the UAVs ability to authenticate command signals, detect threats in real time, and securely relay critical data to ground control stations, thereby compromising mission integrity and safety [88].

The integration of 6G technology into UAV networks promises to overcome these challenges by enabling ultra-fast, ultra-reliable, and low-latency communication channels. Expected to reach peak deployment around the 2030s following ongoing standardization efforts, 6G aims to deliver sub-millisecond latency and near-perfect connectivity, which are essential for real-time threat detection and immediate response in UAV operations. A significant technical advancement within 6G is the incorporation of quantum cryptography-based encryption, which offers unprecedented security by safeguarding UAV-to-ground communications against interception and decryption attempts. This quantum-resistant encryption will ensure data confidentiality and integrity, rendering conventional cyberattack methods ineffective.

Additionally, 6G networks will enable the seamless integration of AI-enhanced security protocols that empower UAVs to autonomously detect and mitigate cyber threats with minimal delay. Recent research, such as the work by Tsegaye [83], has demonstrated prototype AI frameworks that leverage 6Gs high data throughput and low latency to perform real-time anomaly detection, identifying threats like jamming, unauthorized access, and malicious command injections instantaneously. Experimental pilot projects are already underway, exploring 6G-enabled UAV architectures that combine quantum-safe encryption with AI-driven cybersecurity measures, signaling significant progress toward practical implementation.

In summary, the transition to 6G-enabled UAV networks represents a paradigm shift in UAV cybersecurity, offering end-to-end encryption, ultra-fast data transmission, and intelligent autonomous threat detection. While full-scale adoption depends on the maturation of 6G standards and infrastructure, current research and early-stage trials indicate a clear trajectory toward deployment within the next decade, setting the foundation for resilient and secure UAV operations in increasingly complex environments.

7.4.2. Edge AI for Secure UAV Data Processing. Another significant challenge in UAV cybersecurity is the reliance on cloud-based AI models for data processing, threat detection, and decision-making. Traditional UAV systems transmit large volumes of sensor data, flight information, and AI-driven analytics to remote cloud servers for processing and analysis. However, this centralized computing model introduces security vulnerabilities, including data interception risks, latency issues, and potential disruptions from cyberattacks targeting cloud networks [57, 123, 157].

To mitigate these challenges, researchers are exploring Edge AI computing, which enables onboard UAV data processing without relying on external cloud infrastructure. By integrating AI-driven security algorithms directly into UAV hardware, Edge AI allows UAVs to detect cyber threats locally, process mission-critical data in real time, and minimize reliance on remote servers. This approach significantly reduces the risk of cyber intrusions, as UAV data remains within the local network and is not exposed to external cybersecurity vulnerabilities.

One of the key benefits of Edge AI for UAV security is on-device AI learning, where UAVs can continuously adapt to new

cyber threats, analyze security anomalies, and optimize AI threat detection models autonomously. Unlike traditional cloud-based AI models, which require constant internet connectivity and centralized data storage, Edge AI allows UAVs to remain operational even in contested environments where network access is limited or compromised.

A recent research focus has been the development of onboard AI-driven IDS that enable UAVs to autonomously detect cyber threats without requiring an external cloud-based security infrastructure. These self-contained AI security systems analyze network traffic, sensor inputs, and command authenticity to identify potential cyberattacks in real time, ensuring rapid response and threat mitigation without delays caused by cloud server dependencies.

7.5. AI-Empowered UAV Swarm Security. UAV swarms, where multiple drones operate autonomously in a coordinated manner, introduce new security challenges that differ significantly from single-drone operations. These challenges arise due to inter-drone communication dependencies, decentralized AI-based threat detection, and the need for secure mission execution. Ensuring secure communication within UAV swarms is critical, as adversaries can exploit vulnerabilities to intercept signals, manipulate swarm behavior, or introduce unauthorized drones into the network. Additionally, decentralized security frameworks are necessary to prevent single points of failure, ensuring that UAV swarms remain operational even if individual units are compromised. Blockchain technology has also emerged as a key enabler of UAV swarm security, providing tamper-proof mission execution and encrypted communication records to prevent unauthorized drone infiltration or command injection attacks [98, 149].

7.5.1. Decentralized AI for Secure UAV Swarm Operations. One of the primary security challenges in UAV swarm networks is their reliance on centralized control systems, which makes them vulnerable to hacking, signal jamming, and command injection attacks. Cyber adversaries can disrupt swarm coordination by targeting the centralized command node, leading to mission failure, loss of UAV control, or reprogramming of flight objectives. To address these vulnerabilities, decentralized AI-driven security frameworks are being developed to enable independent UAV operation without reliance on a single control center [138].

Future AI-driven UAV swarms will leverage swarm intelligence, where individual UAVs will collaborate dynamically, enabling real-time threat detection, attack response, and mission adaptation without requiring direct human intervention. Instead of relying on fixed pre-programmed security models, these UAVs will employ adaptive AI algorithms that allow them to identify, communicate, and collectively neutralize cyber threats. In the event of a cyberattack, AI-driven UAV swarms will be capable of autonomously reconfiguring flight paths, isolating compromised drones, and reinforcing security protocols to maintain operational stability.

A notable research area in this field is the development of AI-powered UAV swarm threat detection models, which enable drones to communicate securely and autonomously in high-risk environments. These models will integrate RL,

federated AI training, and real-time anomaly detection to strengthen UAV swarm resilience against evolving cyber threats. By adopting decentralized AI-driven security frameworks, UAV swarms will be able to operate effectively in contested airspaces, ensuring that no single point of failure compromises the entire swarm operation.

7.5.2. Blockchain-Based UAV Swarm Communication Security. Another major security concern in UAV swarm operations is unauthorized drone infiltration, where adversaries attempt to spoof communication signals to introduce malicious drones into the swarm network. Attackers can manipulate inter-drone communications, causing system failures, mission deviations, or intelligence leaks. Traditional security mechanisms, such as encrypted command channels, are insufficient to fully secure UAV swarm networks, as they can still be exploited through signal interception or identity spoofing techniques [100, 138].

To address this challenge, researchers are developing blockchain-based UAV security solutions that decentralize swarm communication and ensure cryptographic authentication of UAV-to-UAV transmissions. Blockchain technology offers a tamper-proof, immutable record of UAV communication logs, ensuring that all drone interactions are securely stored, verified, and resistant to unauthorized modifications. By implementing blockchain-powered authentication, UAV swarms can verify the legitimacy of incoming communication signals, preventing adversaries from spoofing commands or introducing unauthorized drones into the network.

Additionally, blockchain-based UAV swarm security frameworks will ensure that each UAV-to-UAV communication exchange is cryptographically signed and validated, preventing MITM attacks that could otherwise allow hackers to intercept and alter drone communication protocols. A promising research area in this domain focuses on enhancing UAV swarm resilience through blockchain-enhanced authentication models, where every drone in the swarm network must verify its identity against a decentralized ledger before executing commands.

By integrating blockchain security frameworks with decentralized AI-driven UAV swarm intelligence, re-searchers aim to develop autonomous, self-secure UAV networks that can operate reliably in contested and high-risk environments. Future studies should continue exploring hybrid AI-blockchain security architectures, optimizing their scalability, efficiency, and integration into real-world UAV swarm applications to ensure highly resilient, self-defending autonomous drone networks.

In conclusion, the future of AI-driven UAV security will be shaped by self-learning AI models, 6G-enhanced encrypted communication, and decentralized UAV swarm protection frameworks, ensuring that UAVs remain secure, autonomous, and resilient against evolving cyber threats, as presented in Table 16. Self-adaptive AI models will allow UAVs to learn from cyberattacks and continuously enhance their security mechanisms, while 6G-powered networks will provide ultra-fast, quantum-secured communication, preventing signal hijacking and data interception. The integration of Edge AI computing will enable UAVs to process security threats locally,

TABLE 16: Future research directions in AI-driven cybersecurity for UAVs.

Research area	Key focus	Challenges and future considerations
Self-Adaptive AI Security Models	• AI-driven real-time learning from cyber threats • RL for adaptive threat detection • Self-supervised learning for autonomous anomaly detection	• High computational overhead for continuous learning • Difficulty in identifying zero-day threats in real-time • Need for scalable and efficient AI-driven security models
AI-Powered Threat Intelligence Systems	• Predictive analytics for emerging cyber threats • Automated software patching and UAV OS security updates • AI-driven UAV cybersecurity datasets for proactive defense	• Requires large-scale real-time data analysis • Potential adversarial attacks on AI training datasets • Difficulty in ensuring low-latency threat mitigation
6G-Enabled Secure UAV Networks	• Ultra-low-latency communication for UAV security • Quantum cryptography-based UAV data encryption • AI-enhanced real-time anomaly detection in UAV networks	• High infrastructure costs for 6G deployment • Security concerns in quantum cryptographic key management • Compatibility with existing UAV network architectures
Edge AI for Secure UAV Data Processing	• AI-driven on-device cybersecurity analysis • Reduction of cloud dependency for security processing • AI-based IDS for UAVs	• Limited computational power of onboard UAV hardware • High energy consumption for real-time AI analysis • Secure firmware updates and AI model retraining challenges
Decentralized AI for Secure UAV Swarm Operations	• AI-driven swarm intelligence for autonomous security • Federated learning and real-time inter-drone threat sharing • Adaptive mission planning for UAV swarm resilience	• Ensuring secure AI model updates across UAV networks • Complexity in implementing real-time inter-drone coordination • Resistance against adversarial AI attacks on swarm behavior
Blockchain-Based UAV Swarm Communication Security	• Decentralized UAV authentication using blockchain • Tamper-proof, immutable UAV communication records • Cryptographic authentication of UAV-to-UAV transmissions	• High storage requirements for blockchain-based UAV data • Network latency issues in decentralized authentication • Scalability concerns in high-volume UAV operations

reducing reliance on cloud networks and minimizing latency vulnerabilities. In swarm operations, AI-powered UAV networks will use decentralized intelligence to detect and neutralize cyber intrusions autonomously, ensuring collective threat mitigation. Additionally, blockchain-based authentication systems will prevent unauthorized drone takeovers, ensuring that only verified UAVs can participate in swarm missions. By advancing research in these areas, UAVs will achieve next-generation cybersecurity resilience, enabling secure, efficient, and autonomous operations in military, commercial, and high-risk environments.

7.6. Ethical Considerations. The expanding deployment of AI-driven UAVs across sectors like surveillance, law enforcement, and delivery services brings forth critical ethical considerations that demand attention to ensure responsible innovation and mitigate potential harm. Future research must proactively address these evolving challenges.

- **Privacy Concerns:** UAVs' capacity for high-resolution surveillance raises significant privacy issues. Mass surveillance capabilities, tracking without consent, and sensitive data collection necessitate robust regulations and privacy-preserving technologies. Future research should focus on developing and integrating privacy-enhancing AI algorithms that can selectively process and anonymize data, minimizing the risk of privacy violations. This includes exploring techniques like FL for collaborative model training without direct data sharing and differential privacy to add noise to data outputs.
- **Autonomous Decision-Making:** The autonomy of AI-driven UAVs, especially in critical applications, brings concerns about accountability and potential errors. Algorithmic biases, incorrect object recognition, and the lack of human oversight can lead to severe consequences. Future research needs to prioritize the

development of XAI models for UAVs, enabling transparency in decision-making processes and facilitating human intervention when necessary. Furthermore, investigating methods for robust verification and validation of AI algorithms used in UAVs is crucial to ensure safety and reliability.

- **Data Security and Misuse:** The vast data collected by UAVs presents risks of breaches and misuse. Strong data protection and secure communication are essential. Future research should explore advanced cryptographic techniques tailored for UAV communication constraints, such as lightweight cryptography and homomorphic encryption, which allows computations on encrypted data. Additionally, research into secure data storage and access control mechanisms within UAV networks is vital.
- **Dual-Use Dilemma:** The potential for AI and UAVs to be used for both beneficial and harmful purposes requires careful consideration. Future research must contribute to developing ethical guidelines and regulatory frameworks that govern the development and deployment of UAVs, preventing their weaponization or misuse. This includes international collaborations to establish standards and protocols that promote responsible innovation and address the dual-use challenge.

In conclusion, addressing the ethical considerations surrounding AI-driven UAVs is not only a necessity for responsible technology adoption but also a crucial area for future research. Interdisciplinary efforts involving technologists, ethicists, policymakers, and the public are essential to navigate these complex challenges and ensure that AI-driven UAVs are used in ways that benefit society while upholding ethical principles.

8. Conclusion

The increasing integration of AI into UAVs has significantly enhanced their operational capabilities across military, commercial, and civilian domains. However, it has simultaneously introduced critical cybersecurity challenges that threaten mission reliability, data confidentiality, and operational integrity. This review comprehensively analyzed the evolving threat landscape faced by AI-driven UAVs, highlighting key vulnerabilities such as cyberattacks, adversarial AI manipulations, GPS spoofing, and communication disruptions.

Theoretically, this research advances the understanding of UAV cybersecurity by identifying gaps in existing AI security models, emphasizing the need for self-adaptive AI defenses, decentralized blockchain-based authentication, and quantum-resistant encryption techniques. It provides a structured taxonomy of threats and counter-measures that can serve as a foundation for future academic research in AI security, UAV swarm resilience, and Edge AI architectures. While the review aimed to provide a holistic view, some findings revealed discrepancies compared to initial expectations, primarily due to the rapid evolution of both UAV technology and cyber threats, leading to a dynamic and shifting research landscape. As shown

in Figure 3, this trend illustrates the rising number of publications in high-impact journals and conferences, highlighting a growing research focus on AI-driven cybersecurity solutions for UAVs.

Practically, the findings of this study have significant implications for real-world UAV operations. They highlight the urgency of deploying multilayered, real-time, and autonomous cybersecurity frameworks to ensure safe and resilient UAV missions in high-risk environments. The adoption of 6G-enabled secure networks, on-board AI threat detection systems, and decentralized swarm security models are essential for maintaining the operational reliability of UAVs in critical sectors such as defense, disaster response, and logistics. These findings emphasize the need for agile and adaptive security strategies, given the pace of technological change. As depicted in Figure 4, the distribution of articles across journals and conferences shows that the majority of the research is published in journals (81%) compared to conferences (19%), indicating a preference for in-depth studies over preliminary findings. A comparative analysis of existing studies is provided in Table 2, highlighting the unique contributions of this review in integrating AI, cryptography, blockchain, and swarm security frameworks. Furthermore, Table 7 compares different FL-based UAV security studies, outlining their methodologies, security mechanisms, and limitations.

By integrating cutting-edge AI security advancements with practical defense strategies, this work not only bridges the gap between academic research and industry needs but also paves the way for the development of next-generation autonomous UAV systems that are both secure and trustworthy. Future research should continue focusing on scalable, lightweight, and XAI solutions, ensuring that UAV systems remain resilient against increasingly sophisticated cyber threats while upholding ethical and regulatory standards. As Figure 5 shows, IEEE and Elsevier are the leading publishers, accounting for 34% and 17% of the publications, respectively, suggesting their significant influence in this domain. Moreover, Figure 6 provides a detailed breakdown of the research categories, with AI-based intrusion detection and prevention being the most prominent area, followed by software and hardware-based countermeasures.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

Conflicts of Interest

The author declares no conflicts of interest.

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References

- [1] A. Oracevic and A. Salman, "Unmanned Aerial Vehicles in Peril: Investigating and Addressing Cyber Threats to UAVs," in *2024 International Conference on Smart Applications*,

- Communications and Networking (SmartNets)*, (IEEE, 2024): 1–7.
- [2] H. H. Chen, “Developing a Custom Communication Protocol for UAVs: Ground Control Station and Architecture Design,” *Internet of Things* 27 (2024): 101319.
 - [3] R. U. Mhapsekar, M. I. Umrani, M. Faizan, O. Ali, and L. Abraham, “Building Trust in AI-Driven Decision Making for Cyber-Physical systems (CPS): A Comprehensive Review,” in *2024 IEEE 29th International Conference on Emerging Technologies and Factory Automation (ETFA)*, (IEEE, 2024): 1–8.
 - [4] K. Cools and C. Maathuis, “Trust or Bust: Ensuring Trustworthiness in Autonomous Weapon Systems,” in *MILCOM 2024-2024 IEEE Military Communications Conference (MILCOM)*, (IEEE, 2024): 182–189.
 - [5] G. Kumar and A. Altalbe, “Artificial Intelligence (AI) Advancements for Transportation Security: in-Depth Insights Into Electric and Aerial Vehicle Systems,” *Environment, Development and Sustainability* (2024): 1–51.
 - [6] A. Masadeh, M. Alhafnawi, H. A. B. Salameh, A. Musa, and Y. Jararweh, “Reinforcement Learning-Based Security/Safety UAV System for Intrusion Detection Under Dynamic and Uncertain Target Movement,” *IEEE Transactions on Engineering Management* 71 (2022): 12498–12508.
 - [7] M. Golam, M. M. Alam, D. S. Kim, and J. M. Lee, “BLM-Chain: AI-Driven Blockchain for UAV Threat Resistance in IOBT,” in *2024 15th International Conference on Information and Communication Technology Convergence (ICTC)*, (IEEE, 2024): 1609–1613.
 - [8] E. Bayhan, Z. Ozkan, M. Namdar, and A. Basgumus, “Deep Learning Based Object Detection and Recognition of Unmanned Aerial Vehicles,” in *2021 3rd International Congress on Human-Computer Interaction, Optimization and Robotic Applications (HORA)*, (IEEE, 2021): 1–5.
 - [9] R. Emani, “Cybersecurity Analysis and Defense of the Mavlink Uas Protocol,” (2024).
 - [10] S. Taheri, A. Khormali, M. Salem, and J. S. Yuan, “Developing a Robust Defensive System Against Adversarial Examples Using Generative Adversarial Networks,” *Big Data and Cognitive Computing* 4, no. 2 (2020): 11.
 - [11] N. Kumar and A. Chaudhary, “Surveying Cybersecurity Vulnerabilities and Countermeasures for Enhancing UAV Security,” *Computer Networks* 252 (2024): 110695.
 - [12] H. Xi, L. Ru, J. Tian, et al., “URAdv: A Novel Framework for Generating Ultra-Robust Adversarial Patches Against UAV Object Detection,” *Mathematics* 13, no. 4 (2025): 591.
 - [13] J. Tian, C. Shen, B. Wang, et al., “LESSON: Multi-Label Adversarial False Data Injection Attack for Deep Learning Locational Detection,” *IEEE Transactions on Dependable and Secure Computing* 21, no. 5 (2024): 4418–4432.
 - [14] Z. Zhao, B. Wang, X. Yao, J. Tian, R. Dong, and P. Zhu, “Dynamic Quasi-Hyperbolic Momentum Iterative Attack With Small Perturbation for 4D-Flight Trajectory Prediction,” *IEEE Internet of Things Journal* 12, no. 12 (2025): 22533–22553.
 - [15] M. J. Page, J. E. McKenzie, P. M. Bossuyt, et al., “The Prisma 2020 Statement: An Updated Guideline for Reporting Systematic Reviews,” *BMJ* 372 (2021).
 - [16] V. Balasubramanian, M. Aloqaily, M. Guizani, and B. Ouni, “Security Challenges and Solutions for Autonomous Vehicles and Drones in the AI Age,” *IEEE Internet of Things Magazine* 8, no. 2 (2025): 129–136.
 - [17] I. Chandran and K. Vipin, “Multi-Uav Networks for Disaster Monitoring: Challenges and Opportunities From a Network Perspective,” *Drone Systems and Applications* 12 (2024): 1–28.
 - [18] Z. Zhou, G. Liu, and Y. Tang, “Multiagent Reinforcement Learning: Methods, Trustworthiness, Applications in Intelligent Vehicles, and Challenges,” *IEEE Transactions on Intelligent Vehicles* 9, no. 12 (2024): 8190–8211.
 - [19] A. T. Olutimehin, A. J. Ajayi, O. C. Metibemu, A. Y. Balogun, T. O. Oladoyinbo, and O. O. Olaniyi, “Adversarial Threats to Ai-Driven Systems: Exploring the Attack Surface of Machine Learning Models and Countermeasures,” (2025).
 - [20] F. Tlili, L. C. Fourati, S. Ayed, and B. Ouni, “Investigation on Vulnerabilities, Threats and Attacks Prohibiting UAVs Charging and Depleting UAVs Batteries: Assessments & Countermeasures,” *Ad Hoc Networks* 129 (2022): 102805.
 - [21] R. Aissaoui, J. C. Deneuille, C. Guerber, and A. Pirovano, “Authenticating Civil UAV Communications With Post-Quantum Digital Signatures,” in *2023 IEEE/AIAA 42nd Digital Avionics Systems Conference (DASC)*, (IEEE, 2023): 1–9.
 - [22] D. Mishra, M. Singh, P. Rewal, et al., “Quantum-Safe Secure and Authorized Communication Protocol for Internet of Drones,” *IEEE Transactions on Vehicular Technology* 72, no. 12 (2023): 16499–16507.
 - [23] J. Cui, X. Liu, H. Zhong, et al., “A Practical and Provably Secure Authentication and Key Agreement Scheme for UAV-Assisted VANETs for Emergency Rescue,” *IEEE Transactions on Network Science and Engineering* 11, no. 2 (2024): 1454–1468.
 - [24] G. Airlangga and A. Liu, “A Study of the Data Security Attack and Defense Pattern in a Centralized UAV–Cloud Architecture,” *Drones* 7, no. 5 (2023): 289.
 - [25] A. Chhabra, A. Sekhari, and P. Mohapatra, “On the Robustness of Deep Clustering Models: Adversarial Attacks and Defenses,” *Advances in Neural Information Processing Systems* 35 (2022): 20566–20579.
 - [26] M. Bakyt, L. La Spada, N. Zeeshan, K. Moldamurat, and S. Atanov, “Application of Quantum Key Distribution to Enhance Data Security in Agrotechnical Monitoring Systems Using UAVs,” *Applied Sciences* 15, no. 5 (2025): 2429.
 - [27] T. Ahamed Ahanger, A. Aldaej, M. Atiquzzaman, I. Ullah, and M. Yousufudin, “Distributed Blockchain-Based Platform for Unmanned Aerial Vehicles,” *Computational Intelligence and Neuroscience* 2022 (2022): 4723124.
 - [28] R. Ferreira, J. Gaspar, P. Sebastião, and N. Souto, “A Software Defined Radio Based Anti-UAV Mobile System With Jamming and Spoofing Capabilities,” *Sensors* 22, no. 4 (2022): 1487.
 - [29] A. Shafique, A. Mehmood, and M. Elhadef, “Detecting Signal Spoofing Attack in UAVs Using Machine Learning Models,” *IEEE Access* 9 (2021): 93803–93815.
 - [30] N. López Vilos, C. Valencia Cordero, R. Souza, and S. Montejo Sánchez, “Clustering-Based Energy-Efficient Self-Healing Strategy for WSNs Under Jamming Attacks,” *Sensors* 23, no. 15 (2023): 6894.
 - [31] T. Sharma, S. A. Soleymani, M. Shojafar, and R. Tafazolli, “Secured Communication Schemes for UAVs in 5G: Crystals-Kyber and Ids,” *ArXiv* (2025).
 - [32] S. Ai, A. S. V. Koe, and T. Huang, “Adversarial Perturbation in Remote Sensing Image Recognition,” *Applied Soft Computing* 105 (2021): 107252.
 - [33] T. Anastasiou, S. Karagiorgou, P. Petrou, et al., “Towards Robustifying Image Classifiers Against the Perils of Adversarial Attacks on Artificial Intelligence Systems,” *Sensors* 22, no. 18 (2022): 6905.
 - [34] C. L. Chang, J. L. Hung, C. W. Tien, C. W. Tien, and S. Y. Kuo, “Evaluating robustness of ai models against adversarial attacks,” in *Proceedings of the 1st ACM Workshop*

- on Security and Privacy on Artificial Intelligence, (Association for Computing Machinery, 2020): 47–54.
- [35] S. Gnatyuk, T. Okhrimenko, and D. Proskurin, “AI-Based Encryption System for Secure UAV Communication,” in *2024 IEEE 7th International Conference on Actual Problems of Unmanned Aerial Vehicles Development (APUAVD)*, (IEEE, 2024): 93–98.
 - [36] T. F. Rahman, A. S. Abdalla, K. Powell, W. AlQwider, and V. Marojevic, “Network and Physical Layer Attacks and Countermeasures to AI-Enabled 6G o-Ran,” *ArXiv* (2021).
 - [37] Z. Wei, W. Hu, J. Zhang, W. Guo, and J. A. McCann, “Explainable Adversarial Learning Framework on Physical Layer Key Generation Combating Malicious Reconfigurable Intelligent Surface,” *IEEE Transactions on Wireless Communications* 24, no. 4 (2025): 3529–3545.
 - [38] A. Yazdinejad, R. M. Parizi, A. Dehghantanha, H. Karimipour, G. Srivastava, and M. Aledhari, “Enabling Drones in the Internet of Things With Decentralized Blockchain-Based Security,” *IEEE Internet of Things Journal* 8, no. 8 (2021): 6406–6415.
 - [39] A. Jain, S. Barke, M. Garg, et al., “A Walkthrough of Blockchain-Based Internet of Drones Architectures,” *IEEE Internet of Things Journal* 11, no. 21 (2024): 34924–34940.
 - [40] Y. Mekdad, A. Aris, A. Acar, et al., “A Comprehensive Security and Performance Assessment of UAV Authentication Schemes,” *Security and Privacy* 7, no. 1 (2024): e338.
 - [41] E. Ntizikira, W. Lei, F. Alblehai, K. Saleem, and M. A. Lodhi, “Secure and Privacy-Preserving Intrusion Detection and Prevention in the Internet of Unmanned Aerial Vehicles,” *Sensors* 23, no. 19 (2023): 8077.
 - [42] B. D. Deebak and S. O. Hwang, “Intelligent Drone-Assisted Robust Lightweight Multi-Factor Authentication for Military Zone Surveil-Lance in the 6g Era,” *Computer Networks* 225 (2023): 109664.
 - [43] S. Goyal, S. Doddapaneni, M. M. Khapra, and B. Ravindran, “A Survey of Adversarial Defenses and Robustness in NLP,” *ACM Computing Surveys* 55, no. 14s (2023): 1–39.
 - [44] N. Ghaffari Laleh, D. Truhn, G. P. Veldhuizen, et al., “Adversarial Attacks and Adversarial Robustness in Computational Pathology,” *Nature Communications* 13, no. 1 (2022): 5711.
 - [45] N. M. Yungaicela-Naula, C. Vargas-Rosales, J. A. A. Pérez-Díaz, and M. Zareei, “Towards Security Automation in Software Defined Networks,” *Computer Communications* 183 (2022): 64–82.
 - [46] M. Keshavarz, M. Gharib, F. Afghah, and J. D. Ashdown, “Uastrustchain: A Decentralized Blockchain-Based Trust Monitoring Framework for Autonomous Unmanned Aerial Systems,” *IEEE Access* 8 (2020): 226074–226088.
 - [47] A. Raja, L. Njilla, and J. Yuan, “Adversarial Attacks and Defenses Toward AI-Assisted UAV Infrastructure Inspection,” *IEEE Internet of Things Journal* 9, no. 23 (2022): 23379–23389.
 - [48] K. Ren, T. Zheng, Z. Qin, and X. Liu, “Adversarial Attacks and Defenses in Deep Learning,” *Engineering* 6, no. 3 (2020): 346–360.
 - [49] V. U. Ihekoronye, S. O. Ajakwe, J. M. Lee, and D. S. Kim, “Droneguard: An Explainable and Efficient Machine Learning Framework for Intrusion Detection in Drone Networks,” *IEEE Internet of Things Journal* 12 (2024): 7708–7722.
 - [50] E. Saldiran, M. Hasanzade, G. Inalhan, and A. Tsourdos, “Towards Global Explainability of Artificial Intelligence Agent Tactics in Close Air Combat,” *Aerospace* 11, no. 6 (2024): 415.
 - [51] C. Wang and N. Aouf, “Explainable Deep Adversarial Reinforcement Learning Approach for Robust Autonomous Driving,” *IEEE Transactions on Intelligent Vehicles* 10 (2024): 2551–2563.
 - [52] A. Bouguettaya, H. Zarzour, A. M. Taberkit, and A. Kechida, “A Review on Early Wildfire Detection From Unmanned Aerial Vehicles Using Deep Learning-Based Computer Vision Algorithms,” *Signal Processing* 190 (2022): 108309.
 - [53] P. Abichandani, D. Lobo, S. Kabrawala, and W. McIntyre, “Secure Communication for Multiquadrotor Networks Using Ethereum Blockchain,” *IEEE Internet of Things Journal* 8, no. 3 (2021): 1783–1796.
 - [54] H. Abulkasim, B. Goncalves, A. Mashatan, and S. Ghose, “Authenticated Secure Quantum-Based Communication Scheme in Internet-of-Drones Deployment,” *IEEE Access* 10 (2022): 94963–94972.
 - [55] Y. Kirsal Ever, “A Secure Authentication Scheme Framework for Mobile-Sinks Used in the Internet of Drones Applications,” *Computer Communications* 155 (2020): 143–149.
 - [56] P. R. Babu, S. A. Kumar, A. G. Reddy, and A. K. Das, “Quantum Secure Authentication and Key Agreement Protocols for Iot-Enabled Applications: A Comprehensive Survey and Open Challenges,” *Computer Science Review* 54 (2024): 100676.
 - [57] O. Adeniyi, A. S. Sadiq, P. Pillai, M. A. Taheir, and O. Kaiwartya, “Proactive Self-Healing Approaches in Mobile Edge Computing: A Systematic Literature Review,” *Computers* 12, no. 3 (2023): 63.
 - [58] G. Kumar and H. Alqahtani, “Machine Learning Techniques for Intrusion Detection Systems in SDN-Recent Advances, Challenges and Future Directions,” *CMES-Computer Modeling in Engineering & Sciences* 134 (2023): 89–119.
 - [59] H. Alqahtani and G. Kumar, “Cybersecurity in Electric and Flying Vehicles: Threats, Challenges, AI Solutions & Future Directions,” *ACM Computing Surveys* 57 (2024): 1–34.
 - [60] G. Kumar and K. Kumar, “A Multi-Objective Genetic Algorithm Based Approach for Effective Intrusion Detection Using Neural Networks,” in *Intelligent Methods for Cyber Warfare*, (Springer, 2014).
 - [61] H. Alqahtani and G. Kumar, “A Deep Learning-Based Intrusion Detection System for in-Vehicle Networks,” *Computers and Electrical Engineering* 104 (2022): 108447.
 - [62] H. Alqahtani and G. Kumar, “Machine Learning for Enhancing Transportation Security: A Comprehensive Analysis of Electric and Flying Vehicle Systems,” *Engineering Applications of Artificial Intelligence* 129 (2024): 107667.
 - [63] H. Alqahtani and G. Kumar, “Advances in Artificial Intelligence for Detecting Algorithmically Generated Domains: Current Trends and Future Prospects,” *Engineering Applications of Artificial Intelligence* 138 (2024): 109410.
 - [64] A. Taneja and G. Kumar, “Attention-CNN-LSTM Based Intrusion Detection System (ACL-IDS) for in-Vehicle Networks,” *Soft Computing* 28, no. 23–24 (2024): 13429–13441.
 - [65] H. Alqahtani and G. Kumar, “Deep Learning-Based Intrusion Detection System for in-Vehicle Networks With Knowledge Graph and Statistical Methods,” *International Journal of Machine Learning and Cybernetics* (2024): 1–17.
 - [66] H. Alqahtani and G. Kumar, “Efficient Routing Strategies for Electric and Flying Vehicles: A Comprehensive Hybrid Metaheuristic Review,” *IEEE Transactions on Intelligent Vehicles* 9, no. 9 (2024): 5813–5852.
 - [67] F. Tlili, S. Ayed, and L. Chaari Fourati, “Advancing UAV Security With Artificial Intelligence: A Comprehensive Survey of Techniques and Future Directions,” *Internet of Things* 27 (2024): 101281.

- [68] R. Shrestha, A. Omidkar, S. A. Roudi, R. Abbas, and S. Kim, "Machine-Learning-Enabled Intrusion Detection System for Cellular Connected UAV Networks," *Electronics* 10, no. 13 (2021): 1549.
- [69] J. Whelan, A. Almeahmadi, and K. El-Khatib, "Artificial Intelligence for Intrusion Detection Systems in Unmanned Aerial Vehicles," *Computers and Electrical Engineering* 99 (2022): 107784.
- [70] J. Tian, B. Wang, R. Guo, Z. Wang, K. Cao, and X. Wang, "Adversarial Attacks and Defenses for Deep-Learning-Based Unmanned Aerial Vehicles," *IEEE Internet of Things Journal* 9, no. 22 (2022): 22399–22409.
- [71] G. Kocher and G. Kumar, "A Hybrid Deep Learning Approach for Effective Intrusion Detection Systems Using Spatial-Temporal Features," *Advanced Engineering Science* 54 (2022): 1503–1519.
- [72] W. Lo, H. Alqahtani, K. Thakur, A. Almadhor, S. Chander, and G. Kumar, "A Hybrid Deep Learning Based Intrusion Detection System Using Spatial-Temporal Representation of in-Vehicle Network Traffic (Best Paper 2023 Award)," *Vehicular Communications* 35 (2022): 100471.
- [73] G. Kocher and G. Kumar, "Machine Learning and Deep Learning Methods for Intrusion Detection Systems: Recent Developments and Challenges," *Soft Computing* 25, no. 15 (2021): 9731–9763.
- [74] I. Sharma, S. K. Gupta, A. Mishra, and S. Askar, "Synchronous Federated Learning Based Multi Unmanned Aerial Vehicles for Secure Applications," *Scalable Computing: Practice and Experience* 24, no. 3 (2023): 191–201.
- [75] Y. Wang, Z. Su, N. Zhang, and A. Benslimane, "Learning in the Air: Secure Federated Learning for UAV-Assisted Crowdsensing," *IEEE Transactions on Network Science and Engineering* 8, no. 2 (2021): 1055–1069.
- [76] N. Nasser, Z. M. Fadlullah, M. M. Fouda, A. Ali, and M. Imran, "A Lightweight Federated Learning Based Privacy Preserving B5G Pandemic Response Network Using Unmanned Aerial Vehicles: A Proof-of-Concept," *Computer Networks* 205 (2022): 108672.
- [77] A. Yazdinejad, R. M. Parizi, A. Dehghantaha, and H. Karimipour, "Federated Learning for Drone Authentication," *Ad Hoc Networks* 120 (2021): 102574.
- [78] J. Liao, B. Jiang, P. Zhao, L. Ning, and L. Chen, "Unmanned Aerial Vehicle-Assisted Federated Learning Method Based on a Trusted Execution Environment," *Electronics* 12, no. 18 (2023): 3938.
- [79] J. Zheng, J. Xu, H. Du, et al., "Trust Management of Tiny Federated Learning in Internet of Unmanned Aerial Vehicles," *IEEE Internet of Things Journal* 11, no. 12 (2024): 21046–21060.
- [80] J. Yao and N. Ansari, "Secure Federated Learning by Power Control for Internet of Drones," *IEEE Transactions on Cognitive Communications and Networking* 7, no. 4 (2021): 1021–1031.
- [81] X. Hou, J. Wang, C. Jiang, X. Zhang, Y. Ren, and M. Debbah, "UAV-Enabled Covert Federated Learning," *IEEE Transactions on Wireless Communications* 22, no. 10 (2023): 6793–6809.
- [82] H. Zhang and L. Hanzo, "Federated Learning Assisted Multi-UAV Networks," *IEEE Transactions on Vehicular Technology* 69, no. 11 (2020): 14104–14109.
- [83] H. B. Tsegaye, "Towards Resilient and Secure Beyond-5G Non-Terrestrial Networks (B5G-NTNS): An End-to-End Cloud-Native Frame-Work," (2024).
- [84] R. Gupta, A. Nair, S. Tanwar, and N. Kumar, "Blockchain-Assisted Secure UAV Communication in 6g Environment: Architecture, Opportunities, and Challenges," *IET Communications* 15, no. 10 (2021): 1352–1367.
- [85] R. Kumar, A. Aljuhani, P. Kumar, A. Kumar, A. Franklin, and A. Jolfaei, "Blockchain-Enabled Secure Communication for Un-manned Aerial Vehicle (UAV) Networks," in *Proceedings of the 5th International ACM Mobicom Workshop on Drone Assisted Wireless Communications for 5G and Beyond*, (ACM, 2022).
- [86] M. Wazid, B. Bera, A. K. Das, S. Garg, D. Niyato, and M. S. Hossain, "Secure Communication Framework for Blockchain-Based Internet of Drones-Enabled Aerial Computing Deployment," *IEEE Internet of Things Magazine* 4, no. 3 (2021): 120–126.
- [87] M. Aloqaily, O. Bouachir, A. Boukerche, and I. A. Ridhawi, "Design Guidelines for Blockchain-Assisted 5G-UAV Networks," *IEEE Network* 35, no. 1 (2021): 64–71.
- [88] S. Aggarwal, N. Kumar, and S. Tanwar, "Blockchain-Envisioned Uav Communication Using 6g Networks: Open Issues, Use Cases, and Future Directions," *IEEE Internet of Things Journal* 8, no. 7 (2021): 5416–5441.
- [89] E. Ghribi, T. T. Khoei, H. T. Gorji, P. Ranganathan, and N. Kaabouch, "A Secure Blockchain-Based Communication Approach for UAV Networks," in *2020 IEEE International Conference on Electro Information Technology (EIT)*, (IEEE, 2020): 411–415.
- [90] S. Hafeez, *Blockchain-Based Secure Unmanned Aerial Vehicles (UAV) in Network Design and Optimization*, (Ph.D. Thesis, University of Glasgow, 2024).
- [91] K. Khullar, Y. Malhotra, and A. Kumar, "Decentralized and Secure Communication Architecture for FANETs Using Blockchain," *Procedia Computer Science* 173 (2020): 158–170.
- [92] D. Saraswat, A. Verma, P. Bhattacharya, et al., "Blockchain-Based Federated Learning in UAVs Beyond 5G Networks: A Solution Taxonomy and Future Directions," *IEEE Access* 10 (2022): 33154–33182.
- [93] R. Sarenche, F. Aghili, T. Yoshizawa, and D. Singelée, "DASLog: Decentralized Auditable Secure Logging for UAV Ecosystems," *IEEE Internet of Things Journal* 10, no. 23 (2023): 20264–20284.
- [94] W. Zhao, I. M. Aldyafiah, P. Gangwani, S. Joshi, H. Upadhyay, and L. Lagos, "A Blockchain-Facilitated Secure Sensing Data Processing and Logging System," *IEEE Access* 11 (2023): 21712–21728.
- [95] R. Alkadi, N. Alnuaimi, C. Y. Yeun, and A. Shoufan, "Blockchain Interoperability in Unmanned Aerial Vehicles Networks: State-of-the-Art and Open Issues," *IEEE Access* 10 (2022): 14463–14479.
- [96] R. Gupta, M. M. Patel, S. Tanwar, N. Kumar, and S. Zeadally, "Blockchain-Based Data Dissemination Scheme for 5g-Enabled Softwarized UAV Networks," *IEEE Transactions on Green Communications and Networking* 5, no. 4 (2021): 1712–1721.
- [97] S. Hafeez, M. A. Shawky, M. Al-Quraan, L. Mohjazi, M. A. Imran, and Y. Sun, "Beta-UAV: Blockchain-Based Efficient Authentication for Secure UAV Communication," *ArXiv* (2024).
- [98] R. Karmakar, G. Kaddoum, and O. Akhrif, "A Blockchain-Based Distributed and Intelligent Clustering-Enabled Authentication Protocol for UAV Swarms," *IEEE Transactions on Mobile Computing* 23, no. 5 (2024): 6178–6195.
- [99] J. C. U. Ortega, J. Rodríguez-Molina, M. Martínez-Núñez, and J. Garbajosa, "A Proposal for Decentralized and Secured Data Collection From Unmanned Aerial Vehicles in Livestock Monitoring With Blockchain and IPFS," *Applied Sciences* 13, no. 1 (2023): 471.

- [100] C. Dong, S. Pal, A. Yao, F. Jiang, S. Chen, and X. Liu, "Optimizing UAV Delivery for Pervasive Systems Through Blockchain Integration and Adversarial Machine Learning," *Computer Communications* 236 (2025): 108113.
- [101] S. Hafeez, H. U. Manzoor, L. Mohjazi, A. Zoha, M. A. Imran, and Y. Sun, "Blockchain-Empowered Immutable and Reliable Delivery Service (Birds) Using Uav Networks," in *2023 IEEE 28th International Workshop on Computer Aided Modeling and Design of Communication Links and Networks (CAMAD)*, (IEEE, 2023): 7–12.
- [102] A. Allouch, O. Cheikhrouhou, A. Kouba^a, K. Toumi, M. Khalgui, and T. Nguyen Gia, "UTM-Chain: Blockchain-Based Secure Unmanned Traffic Management for Internet of Drones," *Sensors* 21, no. 9 (2021): 3049.
- [103] C. Ge, X. Ma, and Z. Liu, "A Semi-autonomous Distributed Blockchain-Based Framework for UAVs System," *Journal of Systems Architecture* 107 (2020): 101728.
- [104] Y. Tan, J. Wang, J. Liu, and N. Kato, "Blockchain-Assisted Distributed and Lightweight Authentication Service for Industrial Unmanned Aerial Vehicles," *IEEE Internet of Things Journal* 9, no. 18 (2022): 16928–16949.
- [105] R. Aissaoui, J. C. Deneuville, C. Guerber, and A. Pirovano, "Evaluating Post-Quantum Key Exchange Mechanisms for UAV Communication Security," in *2024 AIAA DATC/IEEE 43rd Digital Avionics Systems Conference (DASC)*, (IEEE, 2024): 1–10.
- [106] V. K. Ralegankar, J. Bagul, B. Thakkar, et al., "Quantum Cryptography-as-a-Service for Secure UAV Communication: Applications, Challenges, and Case Study," *IEEE Access* 10 (2022): 1475–1492.
- [107] O. A. Hussien, I. S. Arachchige, and H. Jahankhani, "Strengthening Security Mechanisms of Satellites and UAVs Against Possible Attacks From Quantum Computers," in *International Conference on Global Security, Safety, and Sustainability*, (Springer, 2023): 1–20.
- [108] T. Xia, M. Wang, J. He, G. Yang, L. Fan, and G. Wei, "A Quantum-Resistant Identity Authentication and Key Agreement Scheme for UAV Networks Based on Kyber Algorithm," *Drones* 8, no. 8 (2024): 359.
- [109] A. S. Nair, S. M. Thampi, and V. Jafeel, "A Post-Quantum Secure PUF Based Cross-Domain Authentication Mechanism for Internet of Drones," *Vehicular Communications* 47 (2024): 100780.
- [110] S. Goyal, A. S. Rajawat, R. K. Solanki, L. Zhu, and W. Chee, "Enhancing Privacy and Security for UAV and IoT Enabled Drones an Intelligent Integration of Blockchain, AI, and Quantum Computing," in *International Conference on Intelligent Computing & Optimization*, (Springer, 2023): 16–27.
- [111] P. K. Sandanamudi, N. Agrawal, N. Tripathi, and P. K. BN, "Securing UAV Communications: A Comparative Performance Analysis of Post-Quantum Cryptographic Techniques," in *2025 17th International Conference on COMmunication Systems and NETworks (COMSNETS)*, (IEEE, 2025): 1096–1101.
- [112] A. T. Jawad, R. Maaloul, and L. Chaari, "Authentication Communication by Using Visualization Cryptography for UAV Networks," *Computer Standards & Interfaces* 92 (2025): 103918.
- [113] N. Alshaer, A. Moawad, and T. Ismail, "Reliability and Security Analysis of an Entanglement-Based QKD Protocol in a Dynamic Ground-to-UAV FSO Communications System," *IEEE Access* 9 (2021): 168052–168067.
- [114] M. A. El-Zawawy, A. Brighente, and M. Conti, "Authenticating Drone-Assisted Internet of Vehicles Using Elliptic Curve Cryptography and Blockchain," *IEEE Transactions on Network and Service Management* 20, no. 2 (2023): 1775–1789.
- [115] D. Wang, Y. Cao, K. Y. Lam, Y. Hu, and O. Kaiwartya, "Authentication and Key Agreement Based on Three Factors and Puf for UAVs-Assisted Post-Disaster Emergency Communication," *IEEE Internet of Things Journal* 11 (2024): 20457–20472.
- [116] A. S. Khan, M. I. B. Yahya, K. B. Zen, et al., "Blockchain-Based Lightweight Multifactor Authentication for Cell-Free in Ultra-Dense 6G-Based (6-CMAS) Cellular Network," *IEEE Access* 11 (2023): 20524–20541.
- [117] N. Zhang, Q. Jiang, L. Li, X. Ma, and J. Ma, "An Efficient Three-Factor Remote User Authentication Protocol Based on BPV-FourQ for Internet of Drones," *Peer-to-Peer Networking and Applications* 14, no. 5 (2021): 3319–3332.
- [118] M. Usman, R. Amin, H. Aldabbas, and B. Alouffi, "Lightweight Challenge-Response Authentication in SDN-Based UAVs Using Elliptic Curve Cryptography," *Electronics* 11, no. 7 (2022): 1026.
- [119] H. Khalid, S. J. Hashim, S. M. S. Ahamed, F. Hashim, and M. A. Chaudhary, "Secure Real-Time Data Access Using Two-Factor Authentication Scheme for the Internet of Drones," in *2021 IEEE 19th Student Conference on Research and Development (SCoReD)*, (IEEE, 2021): 168–173.
- [120] C. Tian, Q. Jiang, T. Li, J. Zhang, N. Xi, and J. Ma, "Reliable PUF-Based Mutual Authentication Protocol for UAVs Towards Multi-Domain Environment," *Computer Networks* 218 (2022): 109421.
- [121] G. Bansal and B. Sikdar, "Achieving Secure and Reliable UAV Authentication: A Shamir's Secret Sharing Based Approach," *IEEE Transactions on Network Science and Engineering* 11, no. 4 (2024): 3598–3610.
- [122] H. M. Ismael and Z. T. M. Al-Ta'i, "Authentication and Encryption Drone Communication by Using Hight Lightweight Algorithm," *Turkish Journal of Computer and Mathematics Education* 12 (2021): 5891–5908.
- [123] C. Dong, F. Jiang, S. Chen, and X. Liu, "Continuous Authentication for UAV Delivery Systems Under Zero-Trust Security Framework," in *2022 IEEE International Conference on Edge Computing and Communications (EDGE)*, (IEEE, 2022).
- [124] T. Alladi, Naren, G. Bansal, V. Chamola, and M. Guizani, "Secauthuav: A Novel Authentication Scheme for UAV-Ground Station and UAV-UAV Communication," *IEEE Transactions on Vehicular Technology* 69, no. 12 (2020): 15068–15077.
- [125] C. Zhang, X. Costa-Perez, and P. Patras, "Adversarial Attacks Against Deep Learning-Based Network Intrusion Detection Systems and Defense Mechanisms," *IEEE/ACM Transactions on Networking* 30, no. 3 (2022): 1294–1311.
- [126] T. Tsai, K. Yang, T. Y. Ho, and Y. Jin, "Robust adversarial objects against deep learning models," *Proceedings of the AAAI Conference on Artificial Intelligence* 34: 954–962.
- [127] Q. Tian, S. Zhang, S. Mao, and Y. Lin, "Adversarial Attacks and Defenses for Digital Communication Signals Identification," *Digital Communications and Networks* 10, no. 3 (2024): 756–764.
- [128] X. Zhang, X. Zheng, and W. Mao, "Adversarial Perturbation Defense on Deep Neural Networks," *ACM Computing Surveys (CSUR)* 54 (2021): 1–36.
- [129] H. Waghela, J. Sen, and S. Rakshit, "Robust Image Classification: Defensive Strategies Against FGSM and PGD Adversarial Attacks," *ArXiv* (2024): 1–7.
- [130] S. Tiwari, V. Srestha, and A. Srivastava, "The Role of Explainable AI in Cybersecurity: Addressing Transparency Challenges in

- Autonomous Defense Systems,” *International Journal of Innovative Research in Science, Engineering and Technology* 09, no. 3 (2020): 718–733.
- [131] N. Moustafa, N. Koroniotis, M. Keshk, A. Y. Zomaya, and Z. Tari, “Explainable Intrusion Detection for Cyber Defences in the Internet of Things: Opportunities and Solutions,” *IEEE Communications Surveys & Tutorials* 25, no. 3 (2023): 1775–1807.
- [132] M. T. Masud, M. Keshk, N. Moustafa, I. Linkov, and D. K. Emge, “Explainable Artificial Intelligence for Resilient Security Applications in the Internet of Things,” *IEEE Open Journal of the Communications Society* 6 (2024): 2877–2906.
- [133] S. Wu, Y. Li, Z. Wang, Z. Tan, and Q. Pan, “A Highly Interpretable Framework for Generic Low-Cost UAV Attack Detection,” *IEEE Sensors Journal* 23, no. 7 (2023): 7288–7300.
- [134] S. O. Ajakwe and D.-S. Kim, “Facets of Security and Safety Problems and Paradigms for Smart Aerial Mobility and Intelligent Logistics,” *IET Intelligent Transport Systems* 18, no. S1 (2024): 2827–2855.
- [135] B. Chander, C. John, L. Warriar, and K. Gopalakrishnan, “Toward Trustworthy Artificial Intelligence (Tai) in the Context of Explain-Ability and Robustness,” *ACM Computing Surveys* 57 (2025): 1–49.
- [136] C. Maathuis, “Towards Trustworthy AI-Based Military Cyber Operations,” in *19th International Conference on Cyber Warfare and Security: ICCWS 2024*, (Academic Conferences and Publishing Limited, 2024).
- [137] L. Wang, Y. Chen, P. Wang, and Z. Yan, “Security Threats and Countermeasures of Unmanned Aerial Vehicle Communications,” *IEEE Communications Standards Magazine* 5, no. 4 (2021): 41–47.
- [138] D. He, G. Yang, H. Li, S. Chan, Y. Cheng, and N. Guizani, “An Effective Countermeasure Against UAV Swarm Attack,” *IEEE Network* 35, no. 1 (2021): 380–385.
- [139] Y. Jiang, J. Yuan, L. Sun, and H. Song, “Development of a Laboratory Platform for UAV Cybersecurity Education,” in *2021 ASEE Virtual Annual Conference*, (ASEE, 2021).
- [140] C. Jie, “Research on Intelligentized Anti-UAV Command Control Scheme Technology,” in *E3S Web of Conferences*, (EDP Sciences, 2021).
- [141] J. Wang, Y. Liu, and H. Song, “Counter-Unmanned Aircraft System(s) (C-UAS): State of the Art, Challenges, and Future Trends,” *IEEE Aerospace and Electronic Systems Magazine* 36, no. 3 (2021): 4–29.
- [142] C. S. Babu and A. Pal, “Enhancing Security for Unmanned Aircraft Systems in IoT Environments: Defense Mechanisms and Mitigation Strategies,” in *Unmanned Aircraft Systems*, (Wiley, 2024): 429–476.
- [143] N. Malik, H. Sinha, and M. Dahiya, “Security in UAV Ecosystem: An Implementation Perspective,” *Sigma Journal of Engineering and Natural Sciences* 42 (2024): 1986–1994.
- [144] M. Ficco, D. Granata, F. Palmieri, and M. Rak, “A Systematic Approach for Threat and Vulnerability Analysis of Unmanned Aerial Vehicles,” *Internet of Things* 26 (2024): 101180.
- [145] D. Patil and S. Pournouri, “Evaluating the Security Of Open-Source Linux Operating Systems For Unmanned Aerial Vehicles,” in *International Conference on Global Security, Safety, and Sustainability*, (Springer, 2023): 21.
- [146] X. Wei, Y. Xu, H. Zhang, et al., “Sensor Attack Online Classification for UAVs Using Machine Learning,” *Computers & Security* 150 (2025): 104228.
- [147] J. P. Dias, T. B. Sousa, A. Restivo, and H. S. Ferreira, “A pattern-Language for Self-Healing Internet-of-Things Systems,” in *Proceedings of the European Conference on Pattern Languages of Programs 2020*, (Association for Computing Machinery, 2020): 1–17.
- [148] J. M. Qurashi, K. Jambi, F. Alsolami, F. E. Eassa, M. Khemakhem, and A. Basuhail, “Resilient Countermeasures Against Cyber-Attacks on Self-Driving Car Architecture,” *IEEE Transactions on Intelligent Transportation Systems* 24, no. 11 (2023): 11514–11543.
- [149] A. Phadke and F. A. Medrano, “Towards Resilient UAV Swarms—A Breakdown of Resiliency Requirements in UAV Swarms,” *Drones* 6, no. 11 (2022): 340.
- [150] J. P. A. Yaacoub, H. N. Noura, O. Salman, and A. Chehab, “Robotics Cyber Security: Vulnerabilities, Attacks, Countermeasures, and Recommendations,” *International Journal of Information Security* 21, no. 1 (2022): 115–158.
- [151] M. Adil, H. Song, S. Mastorakis, H. Abulkasim, A. Farouk, and Z. Jin, “Uav-Assisted Iot Applications, Cybersecurity Threats, Ai-Enabled Solutions, Open Challenges With Future Research Directions,” *IEEE Transactions on Intelligent Vehicles* 9, no. 4 (2024): 4583–4605.
- [152] B. S. Sarikaya and S. Bahtiyar, “A Survey on Security of UAV and Deep Reinforcement Learning,” *Ad Hoc Networks* 164 (2024): 103642.
- [153] P. Homann, J. Diegelmann, M. Pacher, and U. Brinkschulte, “Evaluation of Trust Metrics in an Artificial Hormone System,” in *2024 IEEE 27th International Symposium on Real-Time Distributed Computing (ISORC)*, (IEEE, 2024): 1–12.
- [154] J. P. A. Yaacoub, *Modular Robotics Meet Internet of Things: Safety, Security and Performance Challenges and Countermeasures*, (Ph.D. Thesis, Université Bourgogne Franche-Comté, 2024).
- [155] I. Al Ridhawi, O. Bouachir, M. Aloqaily, and A. Boukerche, “Design Guidelines for Cooperative UAV-Supported Services and Applications,” *ACM Computing Surveys (CSUR)* 54 (2021): 1–35.
- [156] G. Mani, B. K. Bhargava, J. Kobes, J. King, and J. MacDonald, *Securing Intelligent Autonomous Systems Through Artificial Intelligence (ISIC)*, (2021).
- [157] S. Wei, Z. Fan, G. Chen, E. Blasch, Y. Chen, and K. Pham, “Tadad: Trust AI-Based Decentralized Anomaly Detection For Urban Air Mobility Networks At Tactical Edges,” in *2024 Integrated Communications, Navigation and Surveillance Conference (ICNS)*, (IEEE, 2024): 1–10.